
Supplementary information

**Fischer–Tropsch synthesis to olefins
boosted by MFI zeolite nanosheets**

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Supplementary Information

Fischer-Tropsch synthesis to olefins boosted by MFI zeolite nanosheets

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Materials

Tetraethylorthosilicate (TEOS, >99%), diethoxydimethylsilane (DEMS, 99%), aluminum isopropoxide (99.99%), 1-hexene (99%, AR), 1-octene (98%, AR), and SiO₂ (AR) were obtained from Aladdin Chemical Reagent Company. FeCl₃·6H₂O, FeCl₂·4H₂O, HCl, and NaOH were obtained from Sinopharm Chemical Reagent Co. Ltd. Tetrapropylammonium hydroxide (TPAOH, 25 wt%) was supplied by Jiangsu Kente Chemical Technology Co. Ltd. Tetrapropylammonium hydroxide (TPAOH, 40 wt%) was supplied by Shanghai Cairui Chemical Technology Co. Ltd. Boehmite (AlOOH) was supplied by Changling Catalyst Co. Ltd. Beta, MOR, and SAPO-34 zeolites were purchased from Tianjin Nankai Catalyst Co. Ltd. SSZ-13 and ZSM-22 zeolites were purchased from Zhejiang Taide catalyst Co. Ltd. H₂, CO/Ar, H₂/CO/Ar, H₂/Ar, C₂H₄, C₃H₆, and C₄H₈ were supplied by Hangzhou Jingong special gas Co. Ltd. Tungsten trioxide (WO₃, 99.8%) was obtained from Shanghai Macklin Biochemical Co. Ltd.

Catalyst preparation

Synthesis of Na-Fe₃O₄ and Na-FeC_x catalysts. Typically, 30.0 g of FeCl₃·6H₂O and 11.9 g of FeCl₂·4H₂O were dissolved in a mixture of deionized water (200 mL) and aqueous HCl solution (5.0 mL, 12.0 mol L⁻¹). Then NaOH aqueous solution (2.0 mol L⁻¹) was added dropwise into the solution under vigorous stirring until the pH value reaching to 10. After stirring for another 1 h, the solid phase was separated from the solution *via* centrifugation. After washing with a large amount of deionized water for six times and drying at 100 °C overnight, the Na-Fe₃O₄ sample was finally obtained. The Na-FeC_x was prepared by treating the Na-Fe₃O₄ sample with syngas (molar ratio of CO to H₂ at 1:1) at 300 °C for 24 h with a gas flow rate 50 mL/min, which was denoted as the Na-FeC_x. The Na loading of the Na-FeC_x was 5.35% by ICP analysis. Different Na loading of the Na-FeC_x was prepared by washing with water for different times, denoted as nNa-FeC_x, where the n stands for the Na loadings. The Na-FeC_x means that the Na loading is 5.35 wt% unless otherwise statement.

Synthesis of *n*-ZSM-5 zeolite. The normal ZSM-5 zeolite (denoted as *n*-ZSM-5) was prepared by a conventional hydrothermal method. In a typical run, 3.5 g of TEOS was added into a mixture containing 11.28 g of deionized water and 3.3 g of tetrapropylammonium hydroxide (TPAOH, 40 wt %), followed by adding 0.03 g of AlOOH and stirring at room temperature for 12 h. Then, the mixture was transfer into an autoclave and heated at 180 °C for 3 days. After removal of the organic template by calcination at 550 °C for 4 h, the *n*-ZSM-5 zeolite with Si/Al ratio at 30.5

was finally obtained.

Synthesis of *s*-S1 zeolite. The S-1 zeolite with sheet morphology and *b*-axis thickness at 80-100 nm (denoted as *s*-S1, where *s* stands for sheet) was synthesized from a urea-assisted method. As a typical run, 4.0 g of TPAOH (25% aqueous solution) was mixed with 5.1 g of deionized water under stirring for 0.5 h, then 2.8 g of TEOS was added to the solution under stirring. After adding 0.65 g of urea, the mixture was continuously stirred at room temperature for another 12 h. The resulting gel was transferred into an autoclave for further crystallization at 180 °C for 2 days. The as-synthesized products were collected by filtrating, washing with water, drying in air, and calcining at 550 °C for 4 h to remove the organic template.

Synthesis of *n*-S1 zeolite. The normal S-1 zeolite (*n*-S1) was prepared by a conventional hydrothermal method. In a typical run, 3.5 g of TEOS was added into a mixture containing 11.28 g of deionized water and 3.3 g of TPAOH (40 wt %), followed by stirring at room temperature for 12 h. Then, the mixture was transfer into an autoclave and heated at 180 °C for 3 days. After removal of the organic template by calcination at 550 °C for 4 h, the *n*-S1 zeolite was finally obtained.

Synthesis of *n*-S1-OH zeolite. The S-1 zeolite with silanol groups (*n*-S1-OH) was prepared by solvent-free crystallization. As a typical run, 64 mL of ethanol was added to 80 mL of a water solution containing 6 mL of aqueous ammonia with stirring. After addition of 6.0 g of TEOS and 0.474 g of DEMS, the solution was stirred at room temperature for another 12 h, followed by distillation under vacuum to remove the water and ethanol and drying at 100 °C overnight. Then the solid powder of amorphous silica modified with methyl groups (SiO₂-Me) was obtained. The S-1-OH was synthesized by solvent-free crystallization with SiO₂-Me in the presence of TPAOH. After grinding of 1.6 g of TPAOH and 2 g of SiO₂-Me at room temperature for 0.5 h, the mixture was transferred into an autoclave for crystallization at 180 °C for 3 days to give the methyl group-modified S-1. The as-synthesized products were collected by filtrating, washing with water, drying in air, and calcining at 550 °C for 4 h for removal of the organic template and transformation of the methyl groups into silanol groups. Finally, the *n*-S1-OH zeolite was obtained.

Catalyst characterization

Powder X-ray diffraction (XRD) patterns were collected on a Rigaku D/MAX 2550 diffractometer with CuK α radiation ($\lambda = 1.5418 \text{ \AA}$). Nitrogen sorption isotherms were measured using a Micromeritics ASAP 2020M system. Transmission electron microscopy (TEM) images, scanning transmission electron microscopy (STEM) images, and Energy dispersive spectrometer (EDS) spectra were recorded on a JEM-2100F electron microscope (JEOL, Japan) with an acceleration voltage of 200 kV. Scanning electron microscopy (SEM) experiments were performed with Hitachi SU-8010 and SU-1510 electron microscopes. NH₃ temperature-programmed desorption (NH₃-TPD) was performed on a catalyst analyzer BELCAT II instrument. ²⁹Si MAS NMR was performed on a Bruker AVANCE-III 500 MHz spectrometer operating at Larmor frequencies of 500.57 and 99.44 MHz for the ¹H and ²⁹Si nucleus, respectively. ²⁹Si MAS NMR spectra with high power proton decoupling were recorded using a $\pi/2$ pulse of 3.9 μ s and a recycle delay of 60 s on a 7 mm MAS probe with a spinning rate of 4 kHz. A contact time of 4 ms and a recycle delay of 3 s were used for the ²⁹Si-¹H CP MAS measurement. The chemical shift of ²⁹Si was externally referenced to kaolinite (-91.5 ppm). IR spectra was performed with Perkin-Elmer Frontier system. Metal loadings and Si/Al ratios of the catalysts were determined by inductively coupled plasma analysis (ICP, Perkin-Elmer 3300DV). Gas chromatograph-mass spectrometer (GC-MS) analysis was performed with a Shimadzu GCMS-QP2020-NX system. X-ray photoelectron spectroscopy (XPS) measurements were carried out using a Thermo Fisher Scientific ESCALAB 250Xi with photoelectron spectroscopy system using a monochromatic Al K α (1486.6 eV) X-ray source. The instrument work function was calibrated to give an Au4f_{7/2} metallic gold binding energy of 83.96 eV. The spectrometer dispersion was adjusted to give a binding energy of 368.21 eV for metallic Ag3d_{5/2} and 932.62 eV for metallic Cu2p_{3/2}. The XPS spectra were collected in constant analyzer energy (CAE) mode. Instrument base pressure was 2.0 $\times$ 10⁻¹⁰ mbar and high-resolution spectra were collected using a spot size of 500 μ m, with pass energy of 30 eV. The charge neutralization being monitored using the C1s (284.8 eV) signal for adventitious carbon. The granule-mixed catalysts were grinded into mixed powder before the Na1s XPS characterization.

Temperature-programmed desorption of C₂H₄ (C₂H₄-TPD) was performed on a catalyst analyzer BELCAT II instrument. Typically, the sample was pretreated with He flow at 400 °C for 1 h. Then, the He flow was switched off and a C₂H₄/He (20 vol% C₂H₄) gas mixture was induced for C₂H₄ adsorption. After adsorption at 40 °C for 1 h, the gas feed was switched to He for a swept treatment at 40 °C for 0.5 h. C₂H₄-TPD profiles were collected in flowing He by

raising the temperature from 40 to 700 °C with ramping rate at 10 °C/min. The C₃H₆-TPD tests were performed from the same procedures except for using C₃H₆/He (10 vol% C₃H₆) as the feed gas for pre-adsorption.

C₂H₄(C₃H₆)-desorption DRIFT spectra were recorded using a Nicolet is50 FT-IR spectrometer equipped with an MCT/A detector and ZnSe windows and a high-temperature reaction chamber. As a typical run, the catalyst was localized in the sample cell and pre-treated in pure Ar at 400 °C for 1 h (flow rate at 30 mL/min), and then the temperature was decreased to 30 °C. After maintaining the gas flow for 0.5 h, the background data were collected. Then the pure C₂H₄ was introduced into the sample with a flow rate of 30 mL/min for 0.5 h to ensure the saturated adsorption. Then, the gas flow was switched to pure Ar with a flow rate of 10 mL/min to start the desorption step. After desorption for 4 min, the IR spectra were recorded at intervals of two minutes. The C₃H₆-desorption DRIFT spectra were recorded from the same procedures except for using C₃H₆/He (50 vol% C₃H₆) as the feed gas for pre-adsorption.

C₃H₆ hydrogenation in pulsed experiment

C₃H₆ hydrogenation was performed using a chemisorption instrument (iChem700, PhysiChem Instruments Ltd.) connected to a mass spectrometer (SRD200M, TILON GRP TECHNOLOGY LIMITED). As a typical run, 0.1 g of Na-FeC_x (40-60 mesh) was mixed with 0.3 g of *s*-ZSM-5 (40-60 mesh) or 0.3 g of *n*-ZSM-5 (40-60 mesh) in the reactor tube. The temperature of catalyst bed was raised to 260 °C in a pure Ar flow (200 mL/min). After that, the gas was switched to a flowing hydrogen gas (10%H₂/Ar, flow rate at 200 mL/min). Then, a slight amount of C₃H₆ (1 mL of mixed gas containing 10% C₃H₆ in Ar) was pulsed into the system, the signals of C₃H₆ and C₃H₈ were collected by a mass detector. The pulsed experiment was repeated for multiple times (C₃H₆ was pulsed every 4 min) in each test.

Computational Models

Model A (Fig. 4a, Supplementary Fig. 53) simulating the Na-FeC_x catalyst: The Na₂O/Fe₅C₂ was constructed to represent the Na-FeC_x catalyst¹. The optimized Fe₅C₂ unit cell was a monoclinic crystal and had C2/c crystallographic symmetry, whose lattice constants are *a* = 11.588 Å, *b* = 4.579 Å, *c* = 5.059 Å and β = 97.75°, showing an excellent agreement with experiments, in which *a*, *b*, and *c* equal 11.562, 4.573 and 5.060 Å respectively, and β equals 97.74°(2). To rationalize the experimental observations of this work, we used the 2 × 2 supercell Fe₅C₂ with Na₂O. In particular, the Na-FeC_x surface contains 96 Fe, 48 C, 1 O and 2 Na atoms, in which the bottom

40 Fe and 16 C were fixed, and the rest atoms were relaxed. These models were used for calculating the adsorption energy between ethene and Na-FeC_x.

Model B simulating the Na-FeC_x/s-ZSM-5 catalyst: The simplified Na-FeC_x (Fig. 4b, Supplementary Fig. 54) and Na-FeC_x/s-ZSM-5 surface model (Fig. 4c, Supplementary Fig. 56) were constructed to investigate the diffusion of ethene. The initial framework structure of pure silicon MFI was taken from the International Zeolite Associations database³ and Si(12)-O(26)-Al(12) site was chosen as the locations of Brønsted acidic proton. For surface model, the $2 \times 1 \times 3$, $2 \times 2 \times 3$, $2 \times 4 \times 3$ and $2 \times 8 \times 3$ supercells were selected to represent ZSM-5 with 1-, 2-, 4-, and 8-unit layers. Then, the s-ZSM-5 surface was obtained by disrupting the periodicity of the bulk zeolite and 40 Å vacuum layer were added also along [010] (Supplementary Fig. 61). It should be noted that, the hydroxyl groups were added to the unsaturated atoms over the external surface. Due to the strong interacting between the Fe and ethene, only the Fe atoms have been built to simplify the models of Na-FeC_x surface. The distance between two Fe atoms is about 2.6 Å which could be subtle tuned to match the zeolites periodicity. Six layers of Na-FeC_x were built while two left layers are C (with the similar force field parameter of C from ethene) to get the single-sided Na-FeC_x model. The loading number ethene is 20 and the distance between Na-FeC_x and s-ZSM-5 is about 30 Å. The lattice parameters are $40.18 \times 158.95 \times 39.43$ Å for both Na-FeC_x and Na-FeC_x/s-ZSM-5.

DFT simulation

All geometry optimizations were performed according to the plane-wave-based periodic DFT method as implemented in the Vienna Ab Initio Simulation Package^{4,5}. The electron-ion interaction was described with the projector augmented wave method^{6,7}. The electron exchange and correlation energies were treated within the generalized gradient approximation in the Perdew-Burke-Ernzerhof functional (GGA-PBE)⁸. The energy cutoff of the plane-wave basis was set up to 400 eV. Electron smearing was used *via* the Methfessel-Paxton technique to speed up the convergence of the metallic systems with a width of $\sigma = 0.2$ eV and the $1 \times 1 \times 1$ gamma-point sampling was used for geometry optimization.

The adsorption energy was calculated according to $E_{\text{ads}} = E_{\text{x/slab}} - [E_{\text{slab}} + E_{\text{x}}]$, where $E_{\text{x/slab}}$ was the total energy of the slab with adsorbates in its equilibrium geometry, E_{slab} was the total energy of the bare slab, and E_{x} was the total energy of the free adsorbates in the gas phase. Therefore, the more negative for the value of E_{ads} , the stronger for the adsorption.

Molecular dynamics (MD) simulation

MD simulations were performed in the canonical ensemble (NVT), where the number of particles (N), volume (V), and temperature (T) were kept constant. The simulated temperature was 260 °C and controlled by a Nosé-Hoover thermostat with a coupling time constant of 0.1 ps. The leapfrog Verlet algorithm was used to integrate the Newton's equations of motion with a time step of 0.5 fs. The TraPPE-UA⁹ and TraPPE-zeo¹⁰⁻¹² force field was used for hydrocarbon and zeolites (Supplementary Table 16), respectively. All the Lennard-Jones cross-interaction parameters were determined by the Lorentz-Berthelot mixing rules, while the parameters for Fe₅C₂ and C₂H₄ were got by fitting the adsorption energy between ethene and the outermost layer of Na₂O/Fe₅C₂ ($\epsilon_{\text{Fe-C}} = 0.169$ kcal/mol and $\sigma_{\text{Fe-C}} = 3.675$ Å). Each MD simulation was equilibrated for 2×10^6 steps, then following 2×10^7 to 3×10^7 steps production for studying the diffusion behaviors of adsorbate molecules. The cutoff radius was 14 Å, and the trajectories were recorded every 1000 steps. 10 independent MD simulations were carried out for better statistics. All MD simulation were performed in the DL_POLY 2.0 code¹³.

Mean square displacement and diffusion coefficient.

The mean square displacement (MSD) of ethene was defined as equation in the following:

$$MSD(\tau) = \frac{1}{N_m} \sum_i^{N_m} \frac{1}{N_\tau} \sum_{t_0}^{N_\tau} \left[r_i(t_0 + \tau) - r_i(t_0) \right]^2 \quad (1)$$

where N_m was the number of gas molecules, N_τ was the number of time origins used in calculating the average, and r_i was the coordinate of the i -th molecule. In addition, the slope of the MSD as a function of time determined the self-diffusion coefficient, D_s , which was defined according to the so-called Einstein relation.

$$MSD(\tau) = 2nD_s\tau + b \quad (2)$$

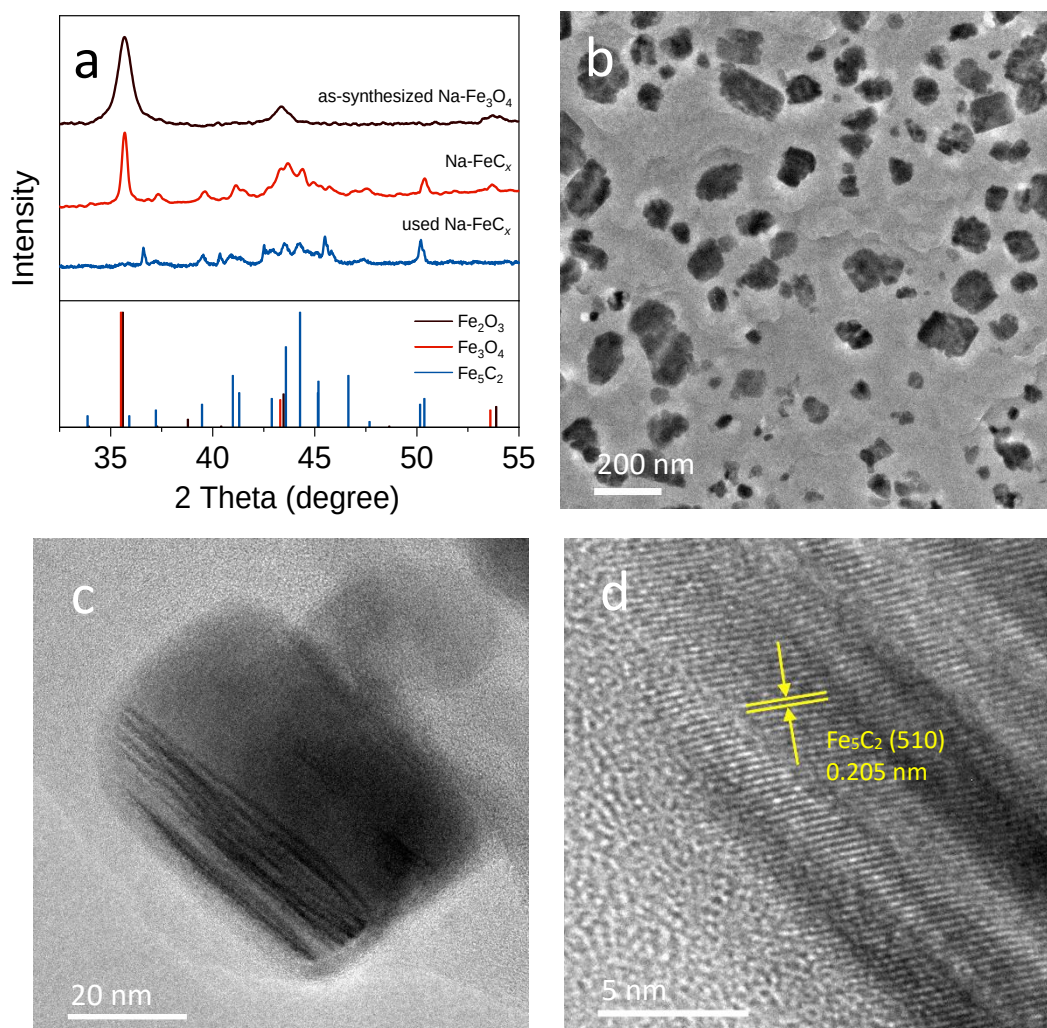
where n was the dimension of zeolites. In this work, $n = 1$ for the diffusion along with straight channel of ZSM-5. The reported MSD and D_s values were calculated as the average of 10 independent MD trajectories.

Interaction energy profiles and energy barriers

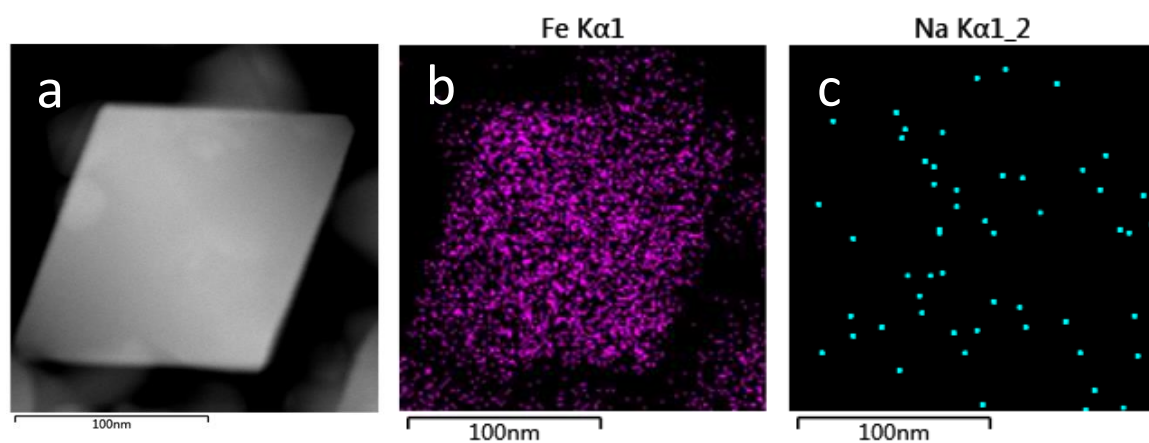
The interaction energy profiles^{14,15} along with straight channel of *s*-ZSM-5 with different layers (*i.e.*, 1, 2, 4 and 8 layers) have been performed (Supplementary Fig. 60). In the calculations, an

ethene molecule was firstly placed near the surface, and then systematically moved along with straight channel following the diffusion path in equidistant steps of 1 Å. The interaction energy between framework and molecule at each point was calculated and the energy barrier for crossing channel was determined by the difference between the local lowest and the highest energy along the diffusion pathway.

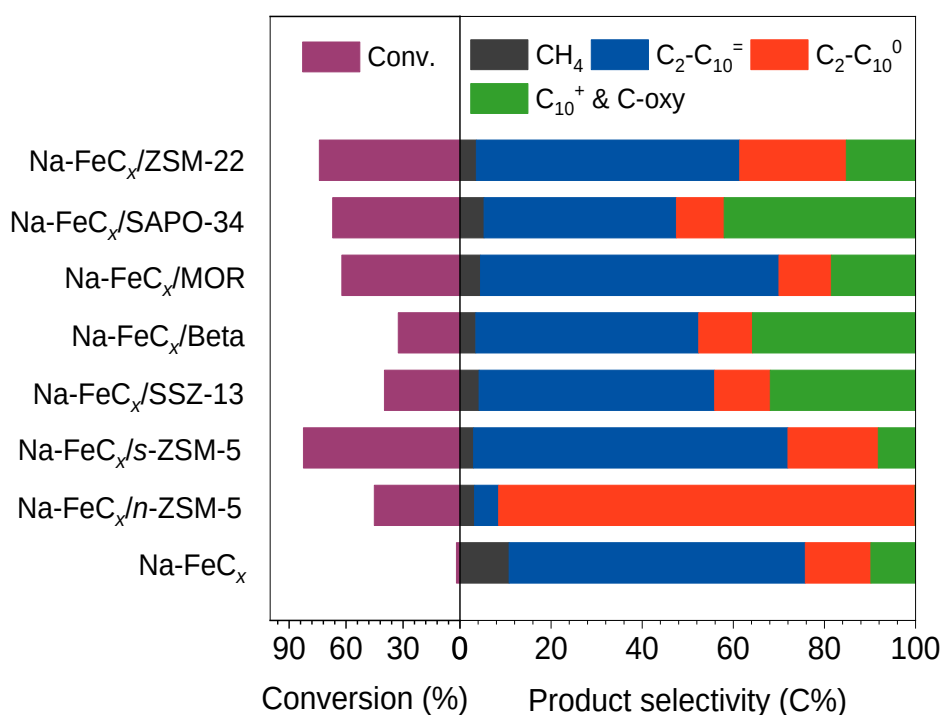
Supplementary Figures



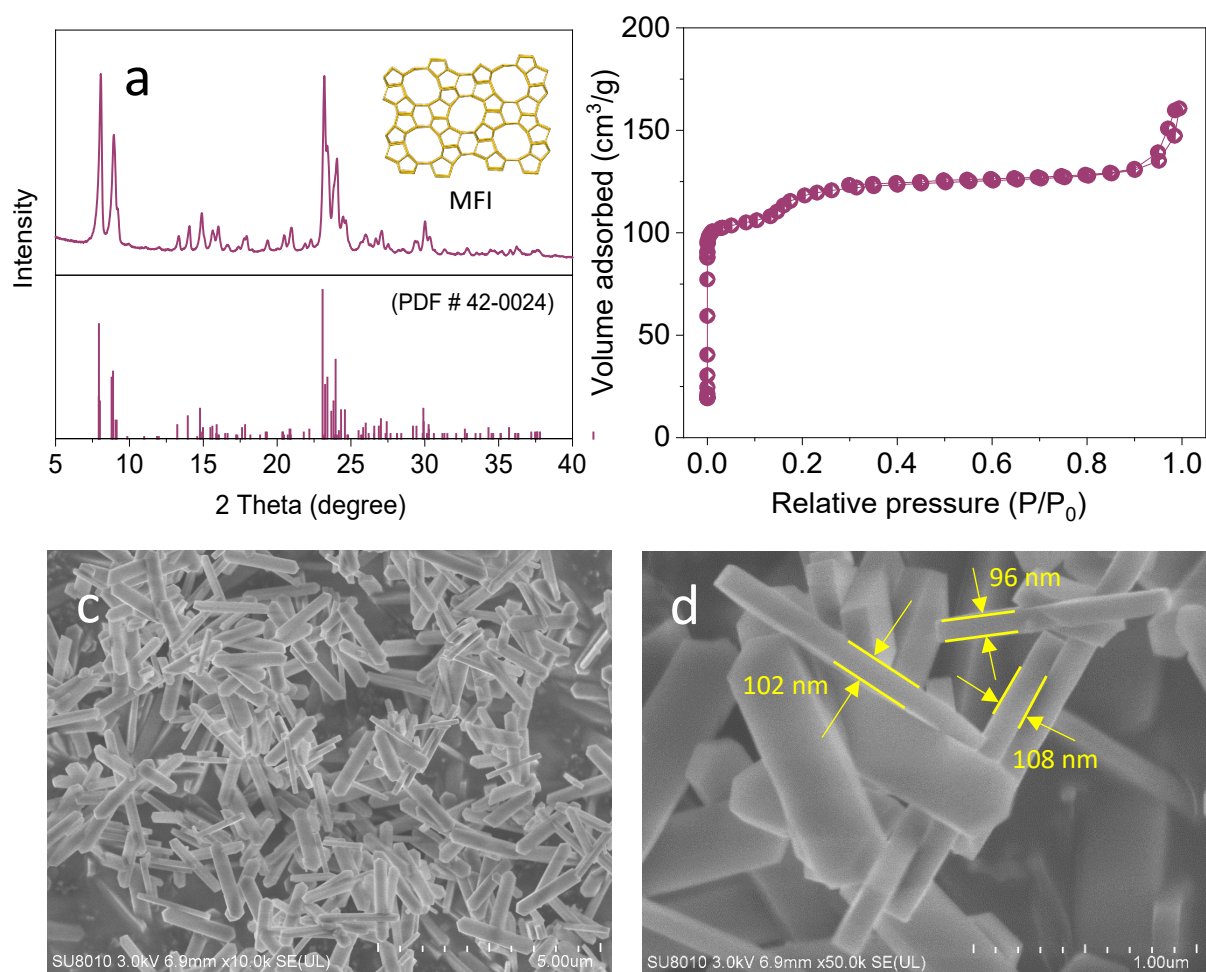
Supplementary Fig. 1. (a) XRD patterns of the as-synthesized Na-Fe₃O₄, Na-FeC_x, and used Na-FeC_x after LT-FTO reaction for 50 h, the standard XRD patterns of Fe₂O₃, Fe₃O₄, and Fe₅C₂ according to PDF # 04-0755, PDF # 03-0863, and PDF # 20-0509, respectively. The Na-FeC_x still has obvious signals of ferric oxides due to the incomplete carbonization. After reaction, the used Na-FeC_x shows sole signals of ferric carbide with almost undetectable oxide phase due to carbonization during the reaction. (b-d) TEM images of the Na-FeC_x.



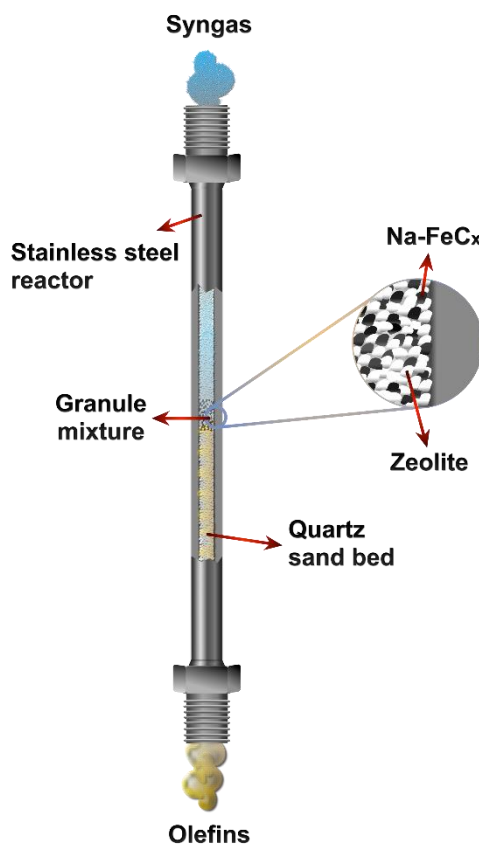
Supplementary Fig. 2. (a) HAADF-STEM image and EDS elemental maps of (b) Fe and (c) Na signals of the Na-FeC_x.



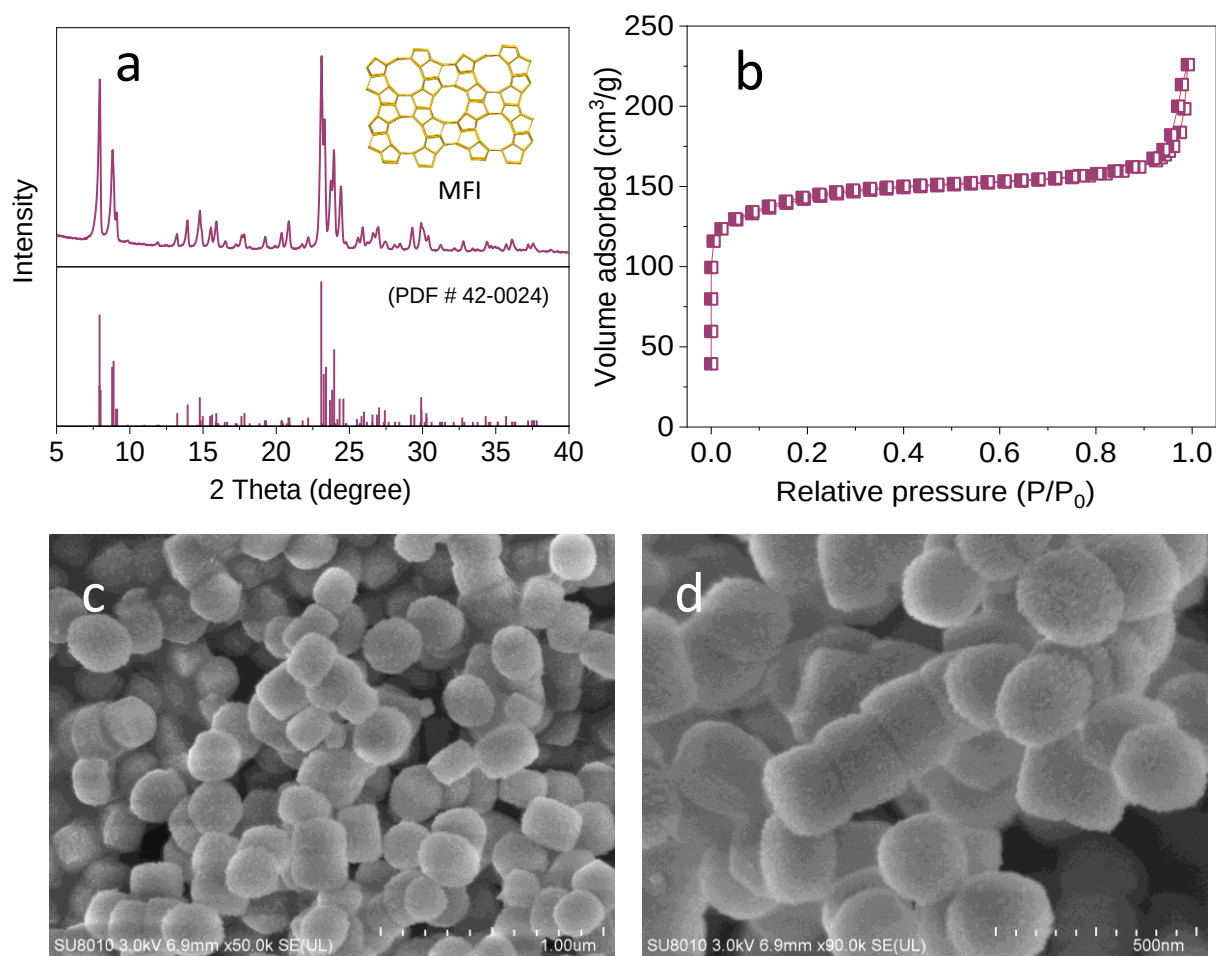
Supplementary Fig. 3. Catalytic data of various catalysts in the LT-FTO. Reaction conditions: weight ratio of zeolite to Na-FeC_x at 3, feed gas with ratio of CO/H₂/Ar at 45/45/10, 2400 mL·h⁻¹·g⁻¹, 2.0 MPa, 260 °C, and granule-mixing of the zeolite and Na-FeC_x individuals. C-oxy is the carbon-containing oxygenates. The aromatic products, if existed, were included in the C₂-C₁₀⁰ and C₁₀⁺ products.



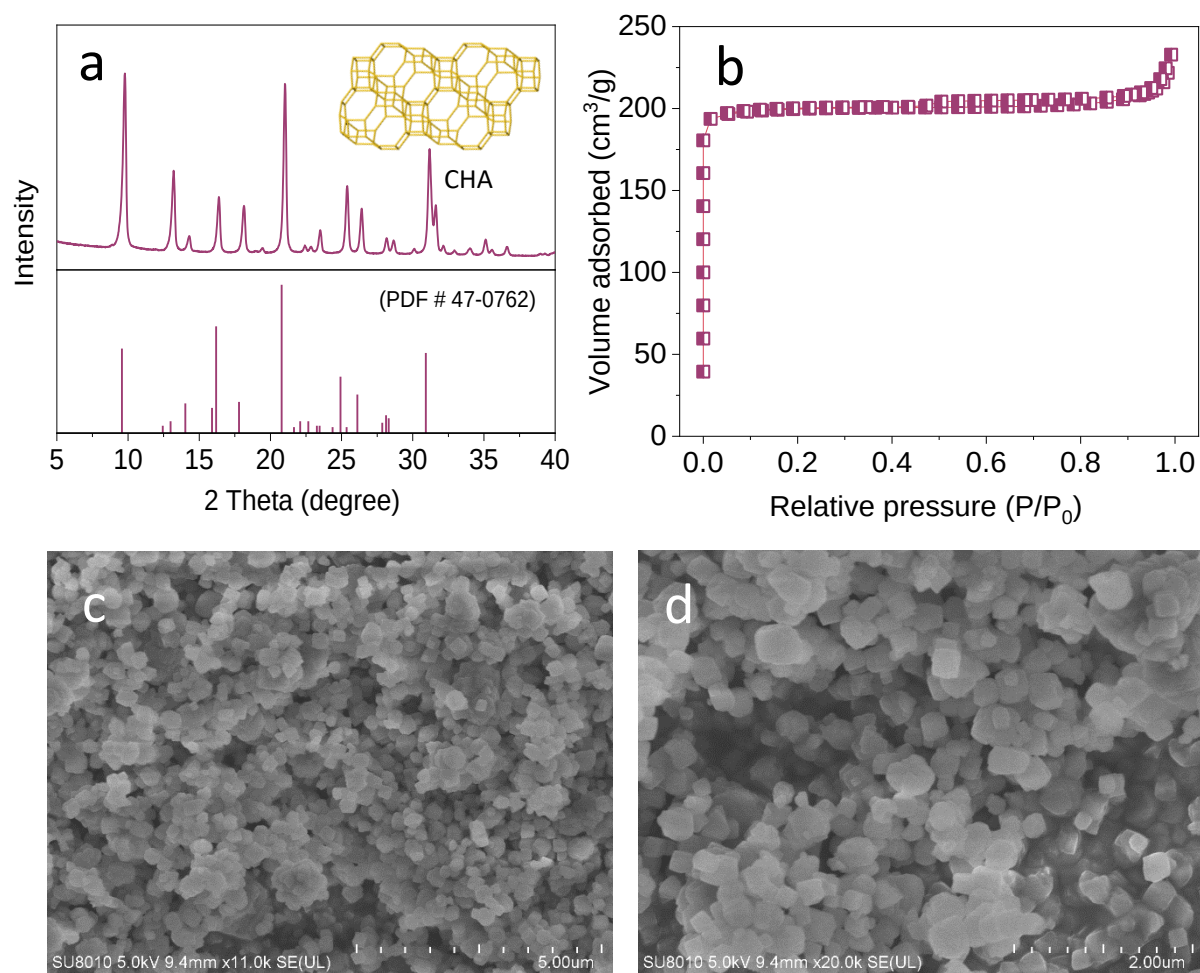
Supplementary Fig. 4. (a) XRD pattern, (b) N₂ sorption isotherms, and (c,d) SEM images of the *s*-ZSM-5 with Si/Al ratio of 26.2. Insets in (a) showed the framework and standard XRD pattern of MFI structure according to PDF # 42-0024.



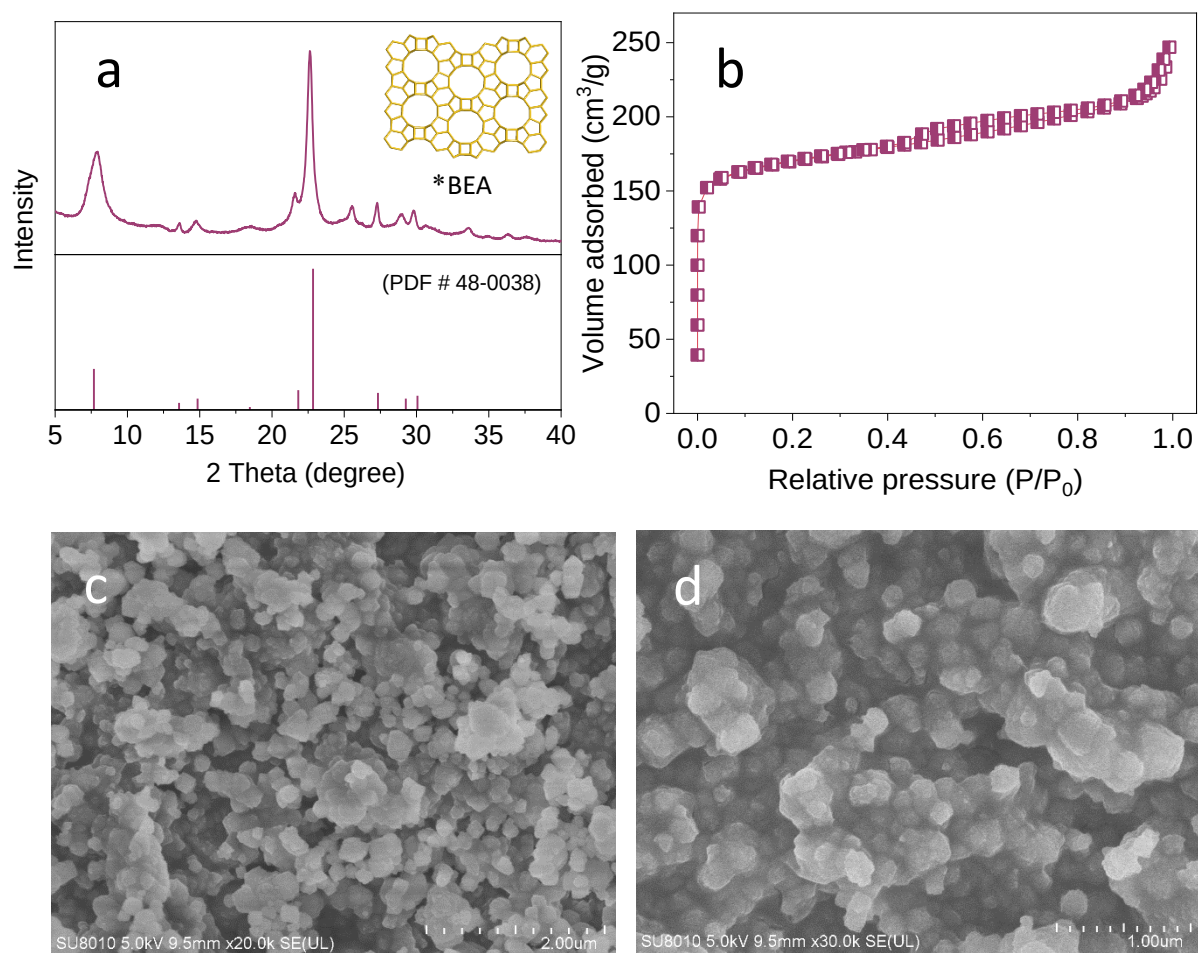
Supplementary Fig. 5. The Na-FeC_x and zeolite were made into granules with sizes at 20-40 mesh and mixed in the reactor (granule mixture).



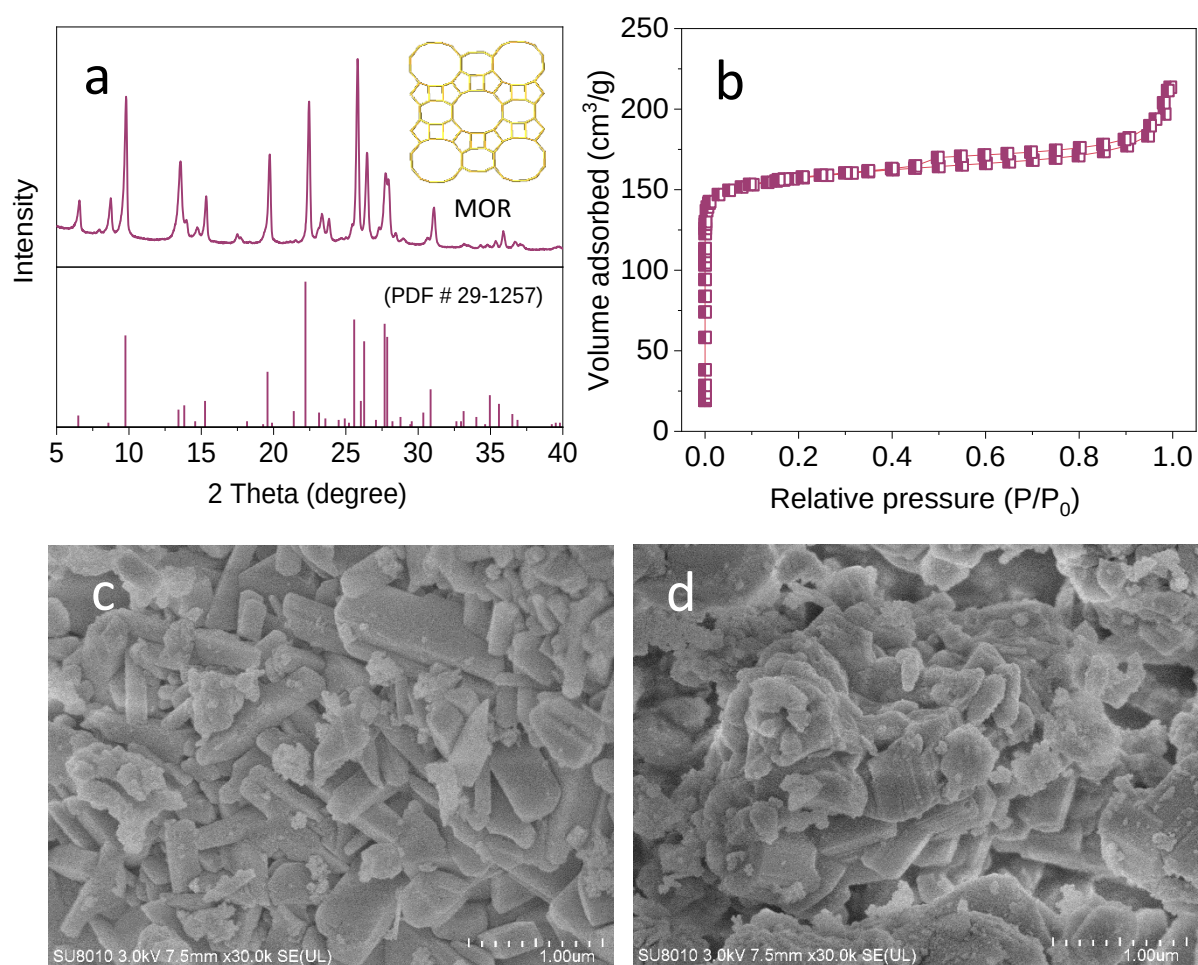
Supplementary Fig. 6. (a) XRD pattern, (b) N₂ sorption isotherms, and (c,d) SEM images of the *n*-ZSM-5 with Si/Al ratio of 30.5. Insets in (a) showed the framework and standard XRD pattern of MFI structure according to PDF # 42-0024.



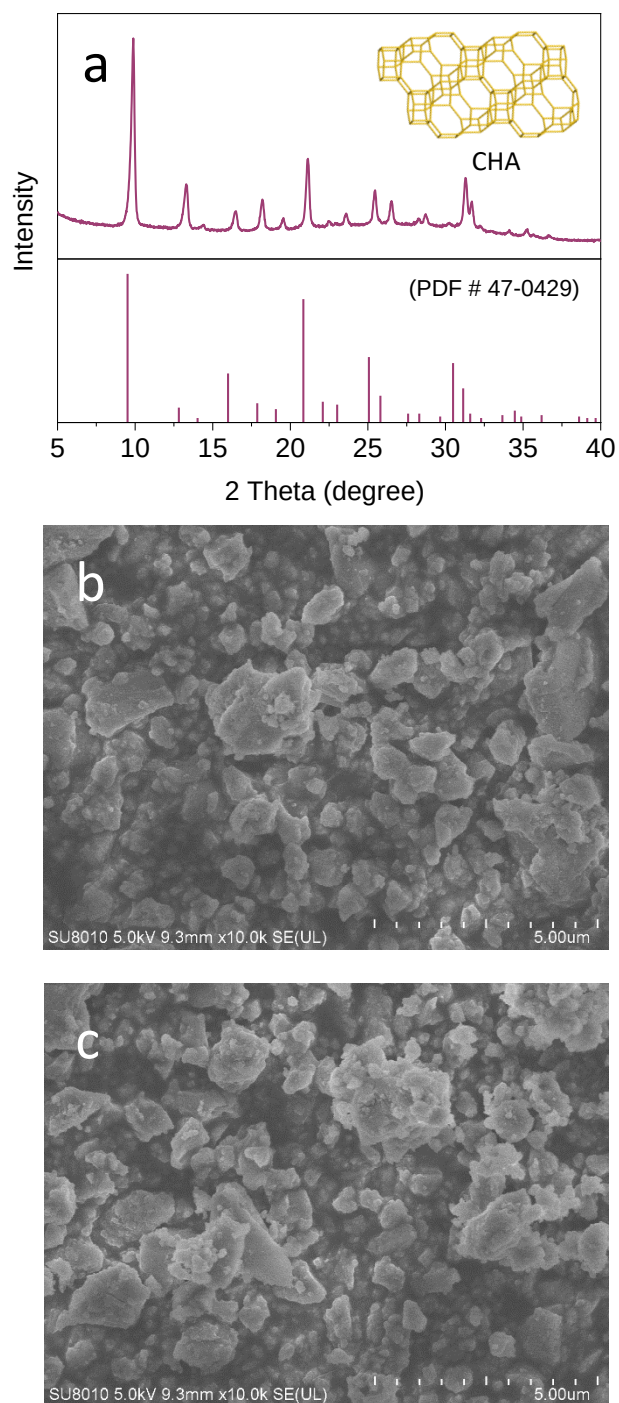
Supplementary Fig. 7. (a) XRD pattern, (b) N₂ sorption isotherms, and (c,d) SEM images of SSZ-13 with Si/Al ratio of 12.6. Insets in (a) showed the framework and standard XRD pattern of CHA structure according to PDF # 47-0762.



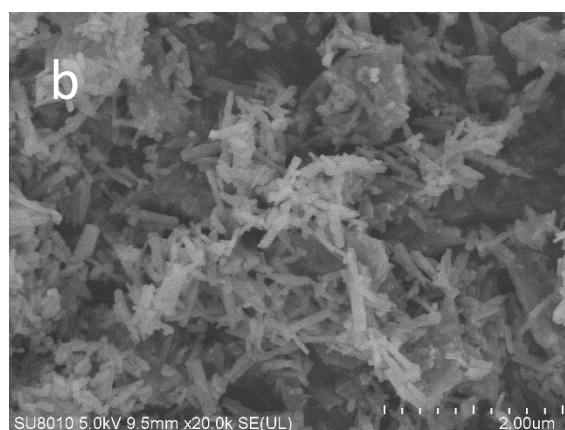
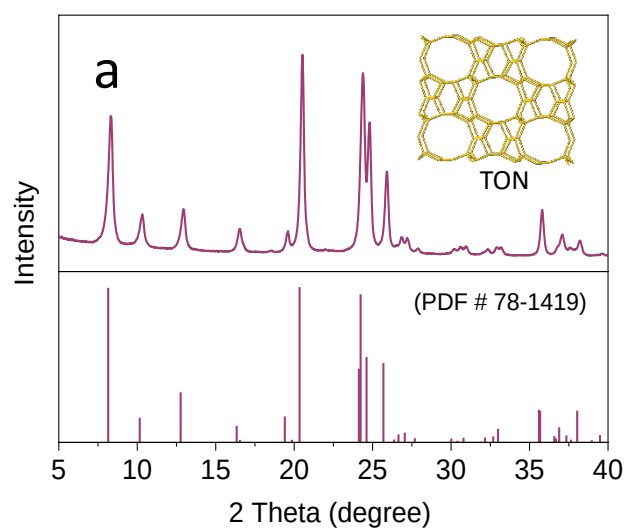
Supplementary Fig. 8. (a) XRD pattern, (b) N₂ sorption isotherms, and (c,d) SEM images of Beta zeolite with Si/Al ratio of 13.1. Insets in (a) showed the framework and standard XRD pattern of *BEA structure according to PDF # 48-0038.



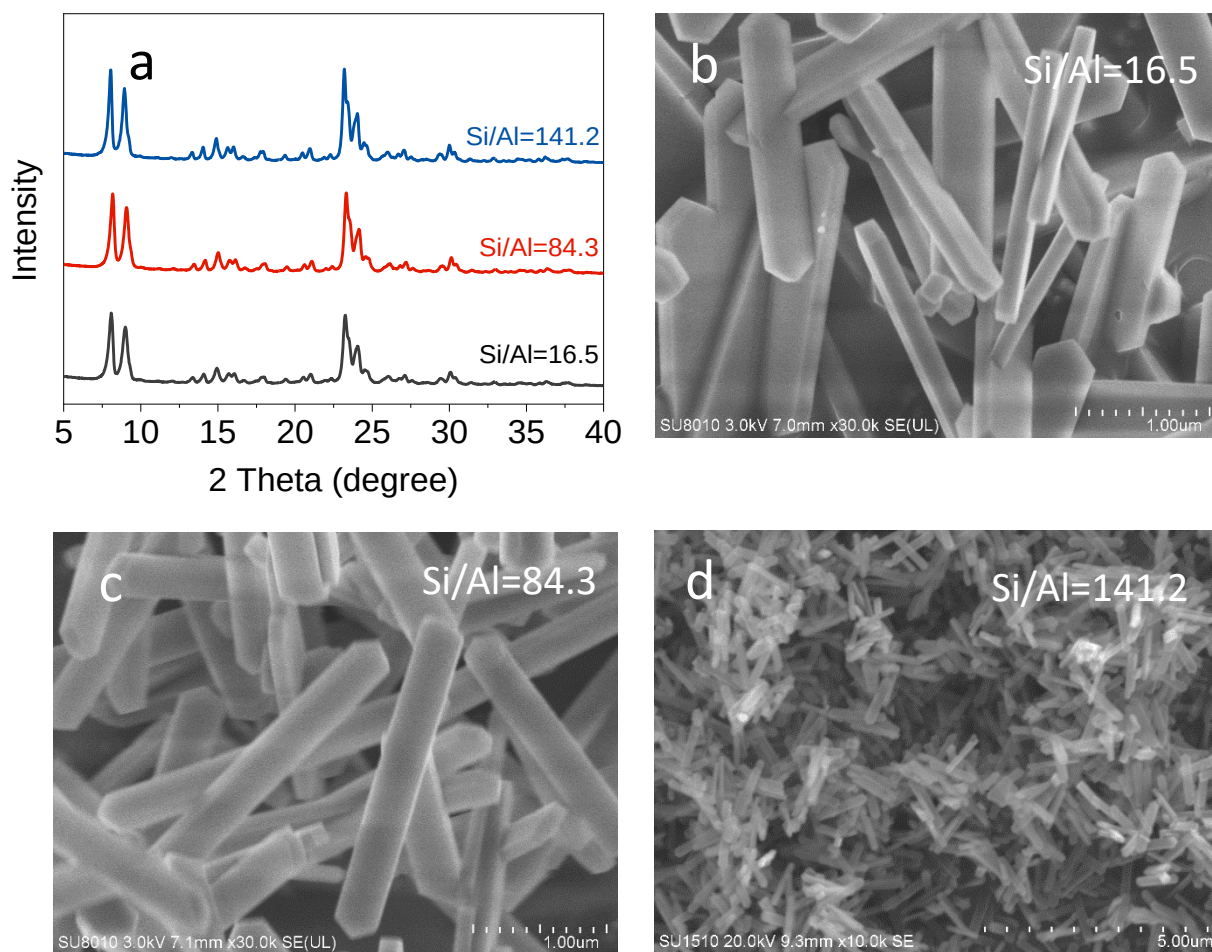
Supplementary Fig. 9. (a) XRD pattern, (b) N₂ sorption isotherms, and (c,d) SEM images of MOR zeolite with Si/Al ratio of 10.4. Insets in (a) showed the framework of MOR and standard XRD pattern of MOR structure according to PDF # 29-1257.



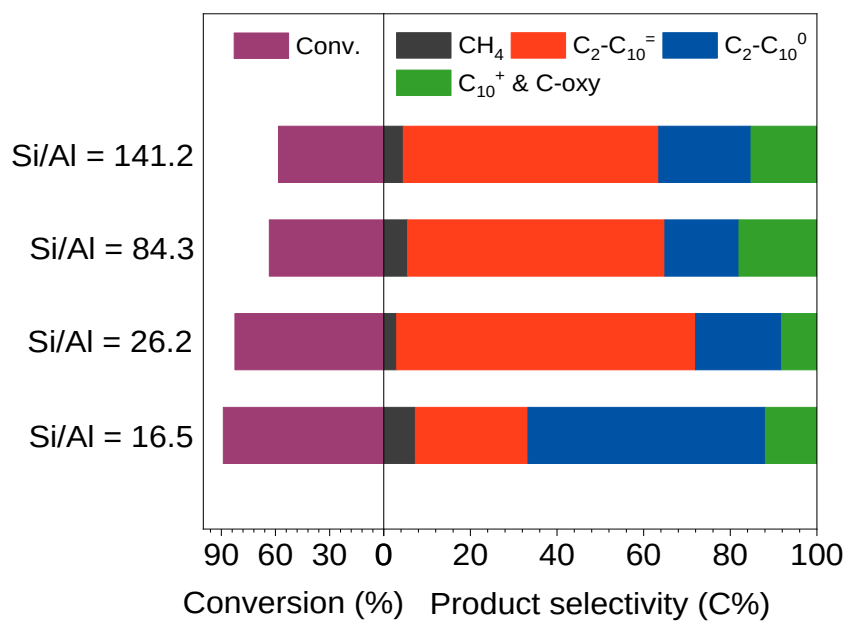
Supplementary Fig. 10. (a) XRD pattern and (b,c) SEM images of SAPO-34 with P/Al ratio of 0.76. Insets in (a) showed the framework of CHA and standard XRD pattern of CHA structure according to PDF # 47-0429.



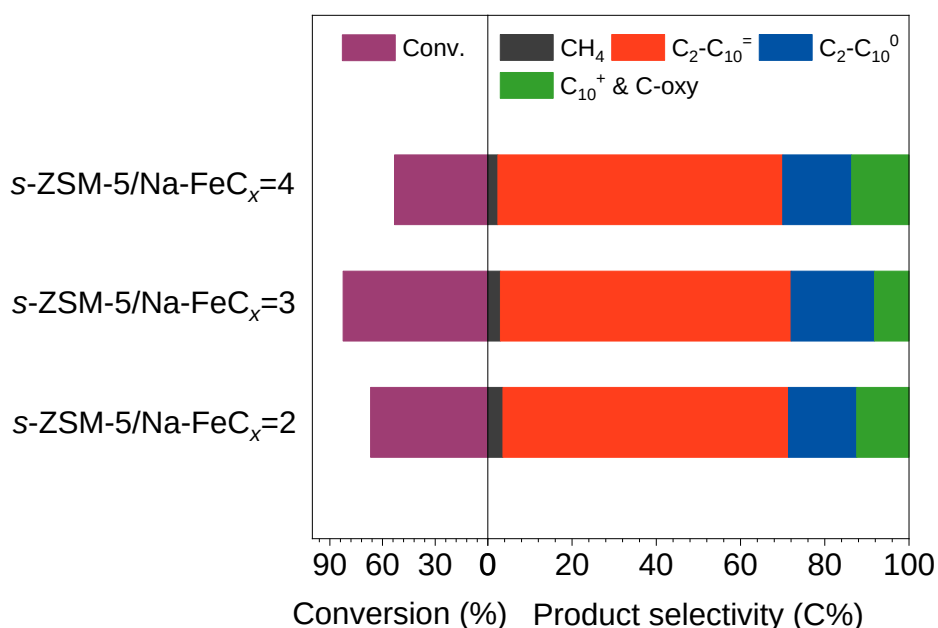
Supplementary Fig. 11. (a) XRD pattern and (b,c) SEM images of ZSM-22 zeolite with Si/Al ratio of 38.4. Insets in (a) showed the framework of TON and standard XRD pattern of TON structure according to PDF # 78-1419.



Supplementary Fig. 12. (a) XRD patterns and (b-d) SEM images of the *s*-ZSM-5 samples with different Si/Al ratios in the LT-FTO.

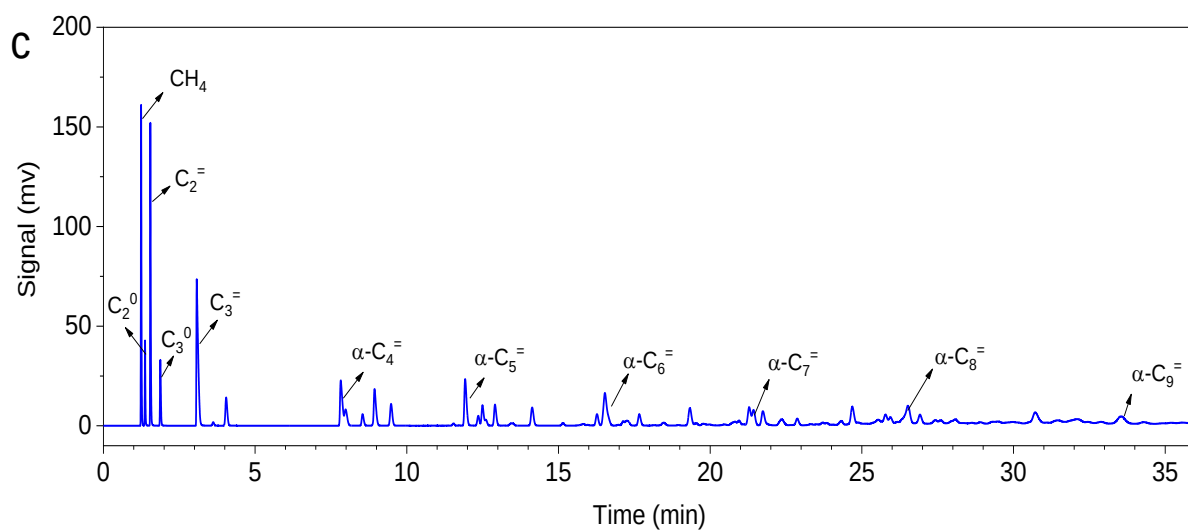
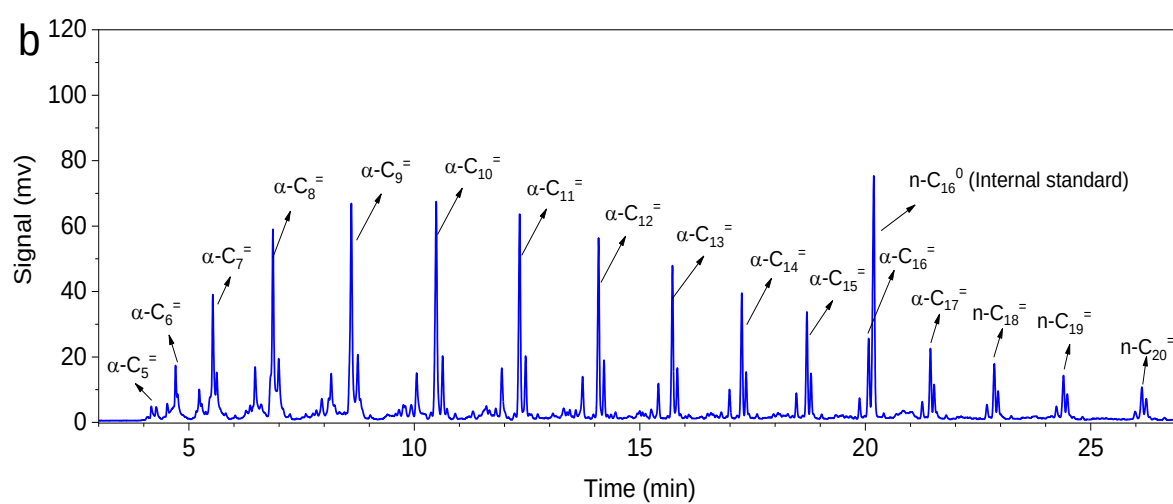
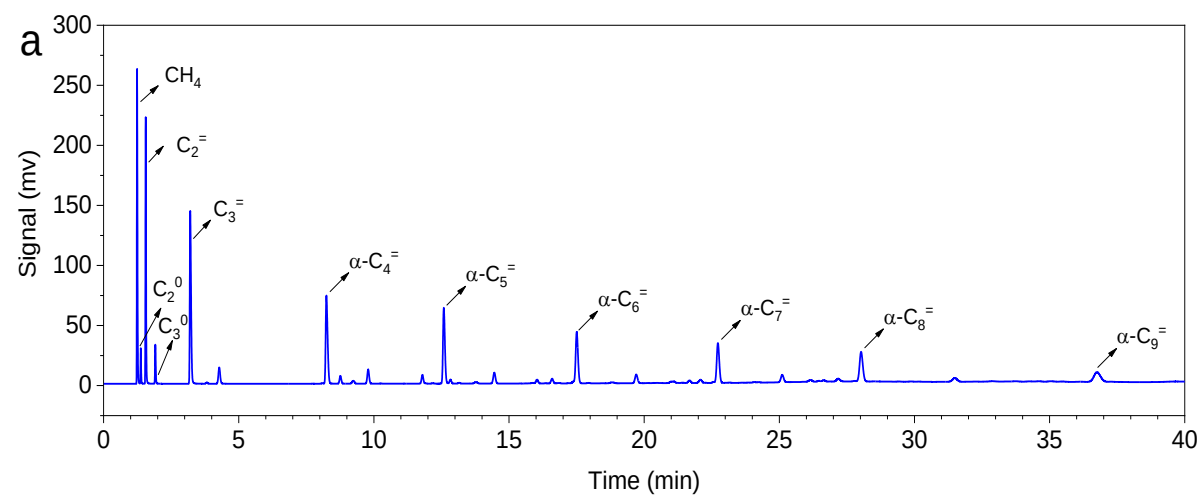


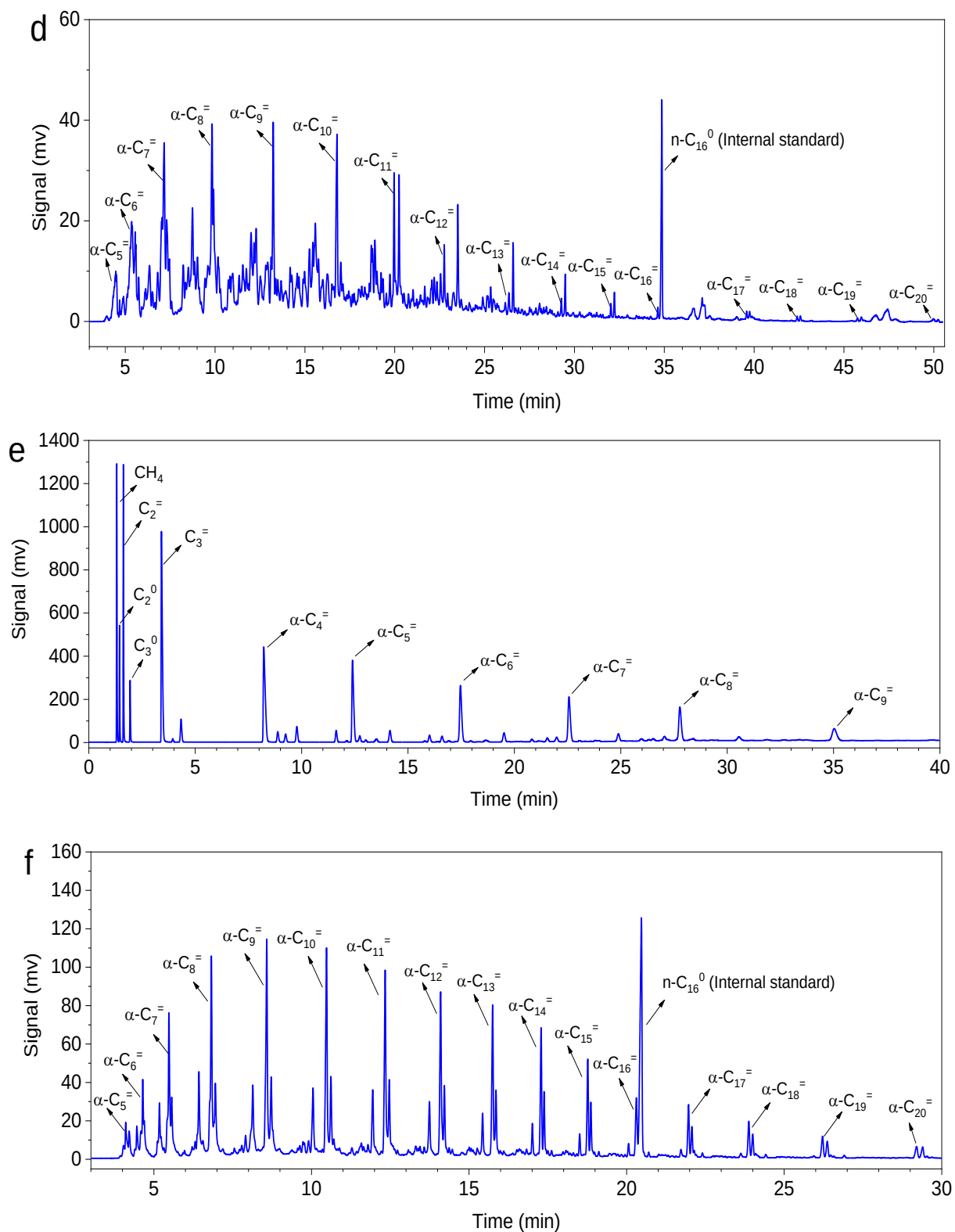
Supplementary Fig. 13. Catalytic performances of the Na-FeC_x/s-ZSM-5 catalyst with different Si/Al ratios in the LT-FTO.



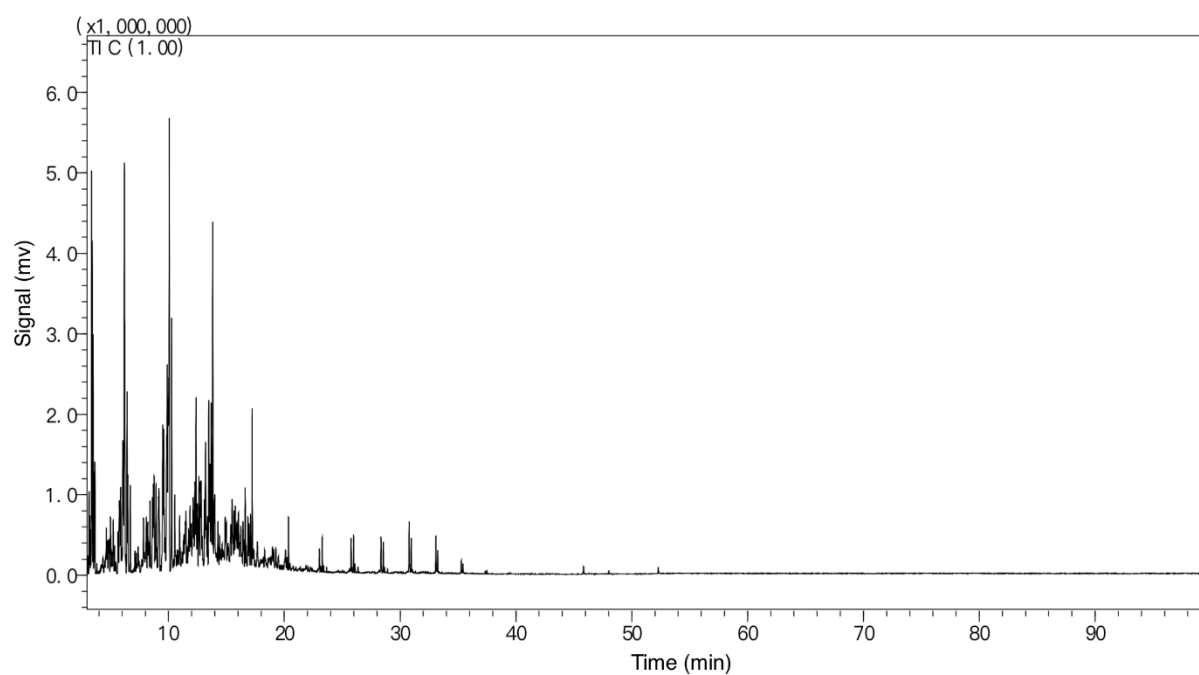
Supplementary Fig. 14. Catalytic data of the Na-FeC_x/s-ZSM-5 catalyst with zeolite/Na-FeC_x mass ratios at 2, 3, and 4. Reaction conditions: CO/H₂/Ar at 45/45/10, 2400 mL·h⁻¹·g⁻¹, 2.0 MPa, 260 °C, and granule-mixing of the s-ZSM-5 and Na-FeC_x individuals.

Note: We studied the influence of zeolite amount to the LT-FTO (Supplementary Fig.14, Supplementary Table 3). When a small amount of zeolite was added in the reaction system such as Na-FeC_x/s-ZSM-5 with mass ratio of zeolite/Na-FeC_x at 0.5, the CO conversion at 27.3% was obtained. More zeolite increased the CO conversion and optimized the product selectivity. For example, Na-FeC_x/s-ZSM-5 with mass ratio of zeolite/Na-FeC_x at 2 showed higher CO conversion of 66.8% and lower C₁₀⁺ selectivity at 12.4%, where the C₂-C₁₀ molecules with o/a at 4.2 were the dominant products. The best performance was obtained on the catalyst with mass ratio of zeolite/Na-FeC_x at 3, where the C₁₀⁺ product selectivity was as low as 8.2%. A further increase of zeolites led to a reduction of the CO conversion, giving 53.2% on the Na-FeC_x/s-ZSM-5 with o/a ratio at 4.1.



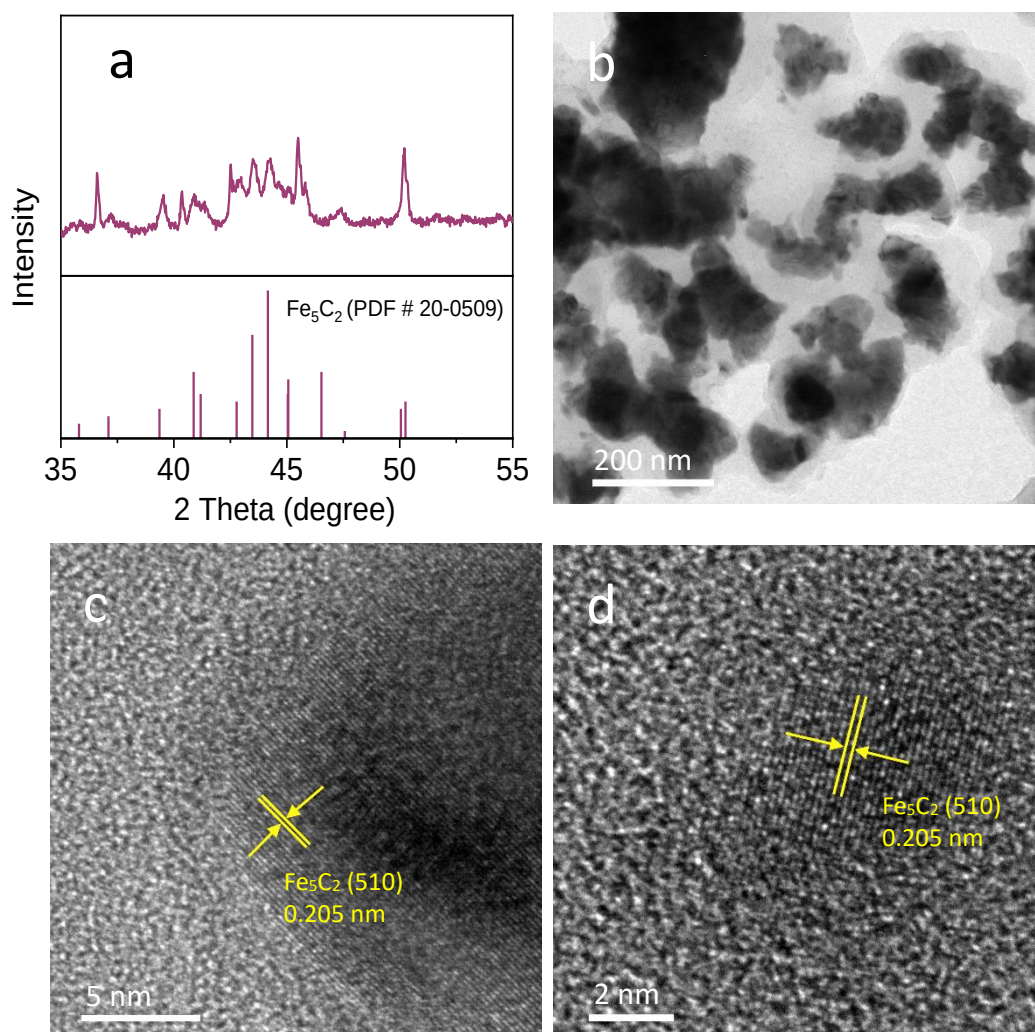


Supplementary Fig. 15. GC curves analyzing the gaseous (online analysis) and liquid products (offline analysis) of (a,b) Na-FeC_x , (c,d) $\text{Na-FeC}_x/\text{s-ZSM-5}$, and (e,f) $\text{Na-FeC}_x/\text{s-S1-OH}$. The reaction conditions were the same as those in Fig. 2a.

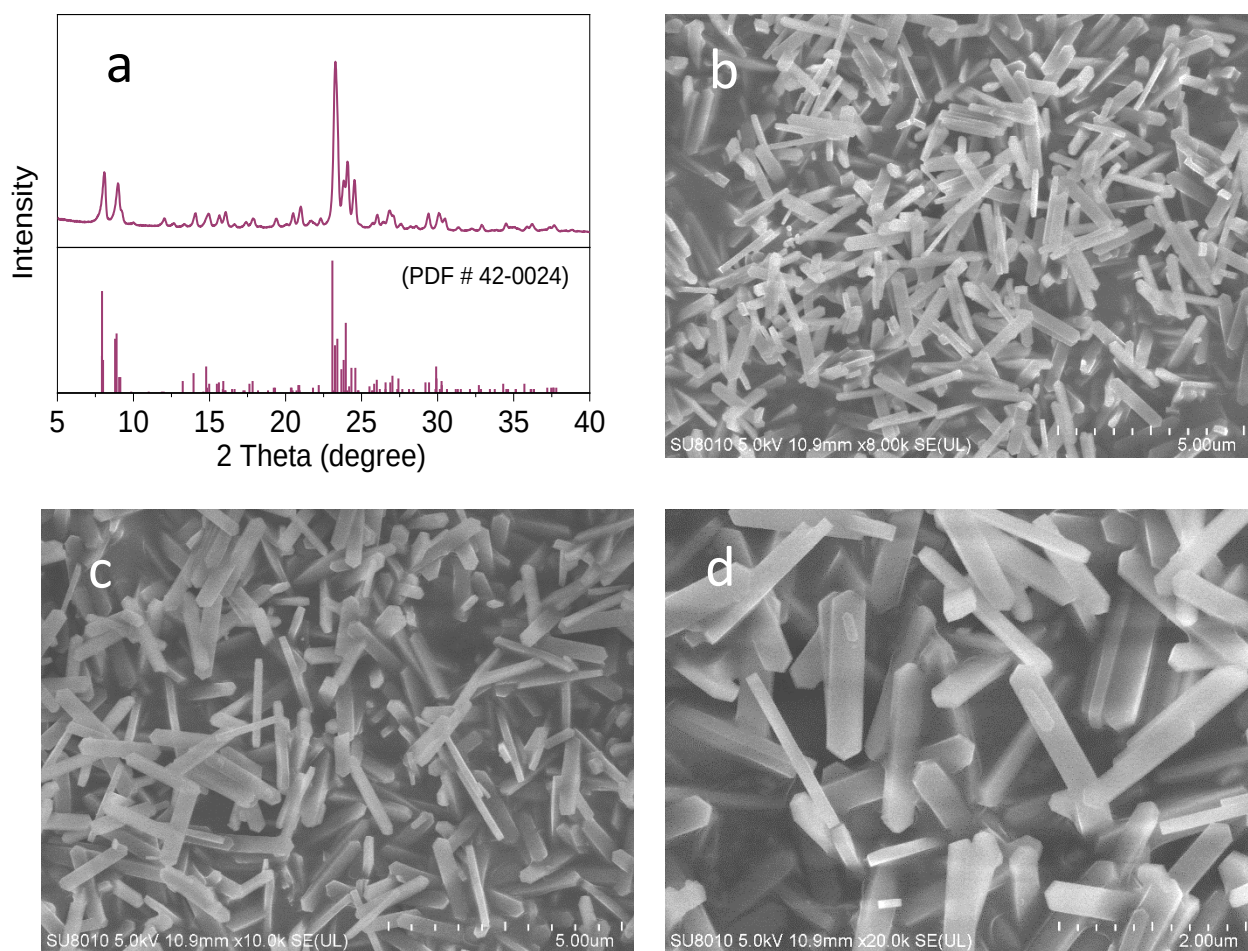


Supplementary Fig. 16. GC-MS analyzing the liquid products of the LT-FTO reaction over the Na-FeC_x/s-ZSM-5 catalyst. The reaction conditions were the same as those in Fig. 2a in the main text.

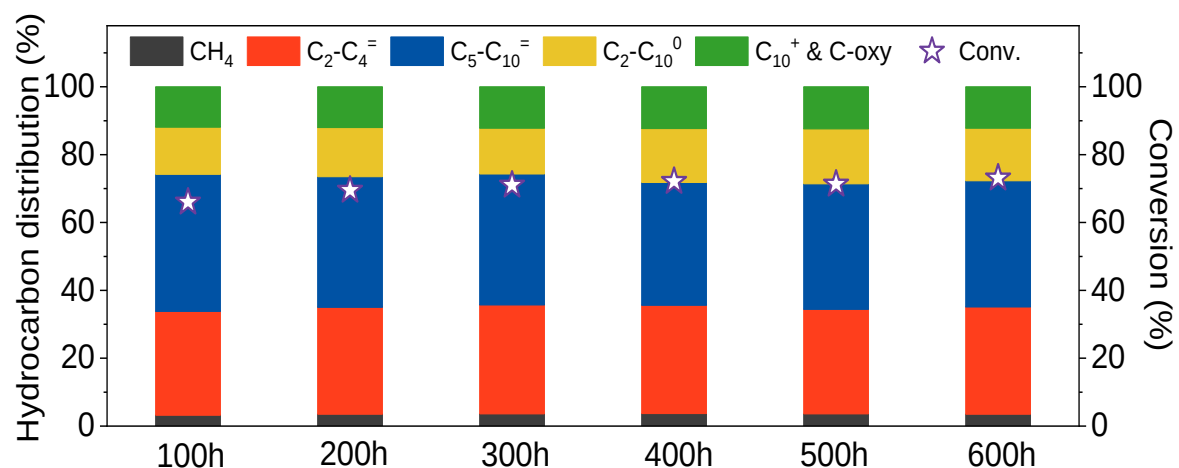
Note: The liquid products of the Na-FeC_x/s-ZSM-5 was analyzed by GC-MS. The aromatic compounds were almost undetectable with selectivity less than 0.5%.



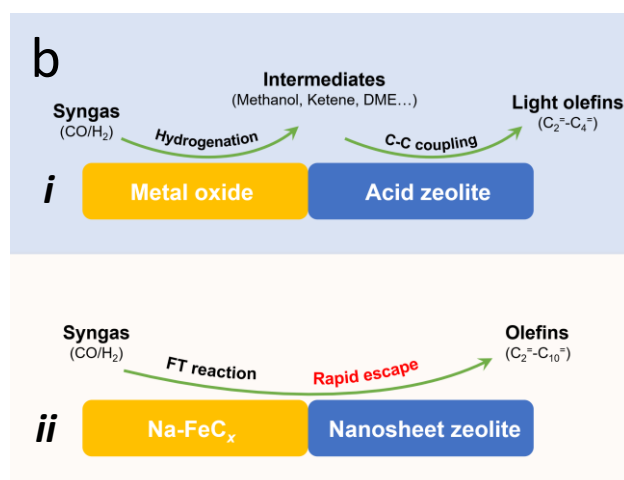
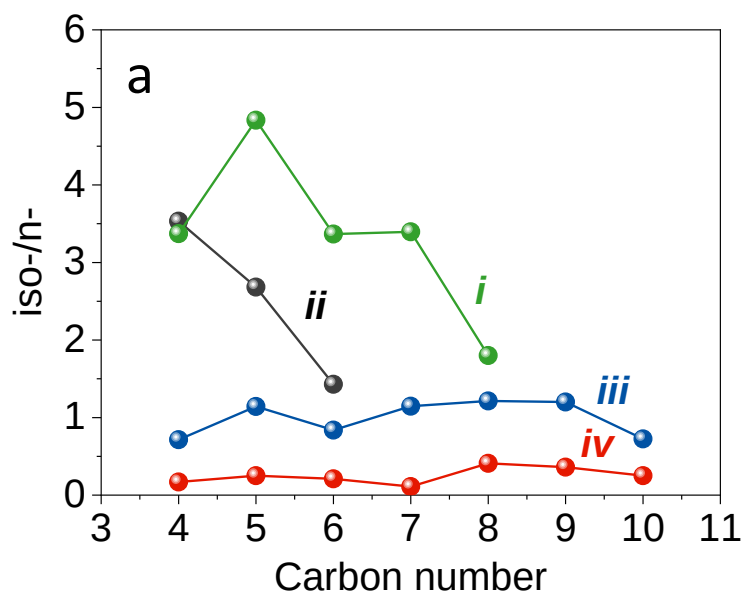
Supplementary Fig. 17. (a) XRD pattern and (b-d) TEM images of used Na-FeC_x after the LT-FTO reaction for 600 h. The reaction conditions were the same to those in Fig. 2c. Inset in (a) showed standard XRD pattern of Fe₅C₂ structure according to PDF # 20-0509. After the reaction, the Na-FeC_x and zeolite individuals were separated for further characterization.



Supplementary Fig. 18. (a) XRD pattern and (b-d) SEM image of used *s*-ZSM-5 after the LT-FTO reaction for 600 h. The reaction conditions were the same to those in Fig. 2c. Inset in (a) showed the standard XRD pattern of MFI structure according to PDF # 42-0024.



Supplementary Fig. 19. Catalytic data at different periods in the LT-FTO characterizing the durability of the Na-FeC_{*x*}/s-ZSM-5 catalyst. The reaction conditions were the same to those in Fig. 2c.

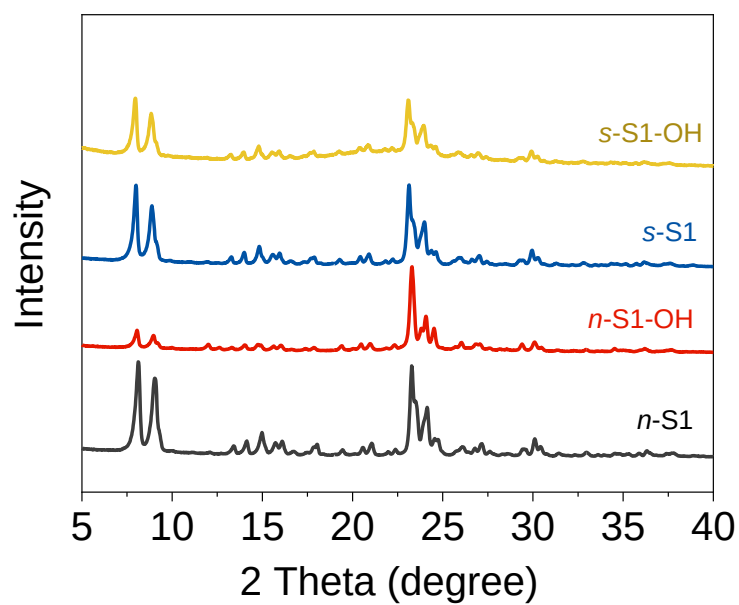


Supplementary Fig. 21. (a) The ratios of isomer to normal hydrocarbons in the products from different reactions. (i) 1-octene isomerization/cracking over *s*-ZSM-5, (ii) 1-hexene isomerization/cracking over *s*-ZSM-5, (iii) LT-FTO over Na-FeC_x/*s*-ZSM-5, and (iv) LT-FTO over Na-FeC_x/*s*-S1-OH. Reaction condition for 1-octene cracking: 0.2 g of catalyst, 260 °C, 0.1 MPa, 1-octene bubbling with a pure N₂ flow at 30 mL/min at 120 °C. Reaction condition for 1-hexene cracking: 0.2 g of catalyst, 260 °C, 0.1 MPa, and 1-hexene bubbling with a pure N₂ flow at 10 mL/min at 30 °C. (b) The strategies on (i) cascade reactions (OX-ZEO catalysts) and (ii) this work (Na-FeC_x/nanosheet zeolite) in FTO reaction.

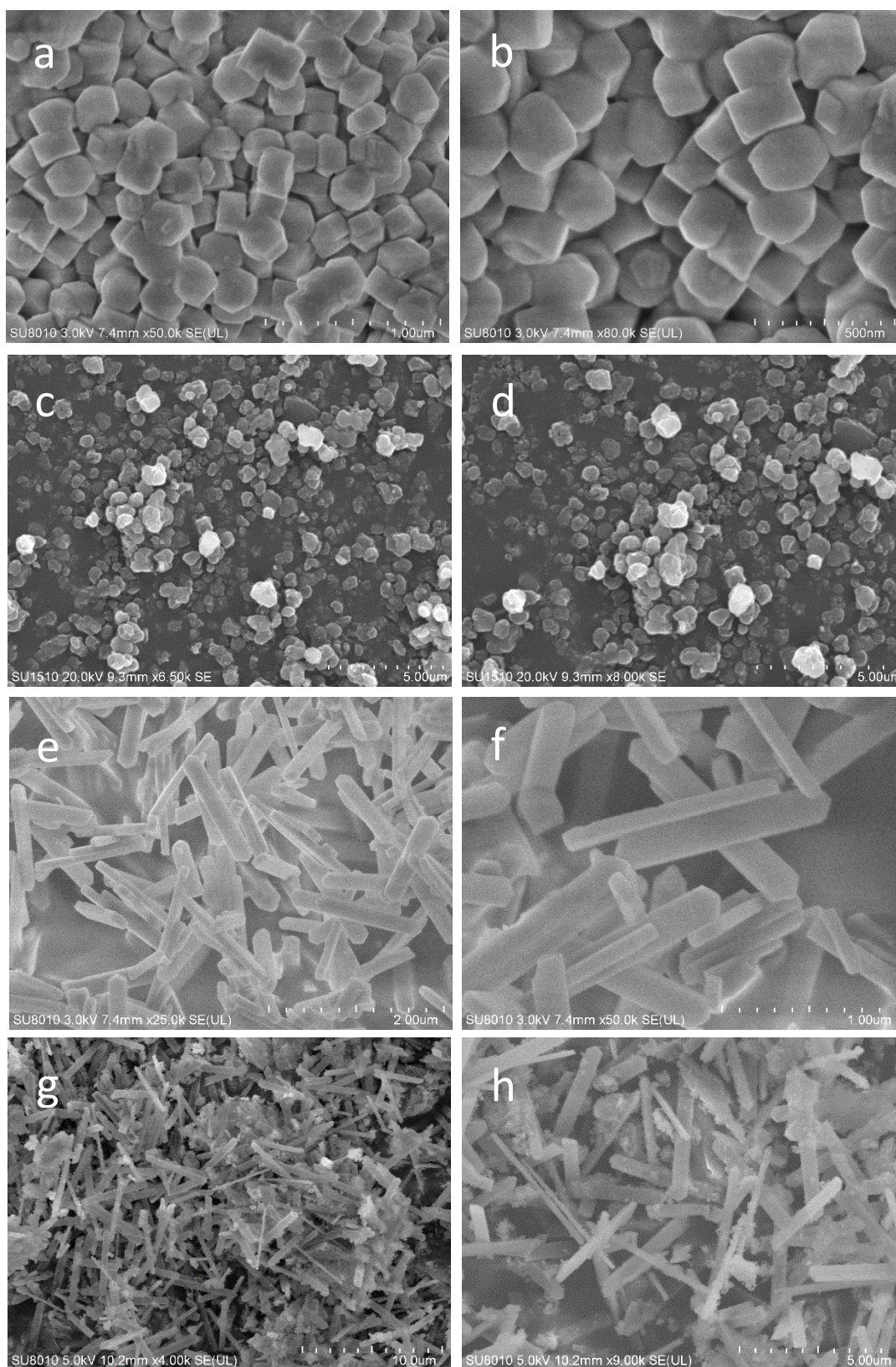
Note: The acidic zeolites could catalyze the side reactions of isomerization and cracking of olefin products. We fed 1-butene, 1-hexene, and 1-octene into the *s*-ZSM-5 zeolite at 260 °C (temperature for the LT-FTO reaction, Supplementary Tables 4-6), where the isomerization

occurred as a dominant reaction with extremely low selectivity to the cracked products. For example, the reaction using 1-butene feed showed 99.1% selectivity to the isomerized butene products with almost undetectable C₁-C₃ products (selectivity < 0.5%). Similar phenomena were also observed on the reaction using 1-hexene and 1-octene feeds, where the 91.5% and 93.4% selectivity to isomerized C₆ and C₈ products were obtained. This feature cannot meet the high α -olefin selectivity in the Na-FeC_x/*s*-ZSM-5 catalyzed LT-FTO (Supplementary Fig. 21a, Supplementary Tables 4-6). In addition, the cracking only occurred slightly because of the low reaction temperature, which would not obviously change the carbon number distribution of the products in LT-FTO.

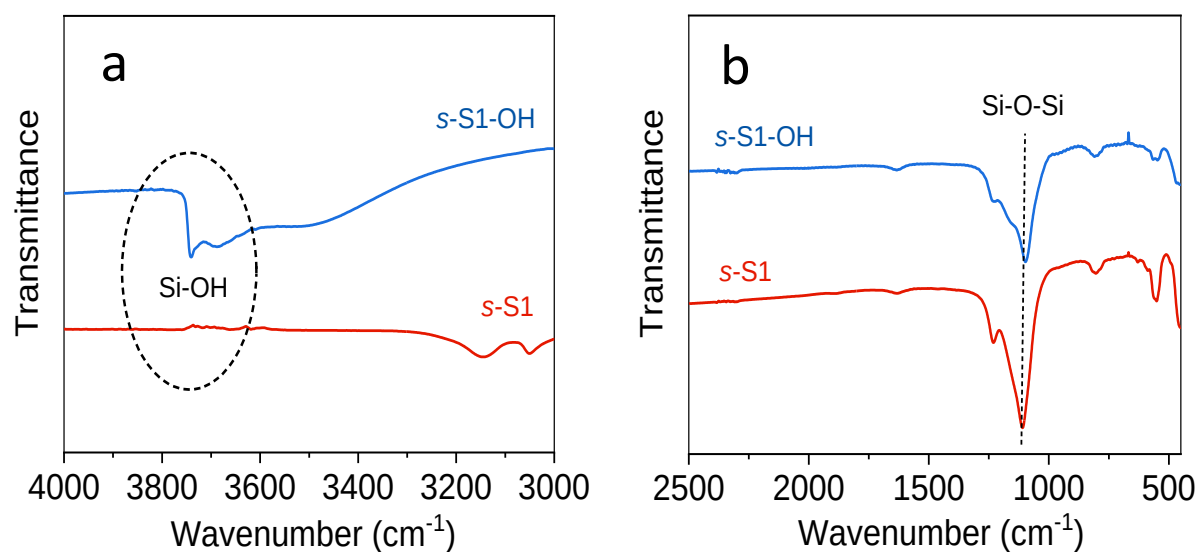
In order to avoid the isomerization/cracking side reaction and improve the selectivity to desired α -olefins, we reasonably developed a siliceous *s*-S1-OH zeolite without acidic sites for the isomerization and cracking (e.g. 1-hexene conversion <1% when fed to *s*-S1-OH). The FeC_x/*s*-S1-OH still exhibited significantly enhanced CO conversions, similar to that using *s*-ZSM-5 catalyst, but giving molar ratio of iso-/normal-olefins (ethylene and propylene were not included in the calculation) at ~0.2 because of the reduced side reactions of isomerization (Supplementary Fig. 21).



Supplementary Fig. 22. XRD patterns of the *n*-S1, *n*-S1-OH, *s*-S1, and *s*-S1-OH.

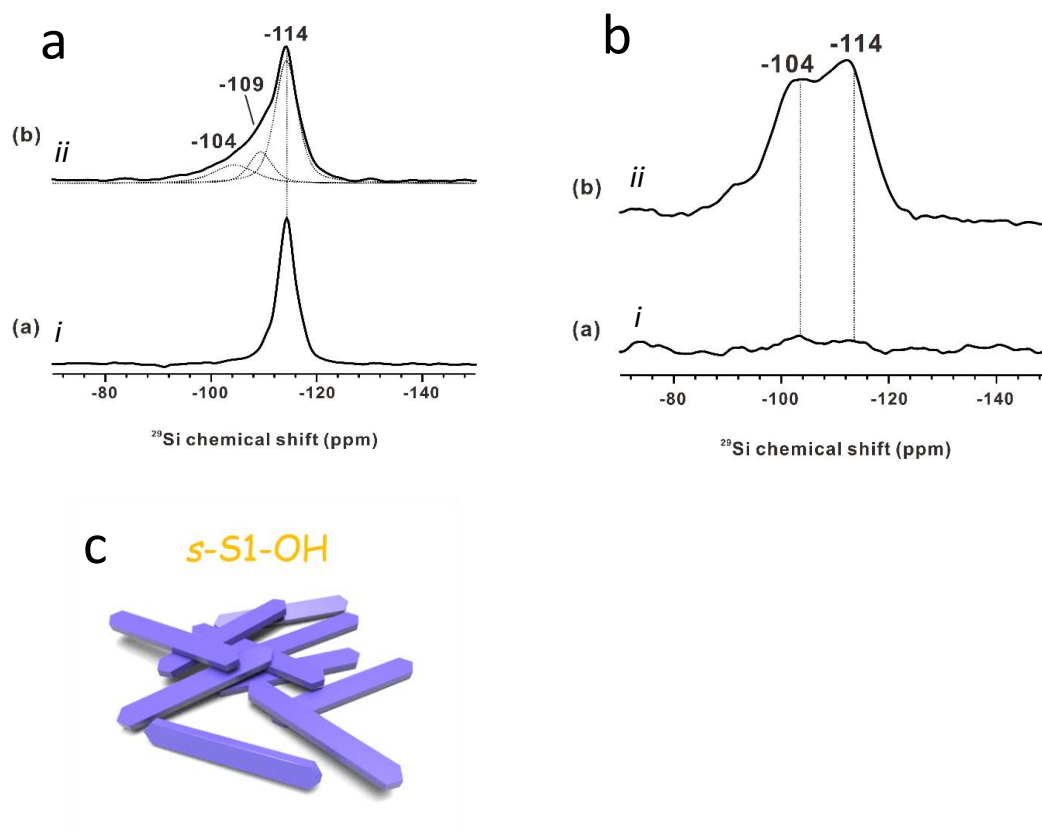


Supplementary Fig. 23. SEM images of the (a,b) *n*-S1, (c,d) *n*-S1-OH, (e,f) *s*-S1, and (g,h) *s*-S1-OH.



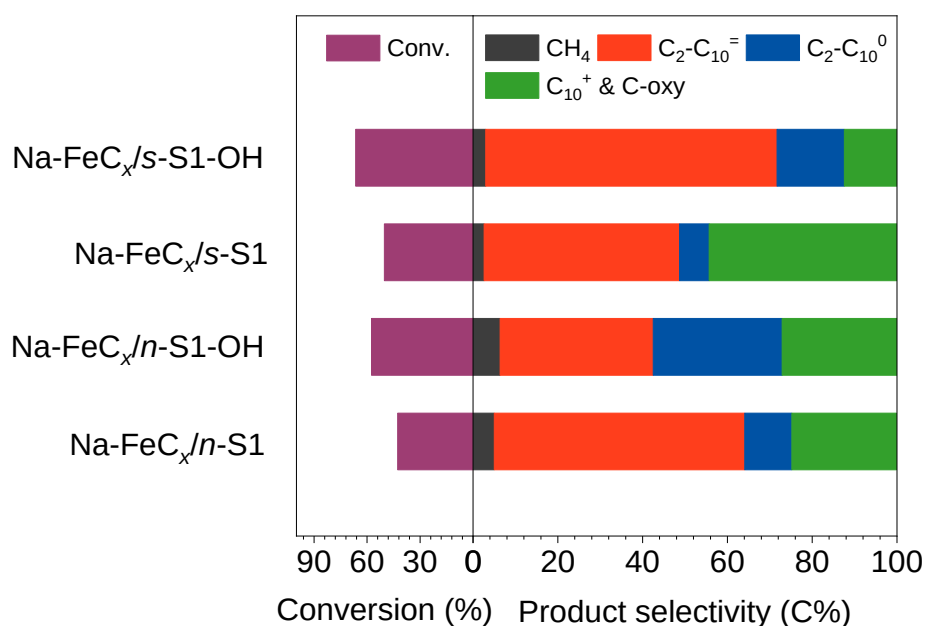
Supplementary Fig. 24. IR spectra of the *s*-S1 and *s*-S1-OH samples in the regions at (a) 3000-4000 cm^{-1} and (b) 450-2500 cm^{-1} .

Note: The *s*-S1 and *s*-S1-OH samples gave a peak at $\sim 1100 \text{ cm}^{-1}$, which is assigned to the Si-O-Si band of zeolite framework. Compared with *s*-S1, the *s*-S1-OH gave much stronger peaks at 3680 and 3730 cm^{-1} in Supplementary Fig. 24a, confirming the abundant silanol species^{16,17}.



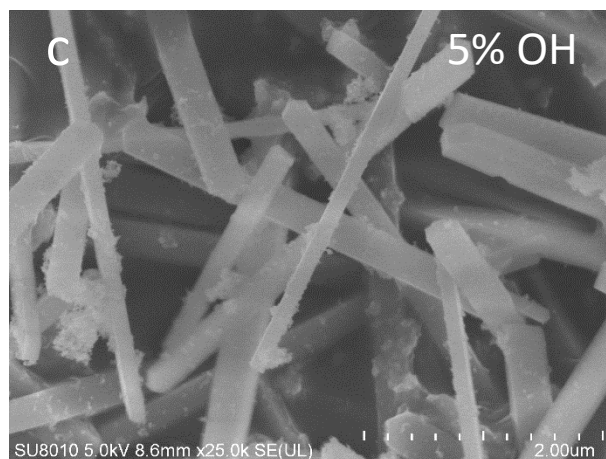
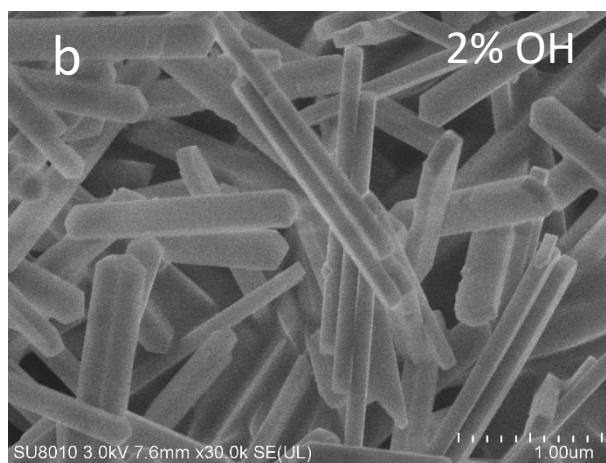
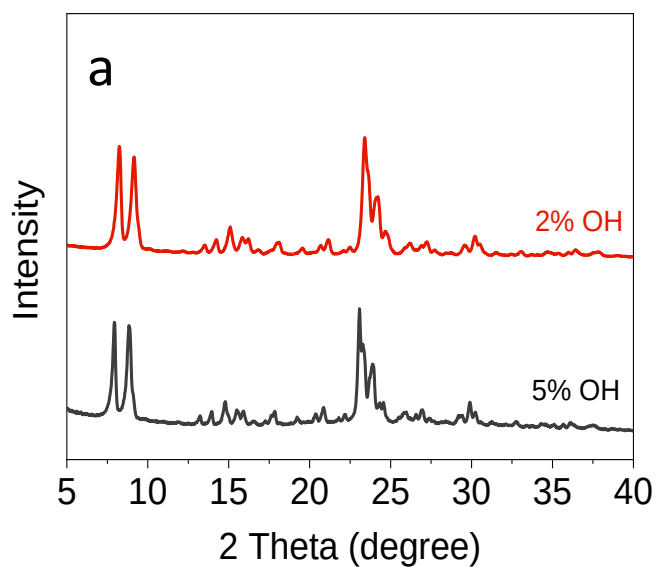
Supplementary Fig. 25. (a) ^{29}Si MAS NMR spectra and (b) ^{29}Si - ^1H CP-MAS spectra of (i) *s*-S1 and (ii) *s*-S1-OH. (c) Model showing the *s*-S1-OH with short *b* axis.

Note: In Supplementary Fig. 25a, both *s*-S1 and *s*-S1-OH displayed Q₄ species $[\text{Si}(\text{OSi})_4]$ at -114 ppm. Compared with *s*-S1, the *s*-S1-OH gave additional signals at -104 and -109 ppm, indicating the presence of Q₃ sites $[\text{Si}(\text{OSi})_3(\text{OH})]$ on siliceous zeolite crystals¹⁷⁻¹⁹. The ^{29}Si - ^1H CP-MAS spectra of *s*-S1-OH showed the signals at -104 and -114 ppm, while no signals were observed on the ^{29}Si - ^1H CP-MAS spectra of *s*-S1 (Supplementary Fig. 25b), indicating the presence of hydrogen atoms around silicon atoms on the *s*-S1-OH zeolite crystals (Q₃ sites for siliceous zeolite), and the absence of hydrogen atoms around silicon atoms on the *s*-S1 zeolite crystals^{18,19}. These results confirm the successful functionalization of hydroxyl groups on the *s*-S1-OH zeolite framework, in good agreement with the IR results.



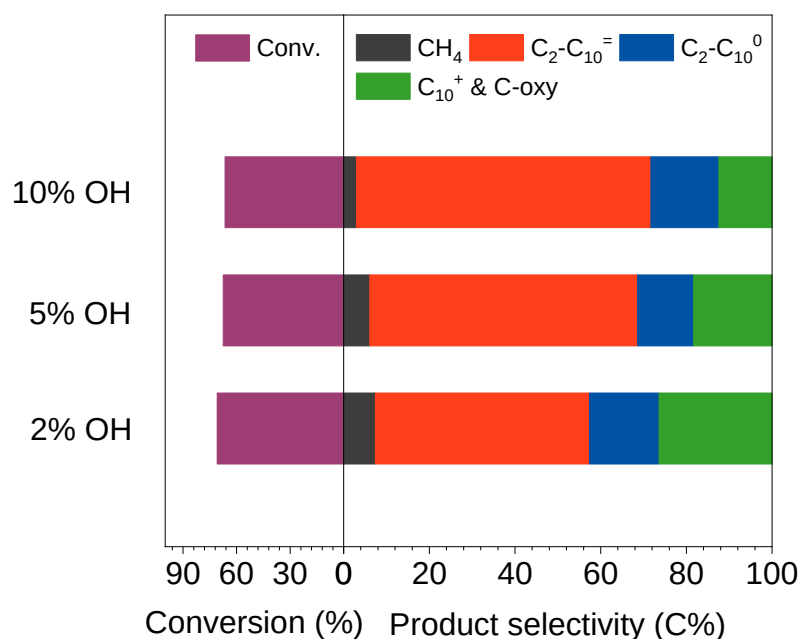
Supplementary Fig. 26. Catalytic data of the Na-FeC_x/n-S1, Na-FeC_x/s-S1, and Na-FeC_x/s-S1-OH in the LT-FTO. Reaction conditions: weight ratio zeolite to Na-FeC_x at 3, feed gas with ratio of CO/H₂/Ar at 45/45/10, 2400 mL·h⁻¹·g⁻¹, 2.0 MPa, 260 °C, and granule-mixing of the zeolite and Na-FeC_x individuals.

Note: The Na-FeC_x/n-S1 and Na-FeC_x/s-S1 showed the CO conversion at 42.7% and 50.2% in the LT-FTO, which were higher than that of the bare Na-FeC_x catalyst under the equivalent reaction conditions. In addition, these catalysts still gave high C₁₀⁺ product selectivity at 24.7% and 44.3%, respectively (Supplementary Fig. 26). In contrast, the Na-FeC_x/s-S1-OH showed enhanced activity and reduced selectivity for C₁₀⁺ products, giving CO conversion and C₁₀⁺ product selectivity at 66.5% and 12.4%, respectively. In this case, the o/a ratio appeared at 4.3 for the C₂-C₁₀ molecules.

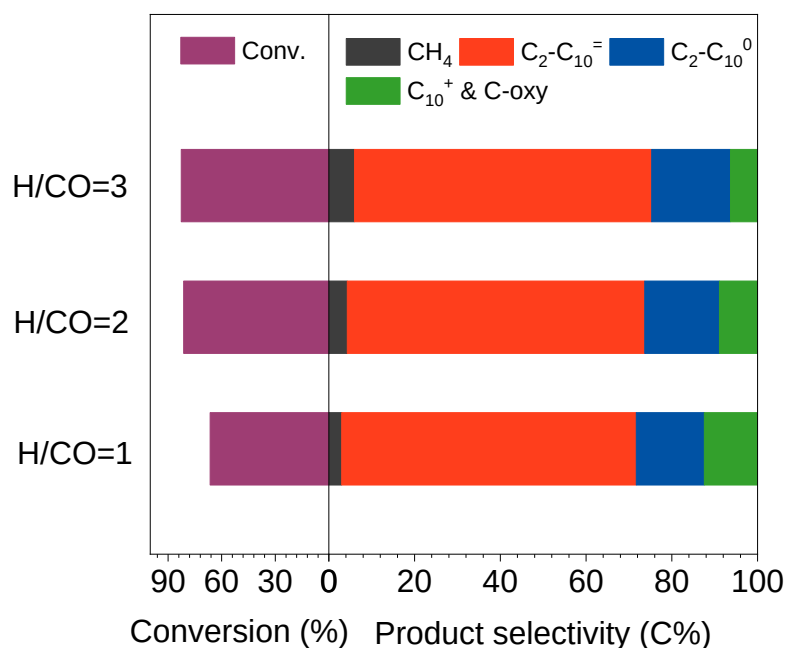


Supplementary Fig. 27. (a) XRD patterns and (b and c) SEM images of *s*-S1-OH zeolites with different silanol concentrations in the LT-FTO.

Note: The silanol concentrations are described by the molar ratio of DEMS to the total amount of silica species in the starting gel for zeolite synthesis.

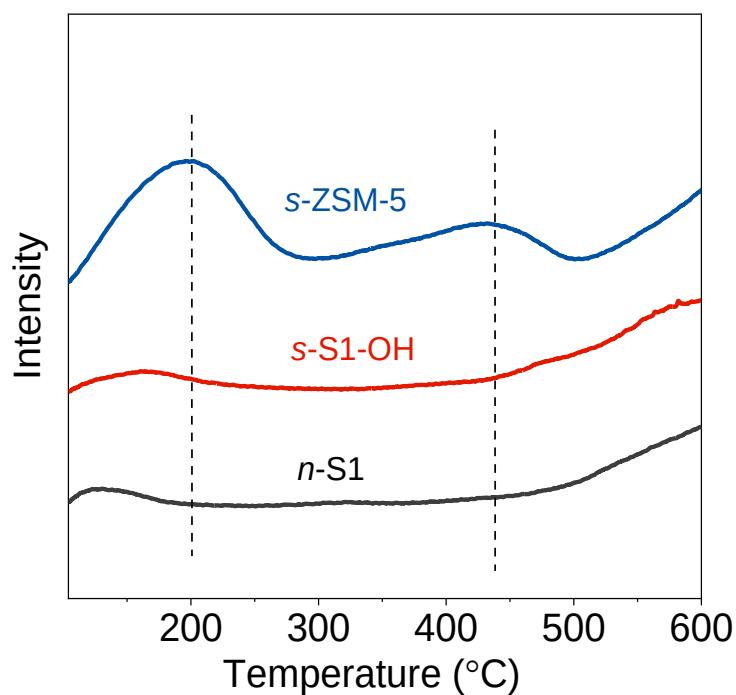


Supplementary Fig. 28. Catalytic performances of Na-FeC_x/s-S1-OH catalyst with different silanol concentrations in the LT-FTO. If it was specially noted, the *s*-S1-OH means the zeolite with silanol concentration at 10%. Reaction conditions: weight ratio zeolite to Na-FeC_x at 3, feed gas with ratio of CO/H₂/Ar at 45/45/10, 2400 mL·h⁻¹·g⁻¹, 2.0 MPa, 260 °C, and granule-mixing of the zeolite and Na-FeC_x individuals.



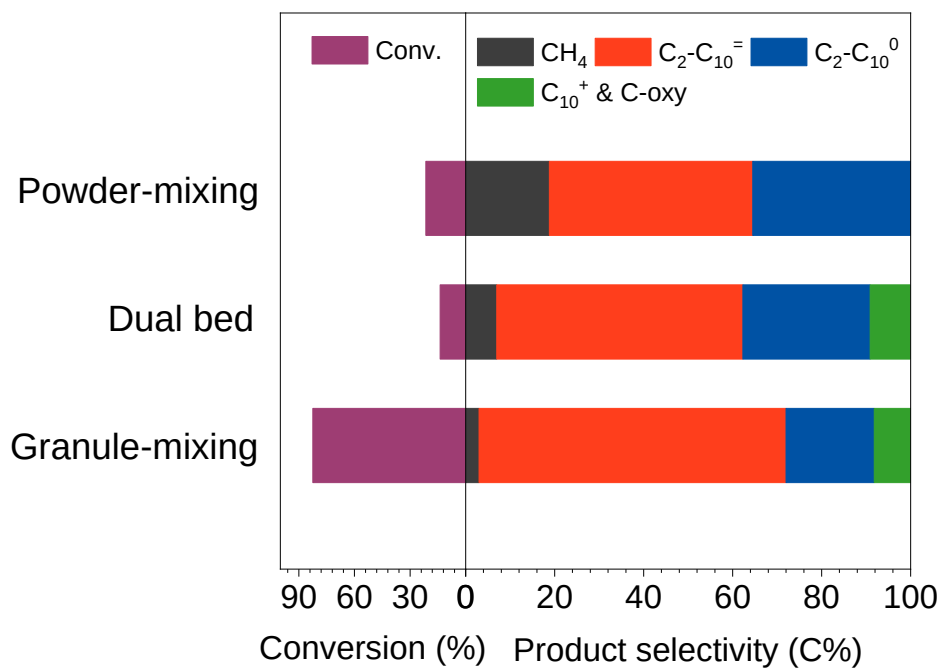
Supplementary Fig. 29. Catalytic data of the Na-FeC_x/s-S1-OH in the LT-FTO with different H₂/CO ratios at 1, 2, and 3. Reaction conditions: weight ratio of s-S1-OH to Na-FeC_x at 3, 2400 mL·h⁻¹·g⁻¹, 2.0 MPa, 260 °C, and granule-mixing of the s-S1-OH and Na-FeC_x.

Note: We optimized the ratio of H₂/CO in the feed gas of LT-FTO over the Na-FeC_x/s-S1-OH. As shown in Supplementary Fig. 29, by adjusting the H₂/CO ratio to 2, the CO conversion could be 81.3%. When this ratio was 3, the CO conversion could reach to 82.6%. More importantly, the Na-FeC_x/s-S1-OH still showed a narrow distribution with dominant C₂-C₁₀ molecules with o/a at 3.8, where the methane and C₁₀⁺ product selectivities are 6.0% and 6.3% (Supplementary Table 8), respectively.

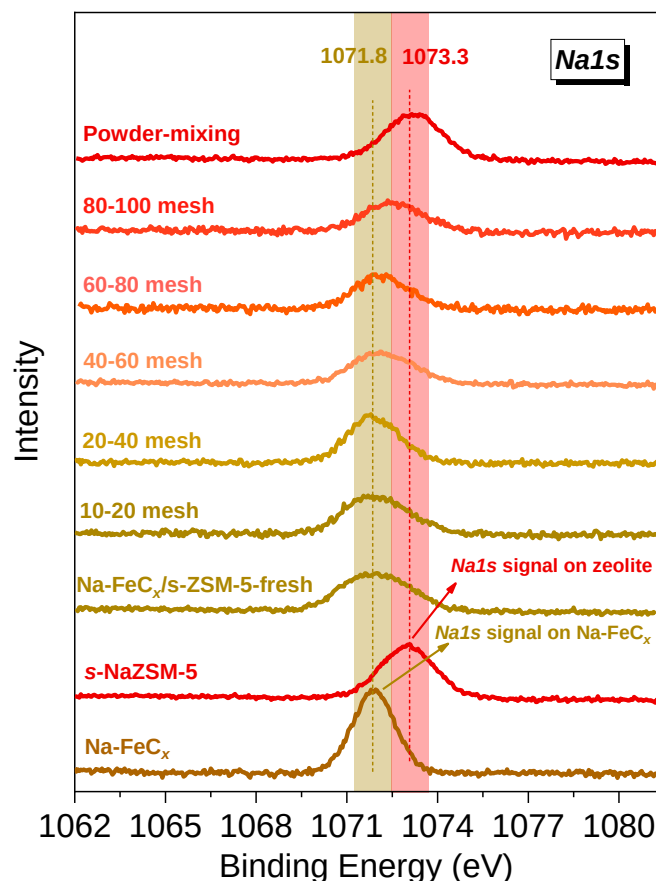


Supplementary Fig. 30. NH₃-TPD profiles of the *n*-S1, *s*-S1-OH, and *s*-ZSM-5.

Note: NH₃-TPD curve of *s*-ZSM-5 showed the peaks centered at 200 and 440 °C, which were assigned to the desorption of ammonia that was adsorbed on the weak and strong acid sites^{20,21}, respectively. In contrast, the signals of desorbed ammonia were almost undetectable on the *n*-S1 and *s*-S1-OH, confirming the absence of acidic sites on the siliceous zeolites.



Supplementary Fig. 31. Catalytic data of the Na-FeC_x/s-ZSM-5 with different mixing manners of Na-FeC_x and s-ZSM-5 individuals. Reaction conditions: weight ratio of s-ZSM-5 to Na-FeC_x at 3, ratios of CO/H₂/Ar at 45/45/10, 2400 mL·h⁻¹·g⁻¹, 2.0 MPa, and 260 °C.



Supplementary Fig. 32. Na 1s XPS spectra of Na-FeC_x, s-NaZSM-5 (Na⁺-exchanged zeolite as standard), and the granule- and power-mixing Na-FeC_x/s-ZSM-5 catalysts. These catalysts were used in LT-FTO for 50 h before the XPS characterization. The Na-FeC_x/s-ZSM-5-fresh is the fresh sample as standard without catalyzing LT-FTO.

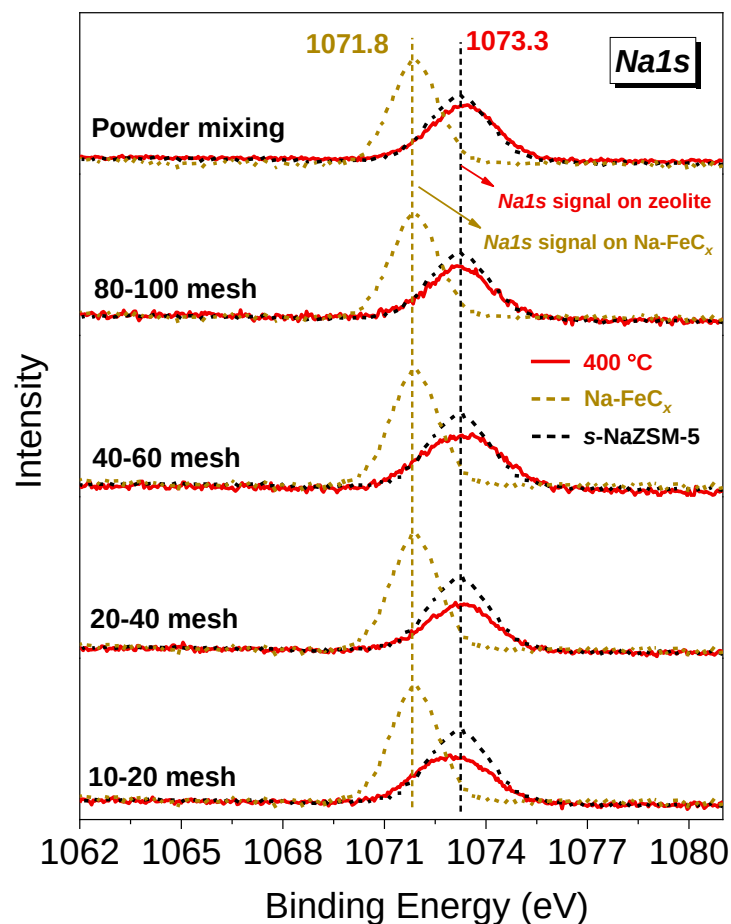
Note: The sodium migration from sodium-modified catalyst to the zeolite might occur in many reactions using physically mixed catalysts reported previously²²⁻²⁷, but this migration is strongly related to the proximity of the catalyst granules. For example, de Jong and co-workers²² reported sodium migration from ferric carbide to the zeolite on the granule-mixing catalyst with granule sizes at ~100 μm for the syngas-to-aromatic conversion at 400 °C, as confirmed by the continuously decreased aromatic selectivity because of poisoning the zeolite acid in the syngas-to-aromatic conversions. For the catalyst with granules sizes at ~500 μm, the catalytic performances were very similar to that over spatially separated dual-bed catalyst, which is regarded as a standard manner with prevented sodium migration in lab research. This result indicate the relatively hindered sodium migration on the catalyst with granules sizes at ~500 μm, which is quite different from the catalyst with granule sizes at ~100 μm. In contrast, the Na-

FeC_x/s-ZSM-5 catalyst in this work with average granule sizes at ~600 μm (~0.6 mm, 20-40 mesh) was performed at 260 °C, where the relatively long distance between the granules and lower reaction temperature should be unfavorable for the sodium migration.

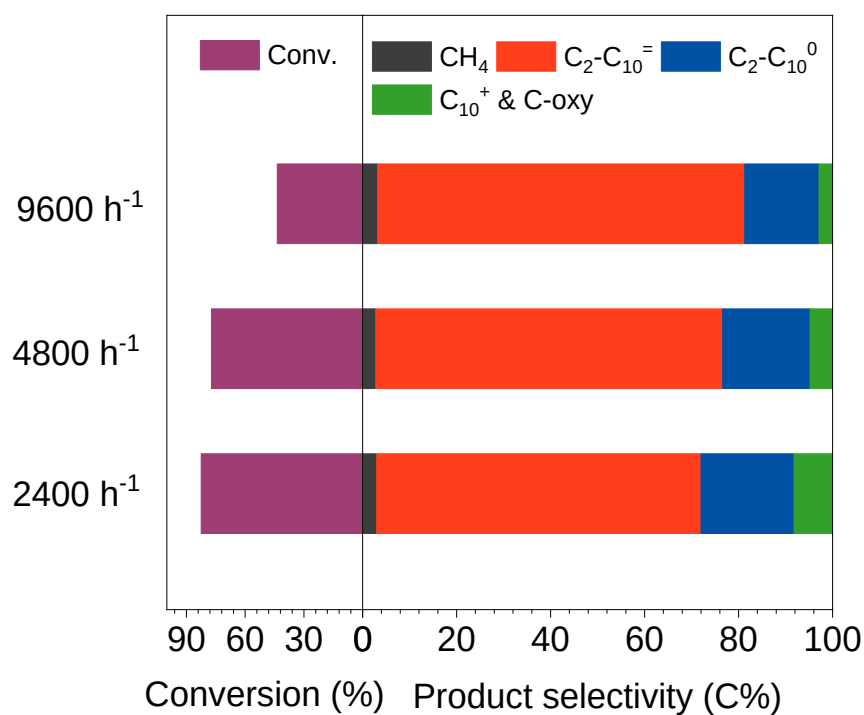
We performed Na1s XPS study to evaluate the sodium migration from Na-FeC_x to zeolite with different proximities, the granule- and powder-mixed Na-FeC_x/s-ZSM-5 catalysts were tested in LT-FTO for 50 h at 260 °C (average diameters: ~1.4 mm for 10-20 mesh, ~0.6 mm for 20-40 mesh, ~0.3 mm for 40-60 mesh, and ~0.16 mm for 80-100 mesh), and then characterized by XPS. Because the Na species on FeC_x and zeolite exhibited distinguishable Na1s peaks at 1071.8 and 1073.3 eV, the sodium migration could be identified by the Na1s peak shift. The catalyst with granules at 10-20 and 20-40 meshes exhibited the Na1s peak centralized at 1071.8 eV, which is very similar to that of the fresh mixture of Na-FeC_x and zeolite, suggesting the undetectable Na migration by XPS technique (Supplementary Fig. 32). For the catalyst with small granules, such as 80-100 mesh, the Na1s peak of the used catalysts appeared at 1072.4 eV with an obvious shift from the peak of Na species on Na-FeC_x (1071.8 eV) because the Na migration occurred. For the Na-FeC_x/s-ZSM-5 catalyst with a very close proximity in powder-mixing manner, the Na1s peak appeared at 1073.3 eV that is similar to that of the Na species on zeolite (Supplementary Fig. 32), suggesting massive sodium migration. Such migration would result into Na-FeC_x catalyst with insufficient sodium species and abundant alkane products in the FTO (Supplementary Table 9).

These results support that the Na migration is strongly influenced by the proximity between the Na-FeC_x and zeolite, where smaller granules would benefit the Na migration, in good agreement with the previous results²². Further studies were performed by treating the Na-FeC_x/s-ZSM-5 catalysts with different granule sizes and powder mixture at 400 °C, giving the Na1s peak at ~1073.3 eV for all the samples (Supplementary Fig. 33). These results confirm the obvious Na migration at higher temperature, even for the catalysts with large granules (e.g. 10-20 mesh).

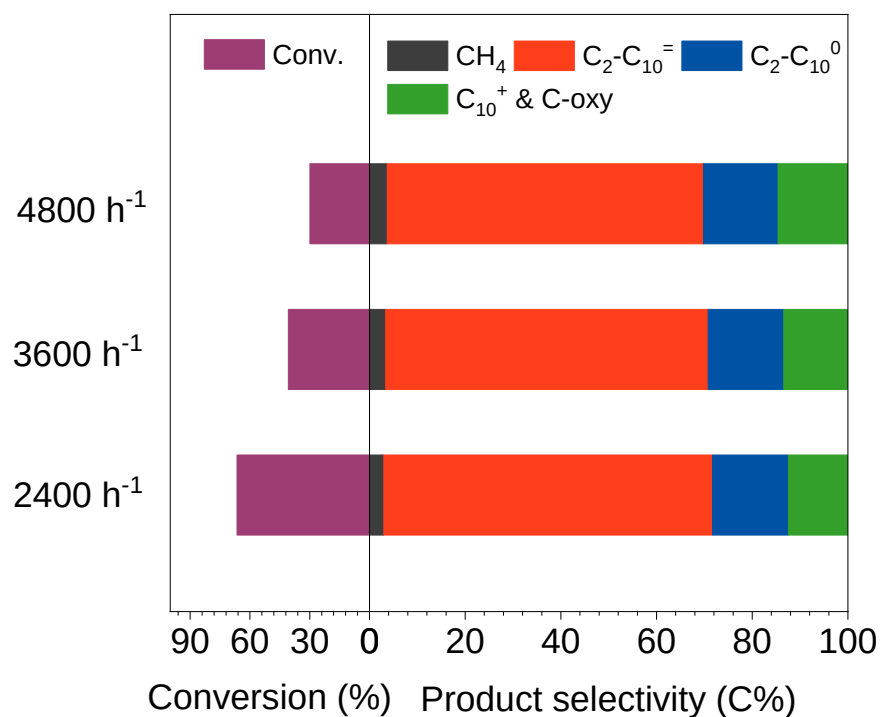
In sum, these data confirm that the sodium migration is strongly related to the proximity of catalyst granules and reaction temperature, where closer proximity and higher reaction temperature would benefit more sodium migration.



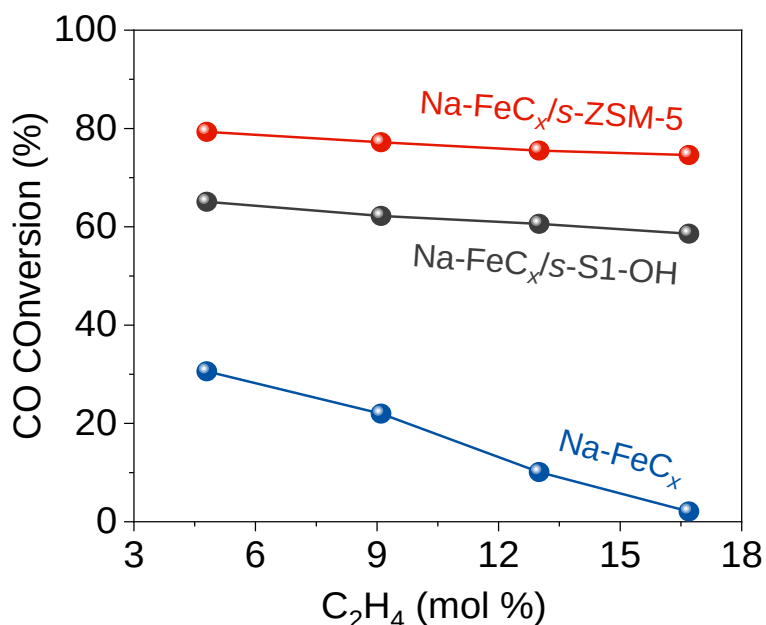
Supplementary Fig. 33. Na_{1s} XPS spectra of Na-FeC_x, *s*-NaZSM-5 (Na⁺-exchanged zeolite as standard), and the granule- and powder-mixing Na-FeC_x/*s*-ZSM-5 catalysts (These samples were treated at 400 °C in flowing nitrogen for 10 h before the XPS analysis). The dotted lines show the standard spectra of Na-FeC_x and *s*-NaZSM-5 for comparison.



Supplementary Fig. 34. Catalytic data of the Na-FeC_x/s-ZSM-5 catalyst in the LT-FTO with different GHSV at 2400, 4800, and 9600 mL·h⁻¹·g⁻¹. Reaction conditions: weight ratio of s-ZSM-5 to Na-FeC_x at 3, ratios of CO/H₂/Ar at 45/45/10, 2.0 MPa, 260 °C, and granule mixture of s-ZSM-5 and Na-FeC_x.



Supplementary Fig. 35. Catalytic data of the Na-FeC_x/s-S1-OH with different GHSV at 2400, 3600, and 4800 mL·h⁻¹·g⁻¹. Reaction conditions: weight ratio of s-ZSM-5 to Na-FeC_x at 3, ratio of CO/H₂/Ar at 45/45/10, 2.0 MPa, 260 °C, and granule mixture of s-S1-OH and Na-FeC_x.

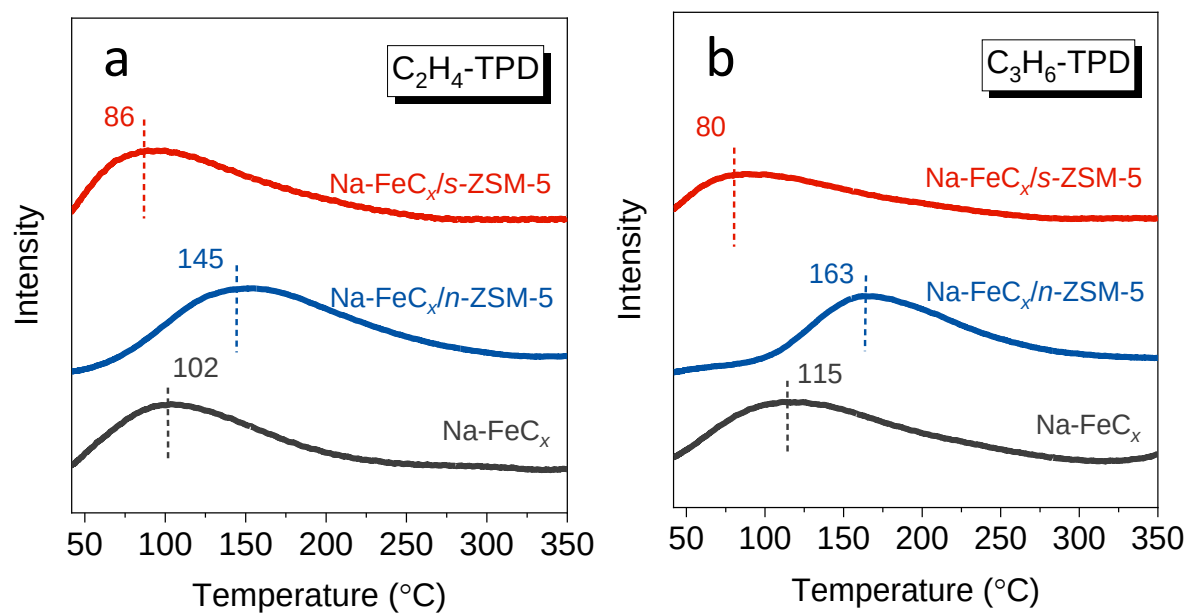


Supplementary Fig. 36. Dependences of CO conversion on ethene concentrations in the feed gas of the LT-FTO reaction over the Na-FeC_x, Na-FeC_x/s-S1-OH and Na-FeC_x/s-ZSM-5 catalysts. Reaction conditions: ratio of H₂/CO at 1.0, 2400 mL·h⁻¹·g⁻¹, 2.0 MPa, reaction temperature at 260 °C for Na-FeC_x/s-S1-OH and Na-FeC_x/s-ZSM-5 catalysts, 300 °C for Na-FeC_x catalyst, and granule mixture of zeolite and Na-FeC_x. Because of poor activity of the Na-FeC_x in the LT-FTO at 260 °C, study over the Na-FeC_x without zeolite was performed at 300 °C for evaluating the influence of ethene to the CO conversion.

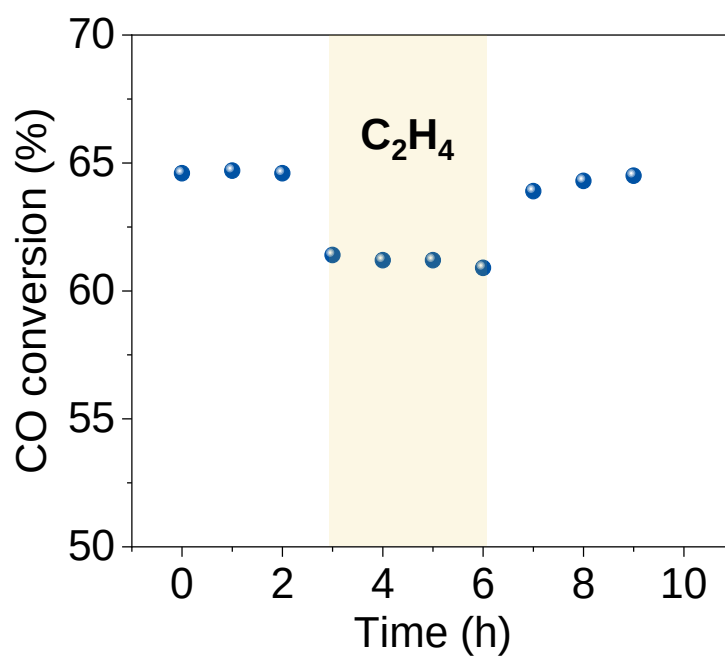
Note: Discussion to exclude the effect of zeolite influenced water diffusion to improve the catalysis. Water is also slightly produced in the syngas conversion reactions, but the zeolite in this work should modulate the olefin sorption rather than water because of the significantly distinguishable phenomenon compared with the previous hydrophobic Fe-based catalysts²². The previous studies have revealed that once the water products were rapidly escaped from the Fe-based catalyst surface, the water-gas shift side reaction would be efficiently hindered to reduce the CO₂ selectivity in FTO. In our case, the CO₂ selectivity was still at 45-50% that is in constant with the general FTO but quite different from the FTO over the hydrophobic catalysts. As well, the rapid water desorption could not obviously increase the CO conversion and adjust the carbon number distribution of the olefin products²², while the zeolite promoter in this work simultaneously boosted the CO conversion and optimize the carbon number distribution. These features are completely different from the hydrophobic catalysts.

Another result to confirm the zeolite influenced olefin sorption more than water is the

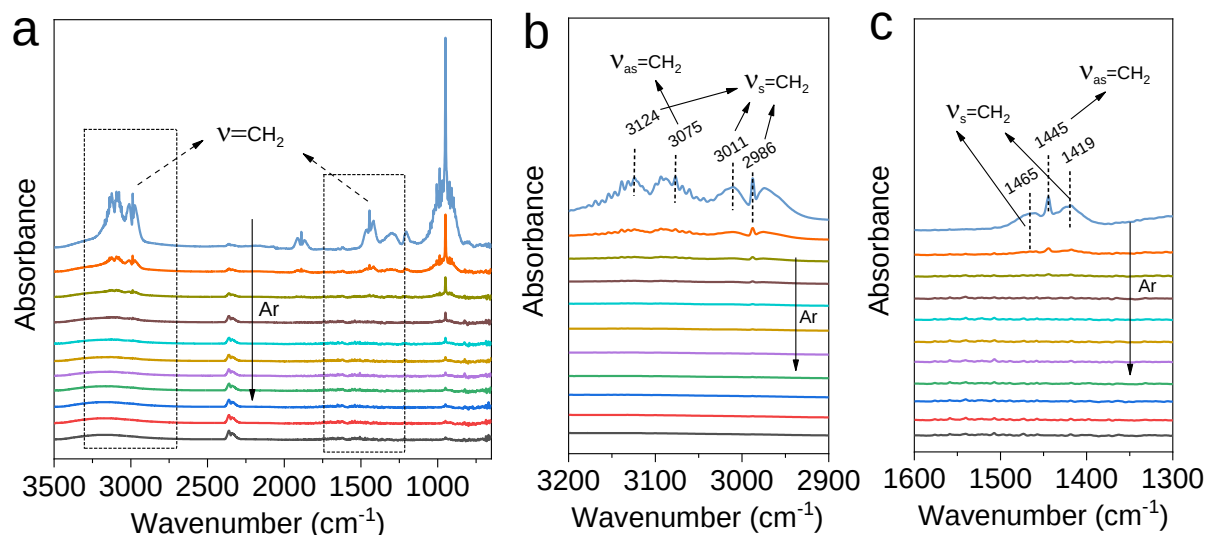
different catalytic behaviors of Na-FeC_x mixed with *s*-ZSM-5 and *n*-ZSM-5. These two zeolites have the same microporous environment but different *b*-axis distance (different diffusion distance for molecules). In the LT-FTO, the two catalysts exhibited quite different product selectivities. The Na-FeC_x/*s*-ZSM-5 gave olefins as dominant products, while the Na-FeC_x/*n*-ZSM-5 favored to produce alkanes (olefin/paraffin molar ratio at only 0.1, Supplementary Table 1). This phenomenon was because of the long retention time of olefins within the micropores of *n*-ZSM-5 that leads to deep hydrogenation to form alkanes, which also confirms the zeolite adjusted olefin diffusion.



Supplementary Fig. 37. Temperature programmed desorption (TPD) curves of (a) ethene and (b) propene over the Na-FeC_x, Na-FeC_x/n-ZSM-5, and Na-FeC_x/s-ZSM-5.

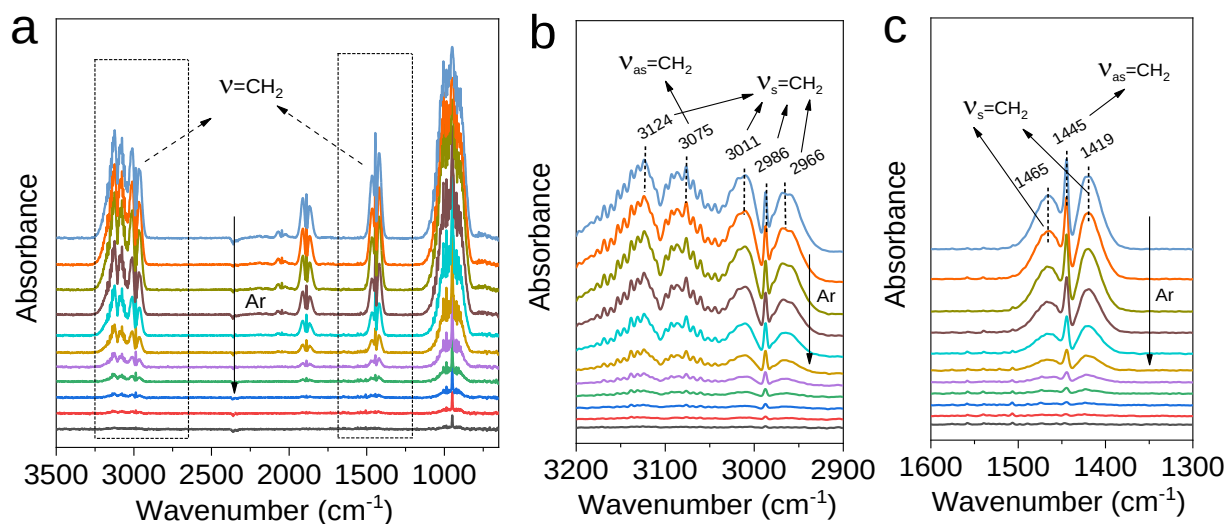


Supplementary Fig. 38. Data showing the influence of ethene to the CO conversion in the LT-FTO over the Na-FeC_x/s-S1-OH catalyst. Reaction conditions: weight ratio of s-S1-OH to Na-FeC_x at 3, GHSV at 2880 mL·h⁻¹·g⁻¹, ratio of ethene/H₂/CO at 0.4/1/1, 2.0 MPa, 260 °C, and granule mixture of Na-FeC_x and zeolite.



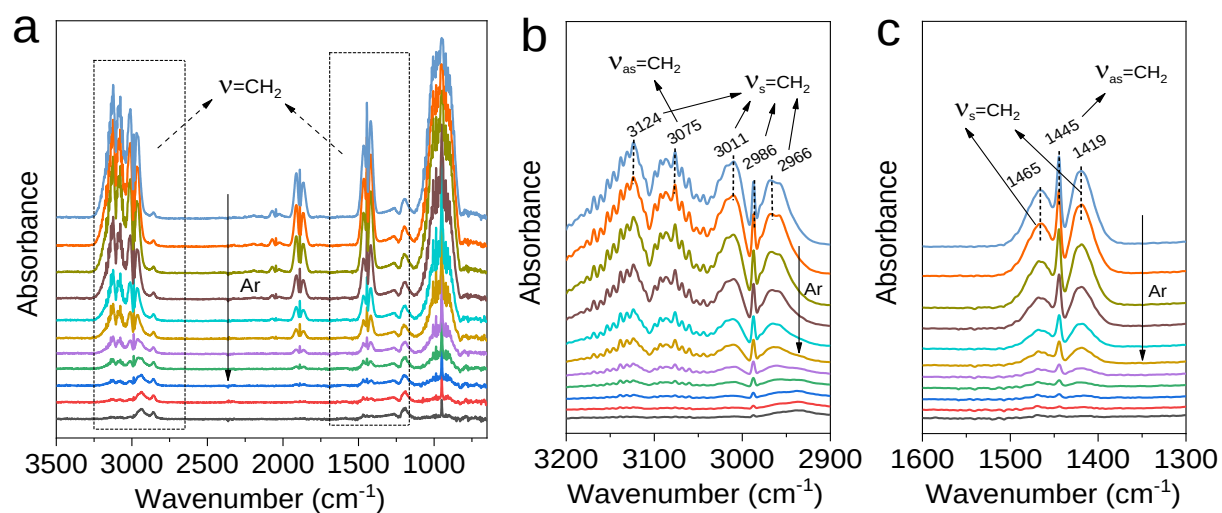
Supplementary Fig. 39. (a) C_2H_4 -desorption DRIFT spectra of the *s*-ZSM-5. (b) Enlarge view of the region at $2900\text{--}3200\text{ cm}^{-1}$. (c) Enlarge view of the region at $1300\text{--}1600\text{ cm}^{-1}$. The IR spectra were recorded at intervals of two minutes.

Note: We preformed the ethene- and propene-desorption DRIFT spectra on *s*-ZSM-5 zeolite without Na-FeC_x . As shown in Supplementary Figs. 39 and 43, most of the ethene and propene molecules would rapidly desorb within the initial several minutes. For example, over 91% of the ethene and 96% of the propene were removed in 4 min (Supplementary Tables 13 and 14). In contrast, their adsorption on Na-FeC_x were much stronger as confirmed by the obvious DRIFT signals after desorption for longer time (Supplementary Figs. 40 and 44). In the olefin desorption test over the $\text{Na-FeC}_x/\text{zeolite}$ samples, a pre-desorption operation was performed for 4 min before recording the signals in Figure 3c in the main text, which would remove the olefins pre-adsorbed in zeolite to avoid their influences to the study.

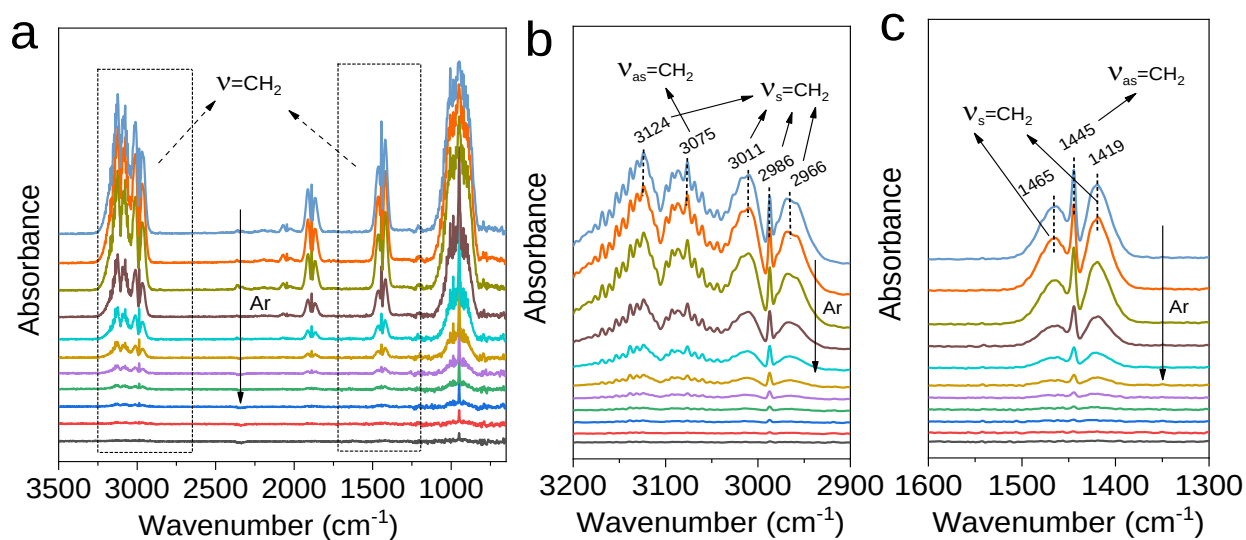


Supplementary Fig. 40. (a) C₂H₄-desorption DRIFT spectra of the Na-FeC_x. (b) Enlarge view of the region at 2900-3200 cm⁻¹. (c) Enlarge view of the region at 1300-1600 cm⁻¹. The IR spectra were recorded at intervals of two minutes.

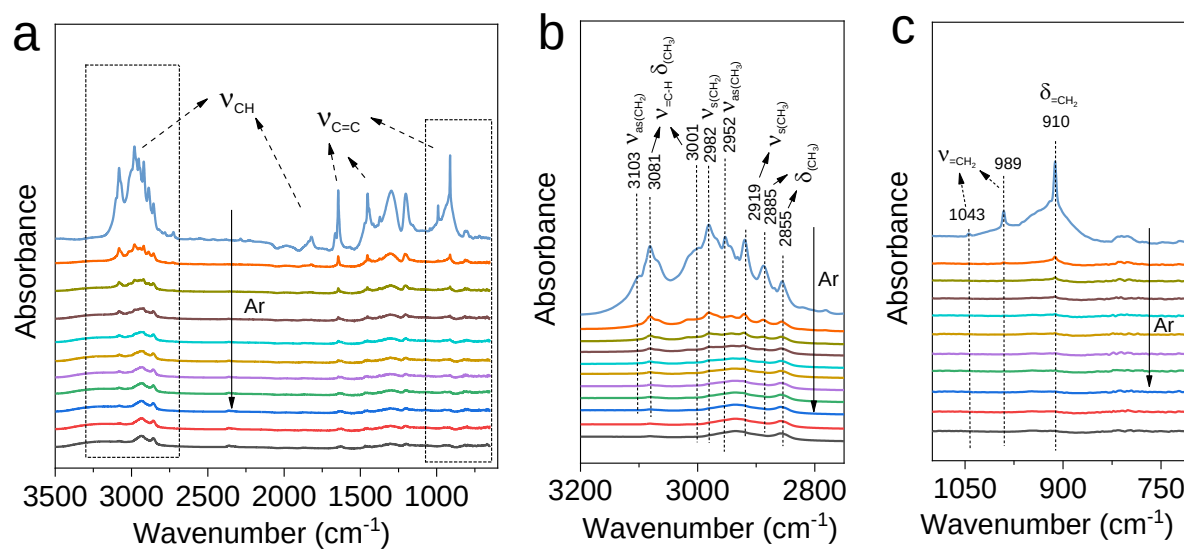
Note: The data in Fig. 3c were selected from the spectra in Supplementary Figs. 40-42.



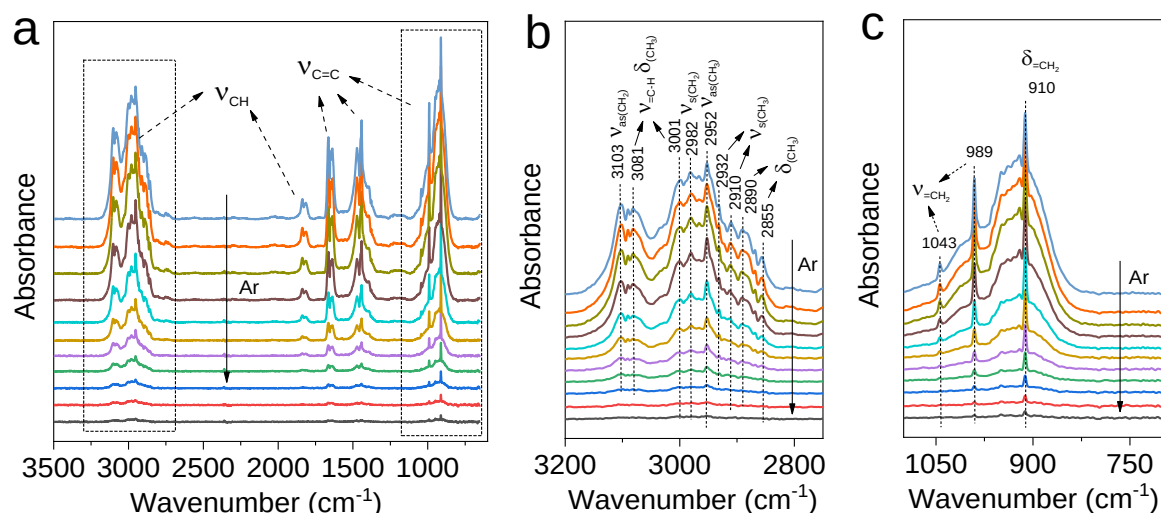
Supplementary Fig. 41. (a) C₂H₄-desorption DRIFT spectra of the Na-FeC_x/n-ZSM-5. (b) Enlarge view of the region at 2900-3200 cm⁻¹. (c) Enlarge view of the region at 1300-1600 cm⁻¹. The IR spectra were recorded at intervals of two minutes.



Supplementary Fig. 42. (a) C_2H_4 -desorption DRIFT spectra of the Na- $\text{FeC}_x/\text{s-ZSM-5}$. (b) Enlarge view of the region at $2900\text{--}3200\text{ cm}^{-1}$. (c) Enlarge view of the region at $1300\text{--}1600\text{ cm}^{-1}$. The IR spectra were recorded at intervals of two minutes.

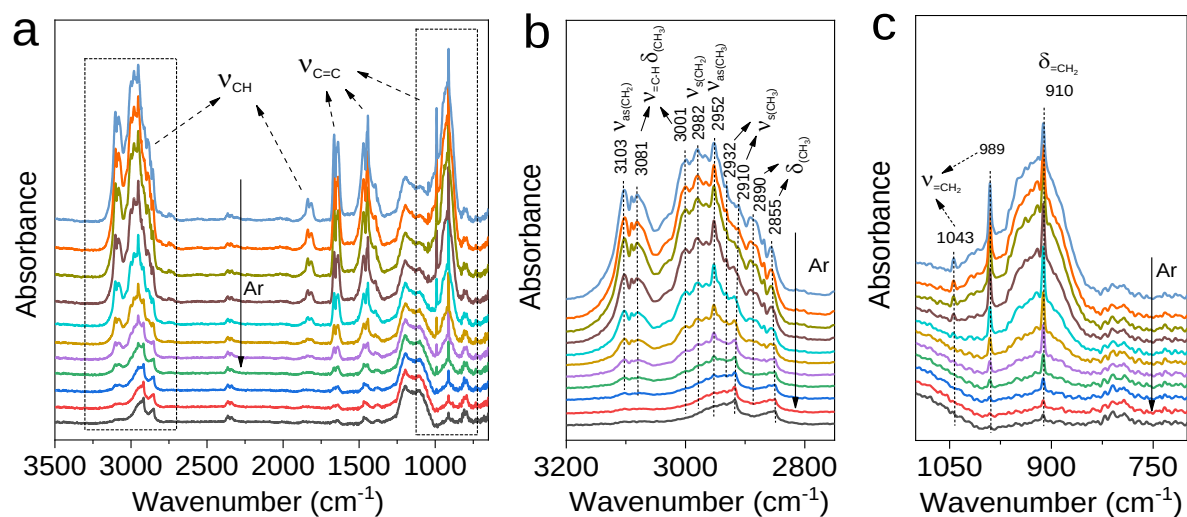


Supplementary Fig. 43. (a) C_3H_6 -desorption DRIFT spectra of the *s*-ZSM-5. (b) Enlarge view of the region at 2900-3200 cm^{-1} . (c) Enlarge view of the region at 1300-1600 cm^{-1} . The IR spectra were recorded at intervals of two minutes.

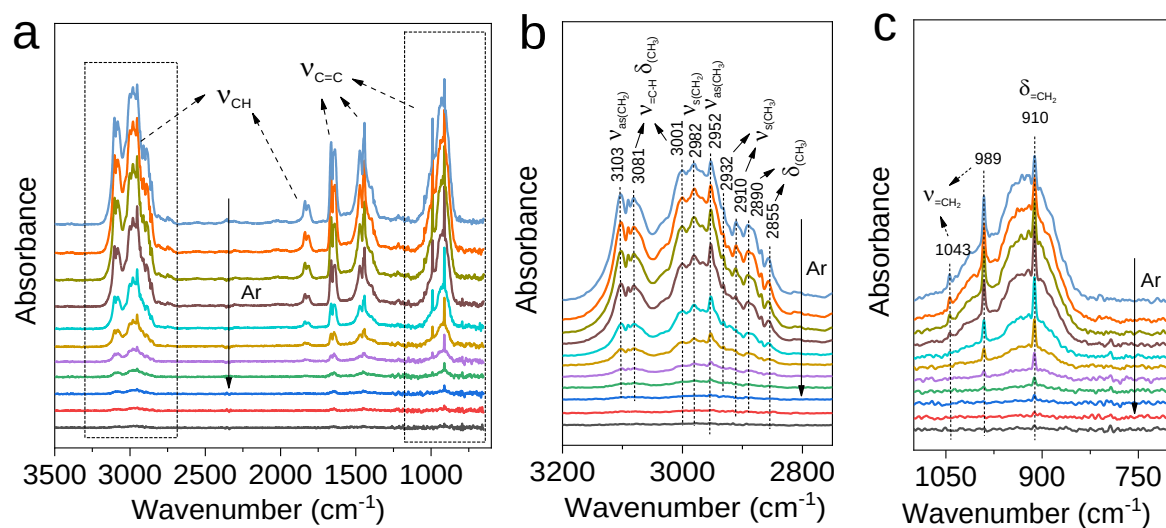


Supplementary Fig. 44. (a) C_3H_6 -desorption DRIFT spectra of the Na-FeC_x . (b) Enlarge view of the region at $2750\text{--}3200\text{ cm}^{-1}$. (c) Enlarge view of the region at $700\text{--}1100\text{ cm}^{-1}$. The IR spectra were recorded at intervals of two minutes.

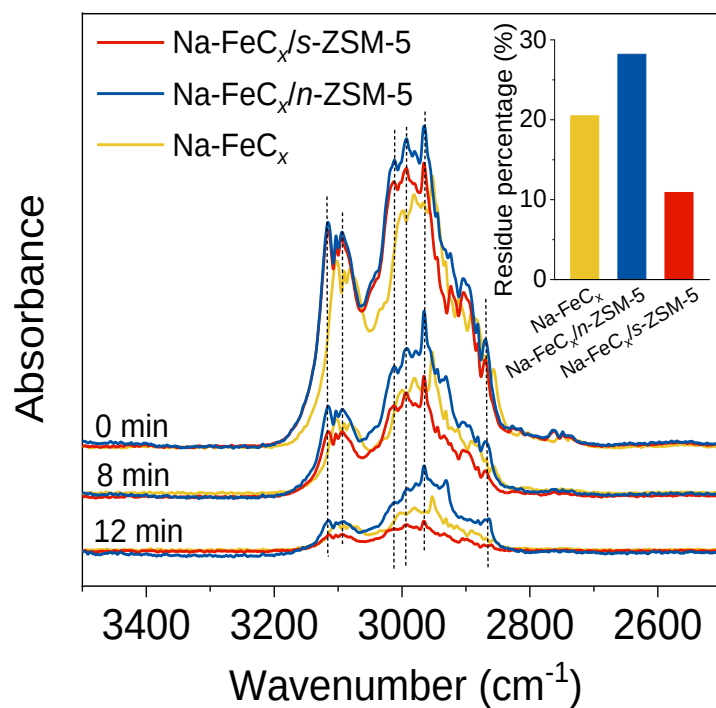
Note: As shown in Supplementary Figs. 43-46, after pre-adsorption treatment, all samples appeared the bands assigning to symmetrical and asymmetrical stretching vibration of $=\text{CH}_2$ (signals at 989 , 1043 , 2982 , and 3103 cm^{-1}), symmetrical and asymmetrical stretching vibration of $-\text{CH}_3$ (signals at 2910 , 2932 , and 2952 cm^{-1}), deformation vibration of CH_2 (signals at 910 cm^{-1}), and deformation vibration of CH_3 (signals at 2855 , 2890 , 3001 , and 3081 cm^{-1}). After flowing Ar to the samples, the IR spectra were recorded every 2 min. According to the peak intensity changes as a function of time, it was evaluated that the propene desorption rates over the $\text{Na-FeC}_x/\text{s-ZSM-5}$ and $\text{Na-FeC}_x/\text{s-S1-OH}$ were higher than Na-FeC_x (Supplementary Figs. 47 and 49).



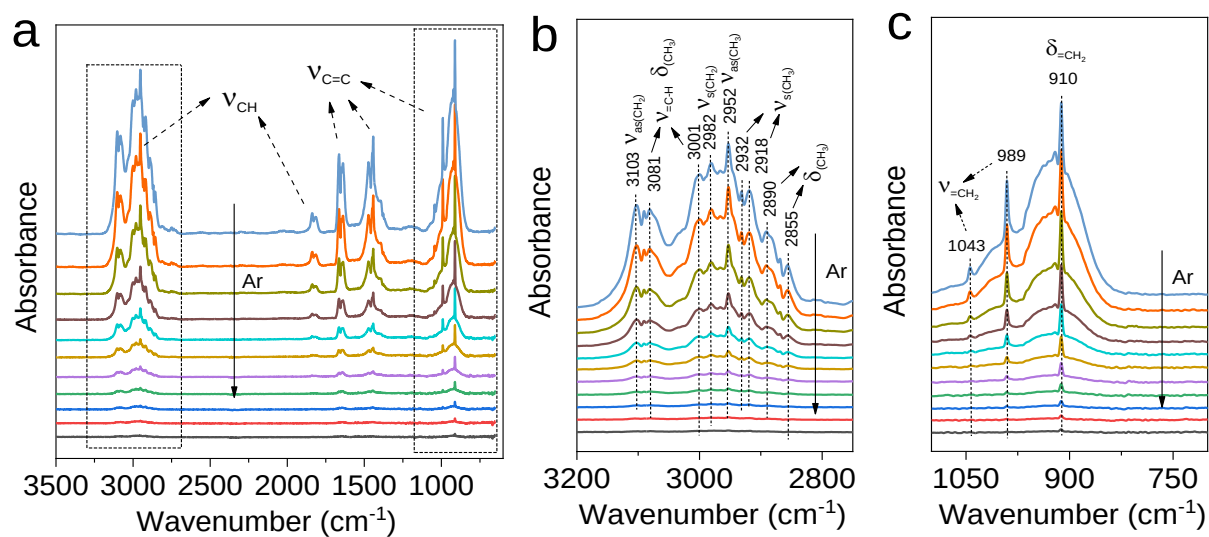
Supplementary Fig. 45. (a) C_3H_6 -desorption DRIFT spectra of the $\text{Na-FeC}_x/\text{n-ZSM-5}$. (b) Enlarged view of the region at 2750-3200 cm^{-1} . (c) Enlarged view of the region at 700-1100 cm^{-1} . The IR spectra were recorded at intervals of two minutes.



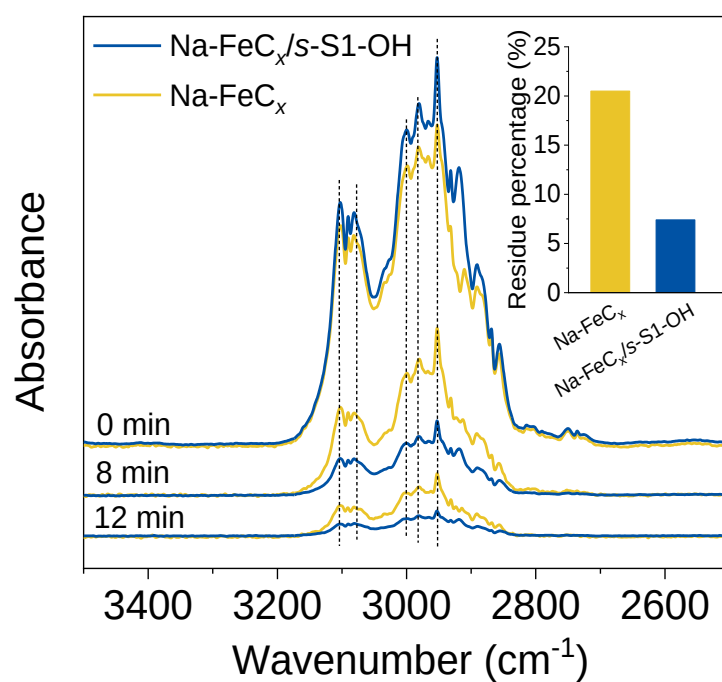
Supplementary Fig. 46. (a) C₃H₆-desorption DRIFT spectra of the Na-FeC_x/s-ZSM-5. (b) Enlarge view of the region at 2750-3200 cm⁻¹. (c) Enlarge view of the region at 700-1100 cm⁻¹. The IR spectra were recorded at intervals of two minutes.



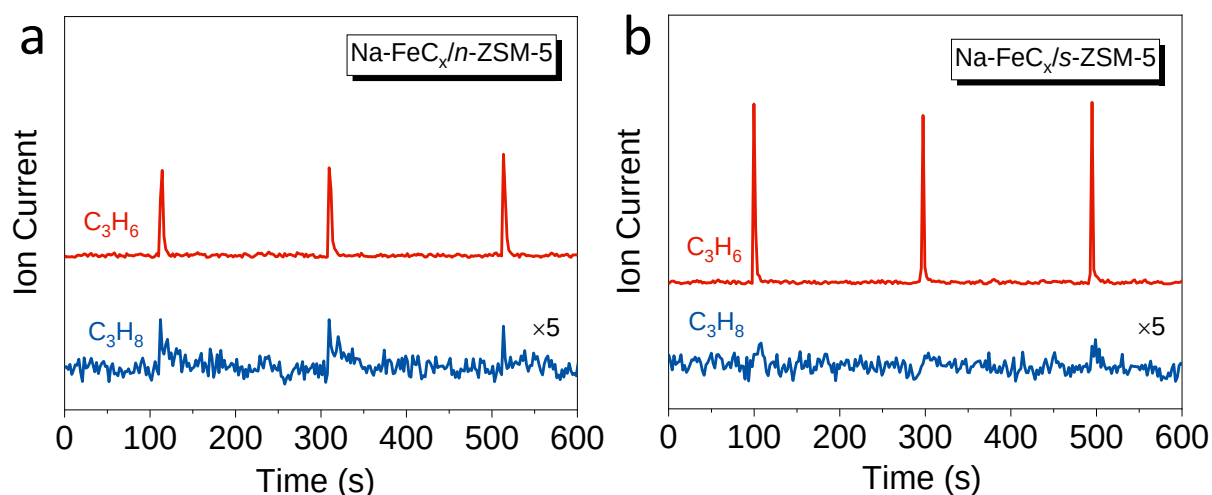
Supplementary Fig. 47. Typical spectra at different time during the C₃H₆-desorption IR tests over the Na-FeC_x/s-ZSM-5, Na-FeC_x/n-ZSM-5, and Na-FeC_x catalysts. These typical spectra were selected from the data in Supplementary Figs. 44-46. The inset figure gave the residue percentage of propene according to the peak intensity changes (signal at 2952 cm^{-1}) over the various catalysts during 0-12 min (data calculated from Supplementary Table 14).



Supplementary Fig. 48. (a) C_3H_6 -desorption DRIFT spectra of the $\text{Na-FeC}_x/\text{s-S1-OH}$. (b) Enlarge view of the region at 2750-3200 cm^{-1} . (c) Enlarge view of the region at 700-1100 cm^{-1} . The IR spectra were recorded at intervals of two minutes.

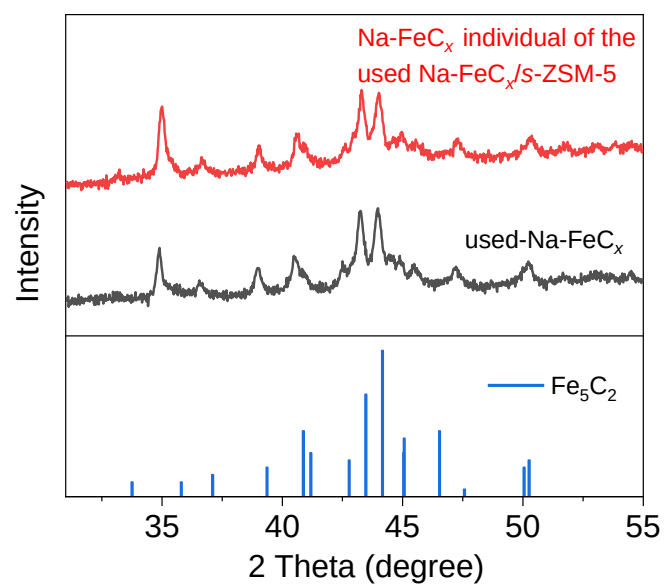


Supplementary Fig. 49. Typical spectra at different time during the C₃H₆-desorption DRIFT tests over the Na-FeC_x/s-S1-OH and Na-FeC_x catalysts. These typical spectra were selected from the data in Supplementary Figs. 44 and 48. The inset figure gave the residue percentage of propene according to the peak intensity changes (signal at 2952 cm⁻¹) over the various catalysts during 0-12 min (data calculated from Supplementary Table 14).

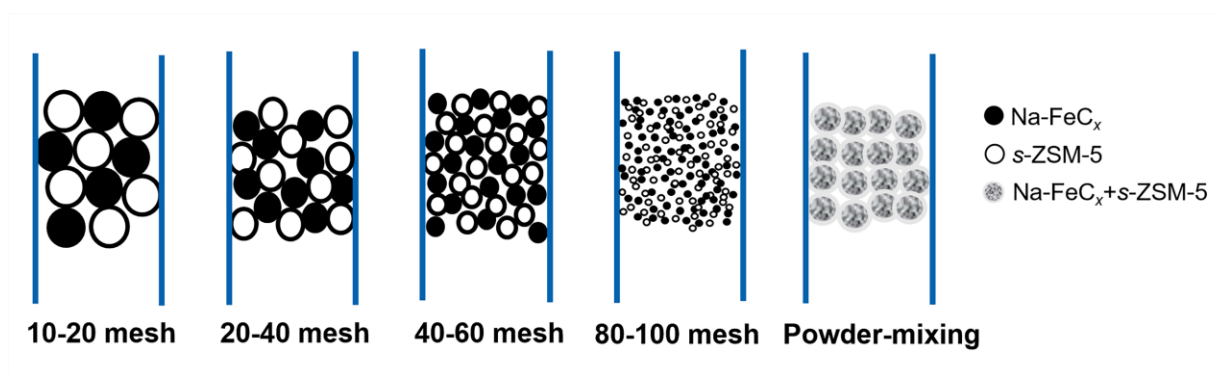


Supplementary Fig. 50. Mass spectra obtained by pulsing 1 mL of 10% C_3H_6/He into a pure H_2 flow (200 mL/min) over the (a) $Na-FeC_x/n-ZSM-5$ and (b) $Na-FeC_x/s-ZSM-5$ catalysts. Reaction conditions: weight ratio of zeolite to $Na-FeC_x$ at 3, 0.1 MPa, 260 °C, and the granule-mixing of the zeolite and $Na-FeC_x$ individuals.

Note: We used pulse experiments to explore the reactivity of olefins on the $Na-FeC_x/n-ZSM-5$ and $Na-FeC_x/s-ZSM-5$. More propane was formed over the $Na-FeC_x/n-ZSM-5$ than that over the $Na-FeC_x/s-ZSM-5$. This phenomenon might be explained by stronger hydrogen spillover and longer retention time of propene molecules in the $n-ZSM-5$ zeolite than that in the $s-ZSM-5$ zeolite, which benefit the deep hydrogenation of propene to propane.

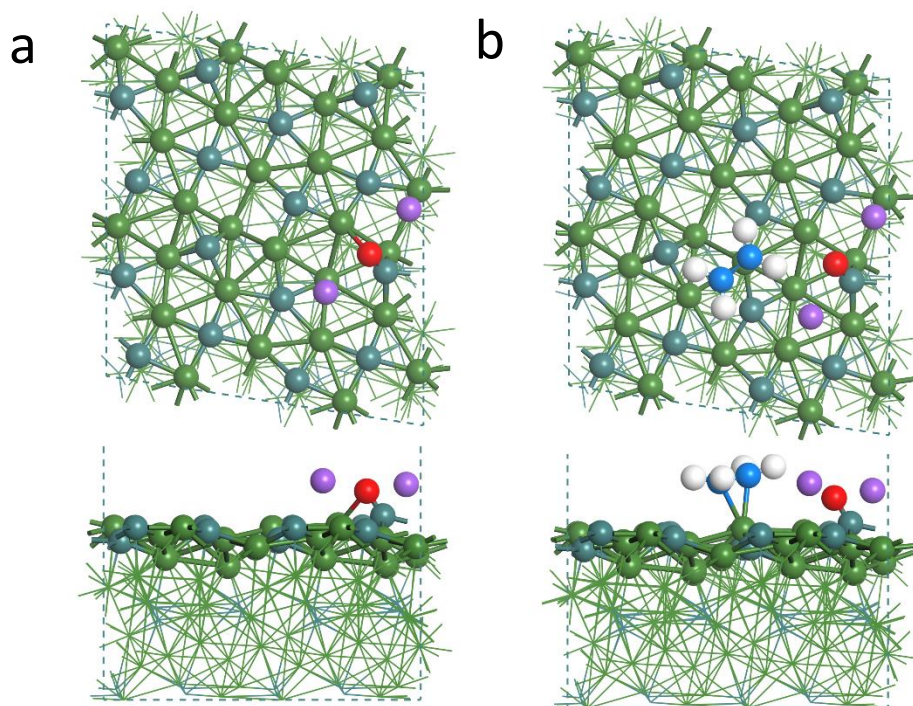


Supplementary Fig. 51. XRD patterns of the used Na-FeC_x in the reactions with and without zeolites. The samples exhibited almost the same XRD patterns with Fe₅C₂ as a dominant phase.

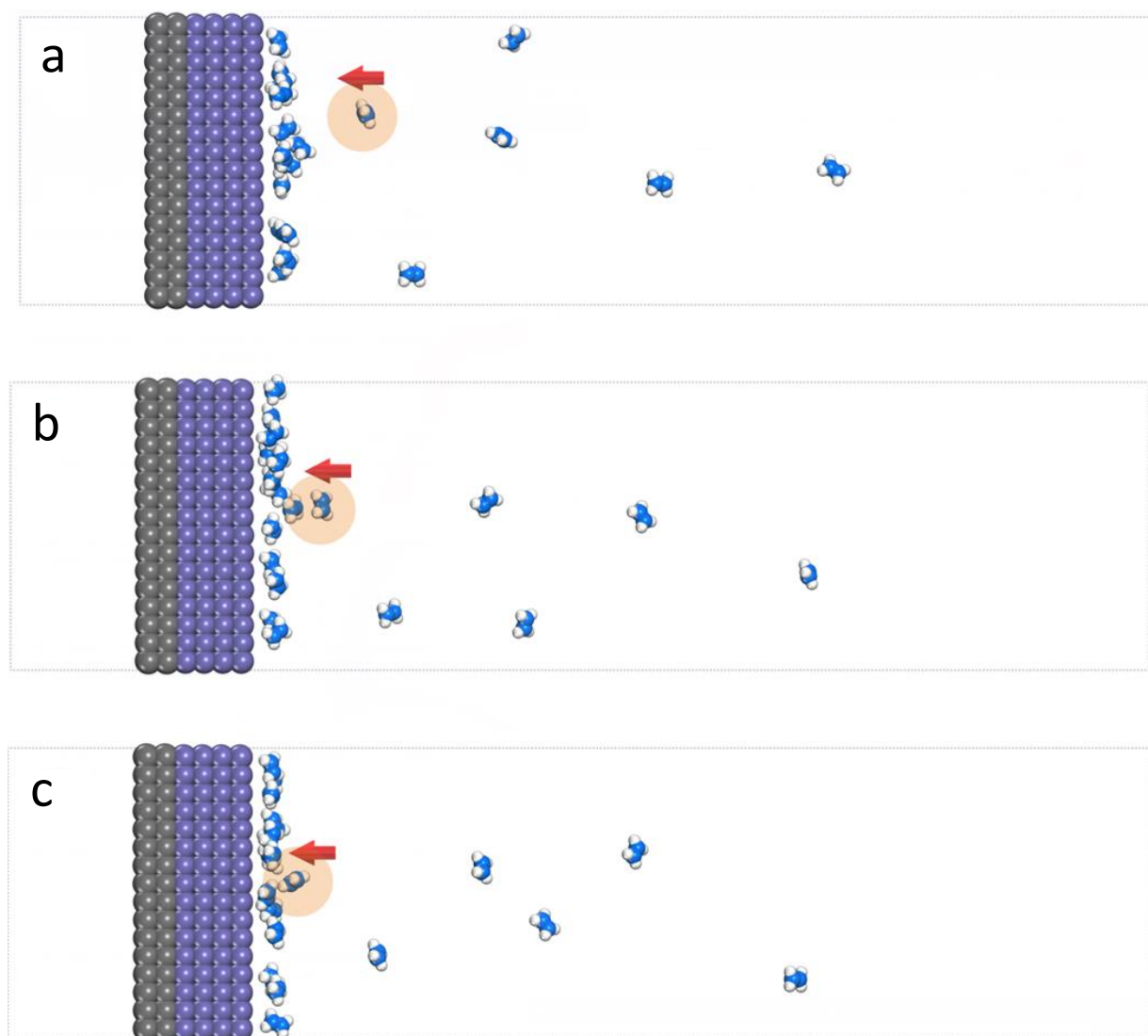


Supplementary Fig. 52. Models of Na-FeC_x/s-ZSM-5 catalysts with different proximities between Na-FeC_x and s-ZSM-5.

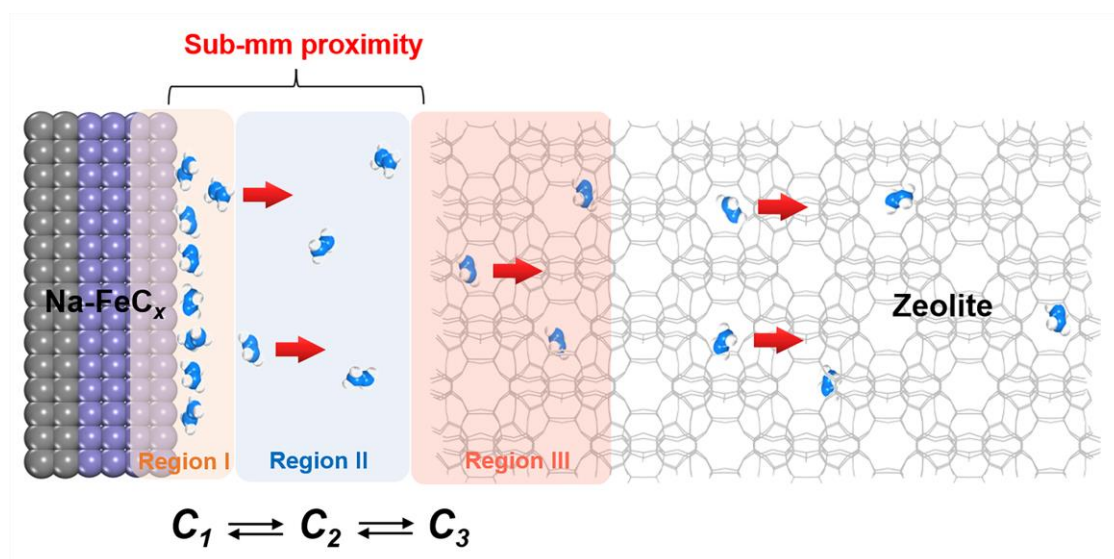
Note : We evaluated the Na-FeC_x/s-ZSM-5 catalysts with different granule sizes (average diameters: ~1.4 mm for 10-20 mesh, ~0.6 mm for 20-40 mesh, ~0.3 mm for 40-60 mesh, and ~0.16 mm for 80-100 mesh) and the powder-mixing Na-FeC_x/s-ZSM-5 catalyst (Supplementary Fig. 52). As shown in Supplementary Table 10, the Na-FeC_x/s-ZSM-5 catalysts with different granule sizes at ~0.3-1.4 mm exhibited the CO conversions at 77.1-88.3%, which are obviously higher than that over the Na-FeC_x catalyst without zeolite promoter. In these reactions over granule-mixing Na-FeC_x/s-ZSM-5 catalysts, the *o/a* ratios were 3.5-4.0. The test using the catalyst with closer proximity with granules at ~0.16 mm (80-100 mesh) still exhibited significant promotion effect of zeolite, giving the CO conversion at 69.8%. This value is lower than that with larger granules, which might be due to stronger sodium migration with close proximity as confirmed by the Na1s XPS study (Supplementary Fig. 32). These data show that the granule mixture with a scale proximity at ~0.16 mm could boost the syngas conversion and maintain the high olefin selectivity, but the proximity at ~0.3 mm is suggested for obtaining better performance.



Supplementary Fig. 53. The top and side view structures of (a) Na-FeC_x, and (b) one ethene adsorbed on the Na-FeC_x (C₂H₄: C atoms in blue and H atoms in white; Na-FeC_x: O atoms in red, Na atoms in purple, Fe atoms in green and C atoms in dark cyan). The adsorption energy values are -0.58 and -0.28 eV for the first and second ethene molecules, respectively.

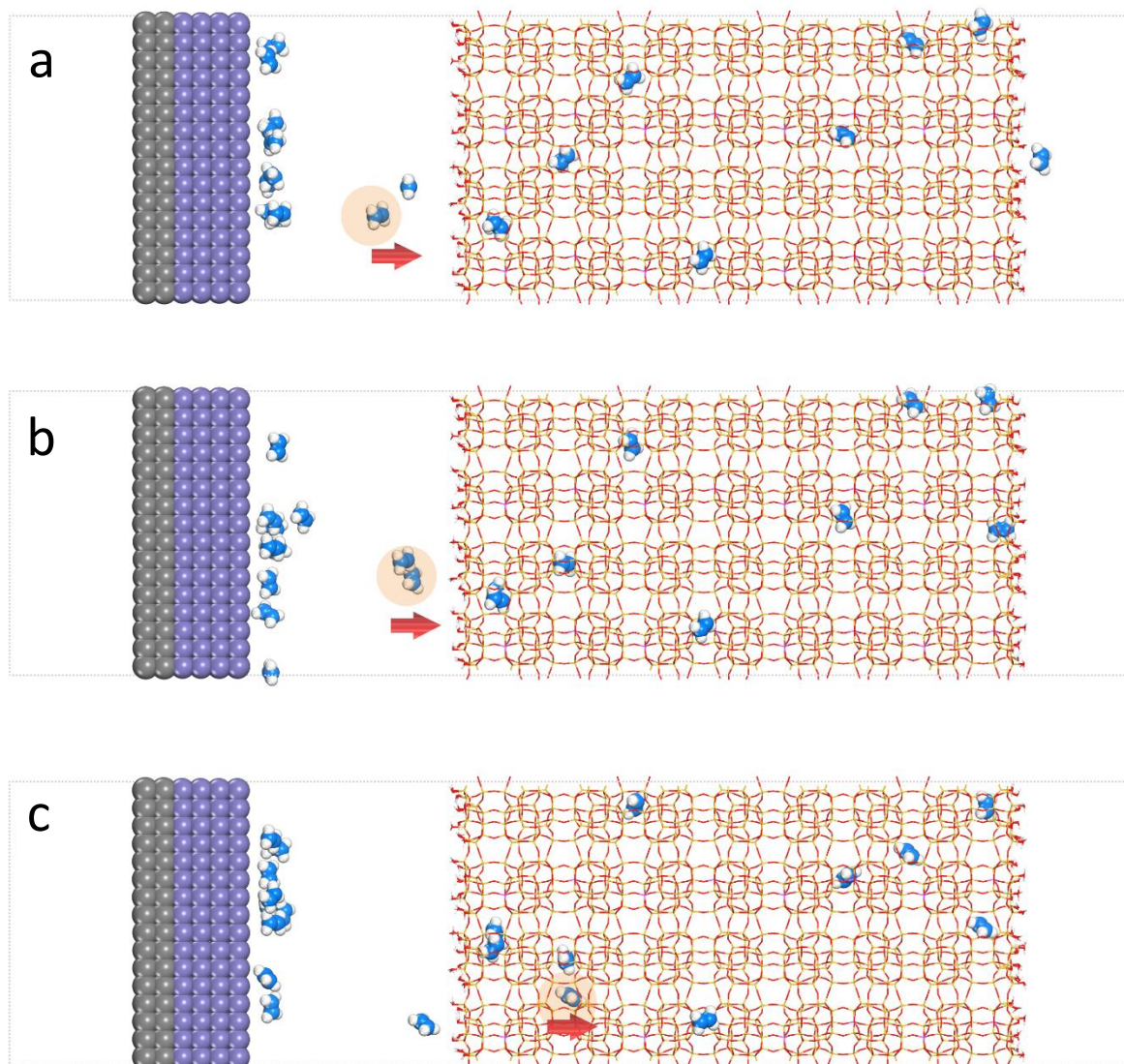


Supplementary Fig. 54. Scheme showing 20 ethene molecules (blue-white balls) desorbing from Na-FeC_x (black-purple surfaces) following the time of a → b → c. It shows free ethene preferring to re-adsorb on the Na-FeC_x surface that hinders the desorption process.

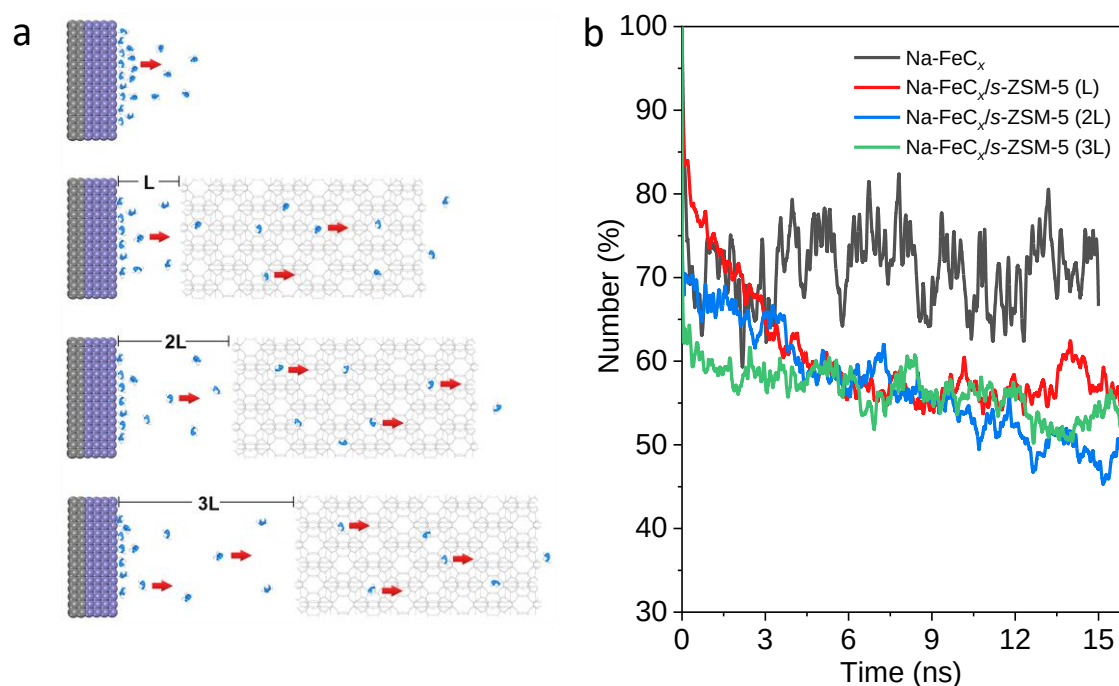


Supplementary Fig. 55. Scheme showing the diffusion of olefins from the Na-FeC_x surface to the zeolite *via* different regions.

Note: With the sub-mm proximity, the diffusion of olefins from the Na-FeC_x surface to the zeolite could be described in the model, the olefins formed on the Na-FeC_x surface (region I) could desorb into the free region between Na-FeC_x and zeolite (region II), where the olefin concentration on Na-FeC_x surface (region I) and in free region (region II) could be described as C₁ and C₂, respectively. On the Na-FeC_x catalyst, the LT-FTO reaction was greatly hindered because of the olefin desorption and re-adsorption on the Na-FeC_x surface described by the equilibrium between C₁ and C₂ ($*C_nH_{2n} \rightleftharpoons * + C_nH_{2n}$, * means the Na-FeC_x surface sites). On the zeolite side, the free olefins in region II could be rapidly adsorbed by zeolite (region III) to obviously reduce C₂, which would efficiently shift the equilibrium of $*C_nH_{2n} \rightleftharpoons * + C_nH_{2n}$ on Na-FeC_x surface.

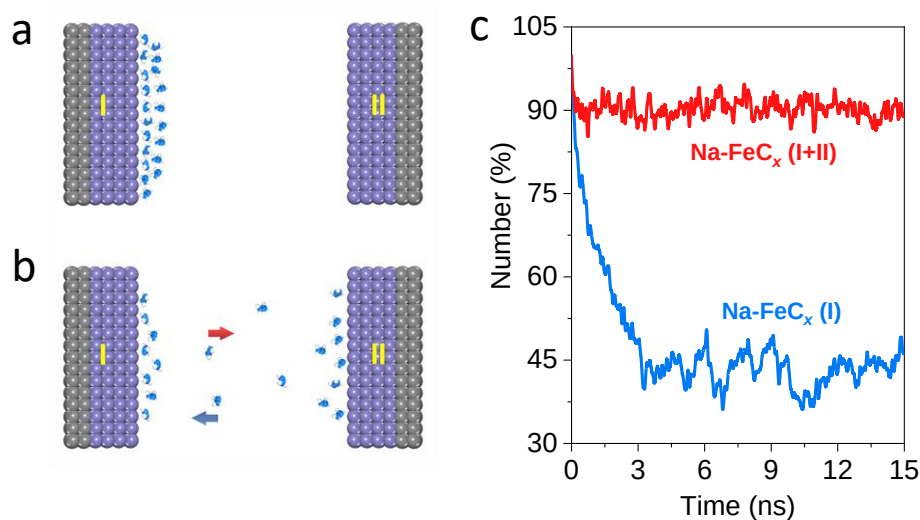


Supplementary Fig. 56. Scheme showing 20 ethene molecules (blue-white balls) desorbing from Na-FeC_x (black-purple surfaces)/*s*-ZSM-5 (red-yellow) following the time of a → b → c. It shows free ethene preferring to adsorb on the *s*-ZSM-5 that facilitates ethene desorption from Na-FeC_x surface.



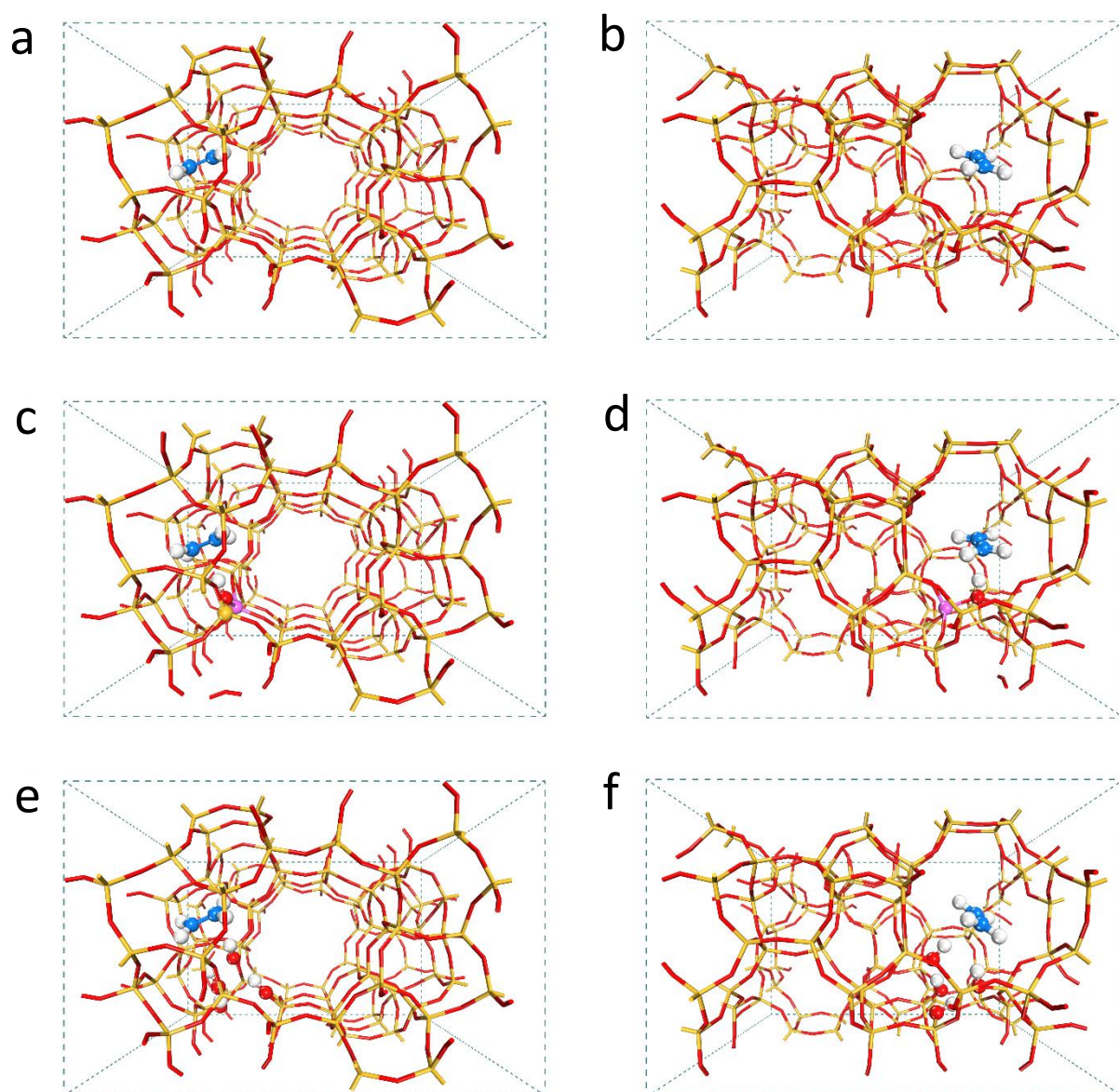
Supplementary Fig. 57. (a) The models and (b) the percentages of ethene molecules adsorption on Na-FeC_x surface during the desorption process with different distance between Na-FeC_x and Na-FeC_x/s-ZSM-5. Ethene molecules: Blue-white balls. Na-FeC_x: Black-purple surfaces. s-ZSM-5: Gray grids.

Note: We have performed the theoretical investigation on ethene desorption/adsorption on the model of Na-FeC_x and zeolite component with varied distance. 20 ethene molecules were localized on the Na-FeC_x surface, and their desorption processes were qualitatively predicted as a function of desorption time by the MD simulation at 260 °C. As shown in Supplementary Fig. 57, the Na-FeC_x/s-ZSM-5 with different proximities exhibited the comparable ethene desorption efficiency, as suggested by the residue ethene molecules on the Na-FeC_x surface with a quantitative equilibrium at ~50%. This result is well consistent with the experimental results, where the Na-FeC_x/s-ZSM-5 catalysts with different proximities (e.g. 10-20, 20-40, 40-60, and 80-100 mesh) could boost the CO conversion, because the zeolite optimized the olefin escape from Na-FeC_x surface (region I in Supplementary Fig. 55) through capturing the olefins from the free region (region II) rather than directly interact with the Na-FeC_x. The proximities between Na-FeC_x and zeolite (diameter of region II) within sub-mm level are enough for olefin escaping, but the Na migration would deactivate the catalysts with a very close proximity (e.g. powder mixing). Therefore, an appropriate proximity is still suggested in the experimental tests.

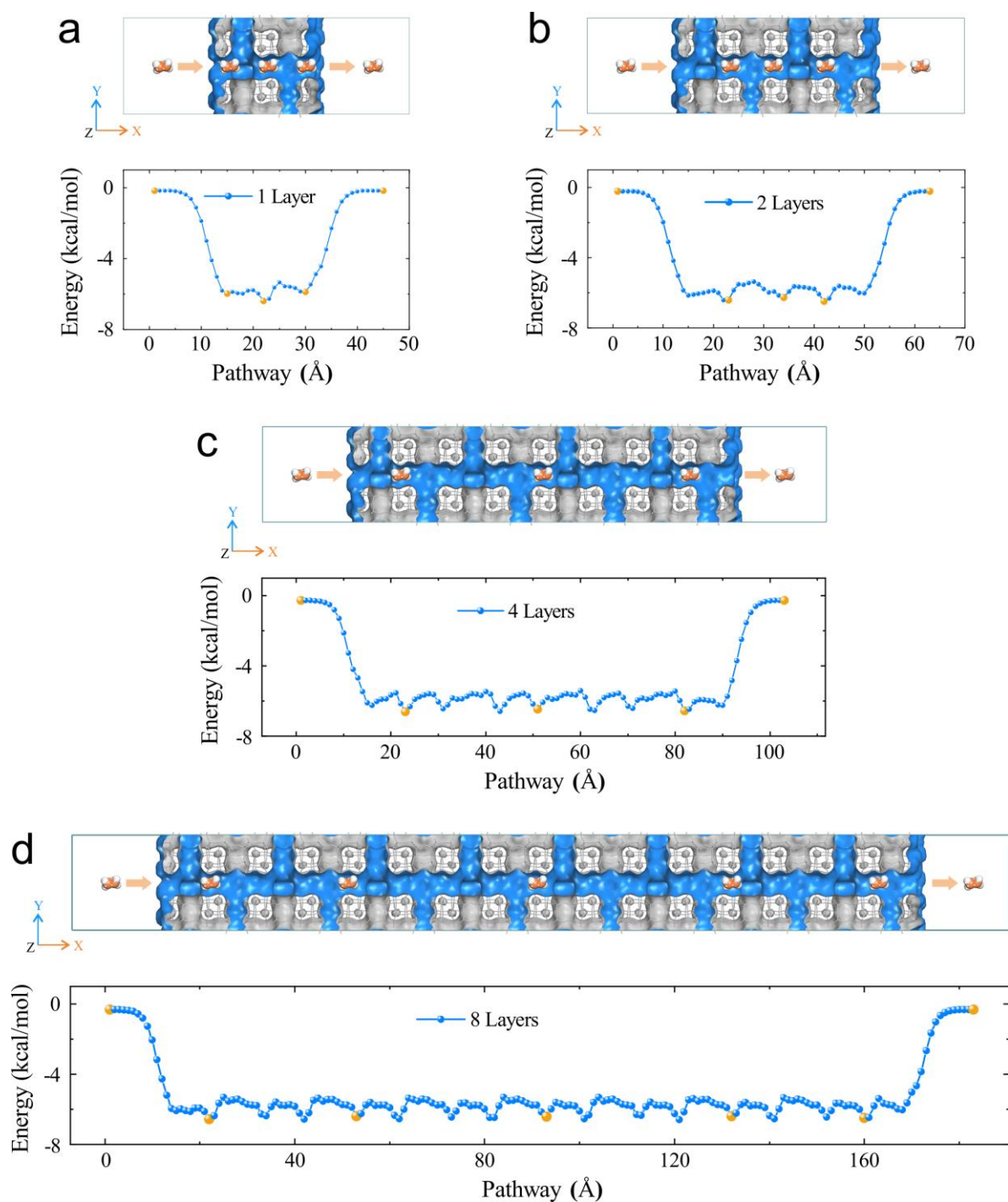


Supplementary Fig. 58. (a) The initial state of 20 ethene molecules adsorbed on Na-FeC_x (I) surface with Na-FeC_x (II). (b) Scheme showing ethene molecules desorbing from Na-FeC_x surface (I) to the other (II). Ethene molecules: Blue-white balls. Na-FeC_x: Black-purple surfaces. (c) Percentages of ethene molecules adsorption on Na-FeC_x surface (I) and (I+II) during the desorption process.

Note: We performed the theoretical investigation on ethene desorption/adsorption on models of Na-FeC_x and Na-FeC_x (20 molecules were localized on the Na-FeC_x surface, and their desorption processes were qualitatively predicted as a function of desorption time by the MD simulation at 260 °C). The results showed that the number of ethene molecules on the Na-FeC_x surface (I in Supplementary Fig. 58) would desorb rapidly and re-adsorb on another Na-FeC_x surface (II in Supplementary Fig. 58). In this case, the quantitative equilibrium gave about 90% of the ethene molecules adsorbing on the Na-FeC_x surface (I and II in Supplementary Fig. 58), which is much higher than 48% in the case using zeolite (Supplementary Fig. 62, Fig. 4d). This result demonstrates the importance of zeolite for the product escape from the Na-FeC_x surface.



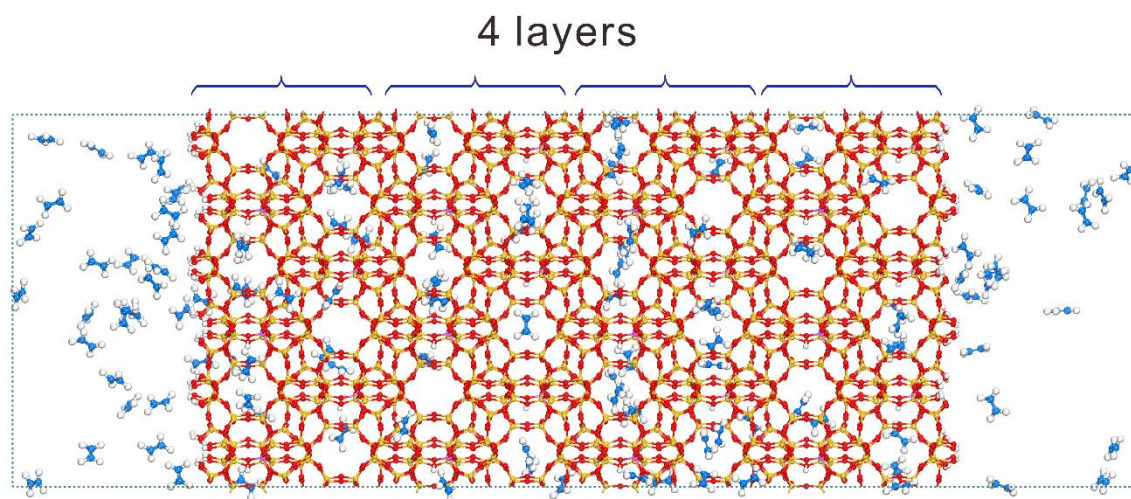
Supplementary Fig. 59. Structures showing ethene adsorption within the micropores of MFI structures. Siliceous MFI (S-1) along (a) Y and (b) X directions, proton-form aluminosilicate MFI (ZSM-5) along (c) Y and (d) X directions, and silanol modified siliceous MFI (S1-OH) along (e) Y and (f) X directions. Zero damping DFT-D3 method of Grime was used for calculating the adsorption energy.



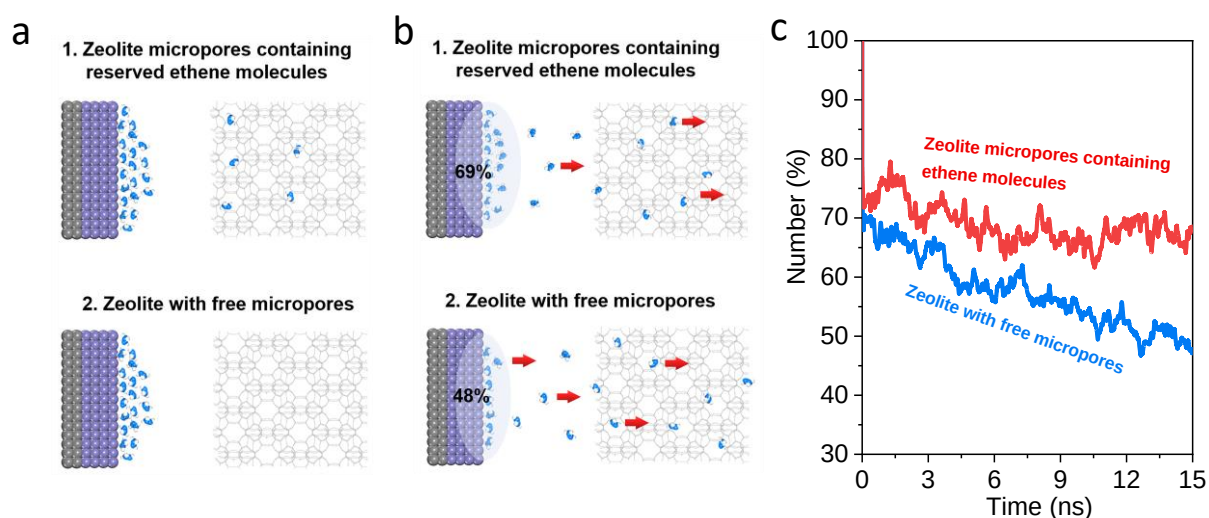
Supplementary Fig. 60. Definition of ethene diffusion pathways and according interaction energy profiles along with straight channels of *s*-ZSM-5 with (a) 1, (b) 2, (c) 4, and (d) 8 layers of units. These data were obtained using a molecular dynamics simulation.

Note: In this work, the efficacy of olefin diffusion within zeolite micropores is important for the rapidly olefin product escaping. Theoretical results on MFI zeolite structures with the same components have revealed that the olefin diffusion within zeolite micropores proceeds a site-by-

site process, and the migration from one site to another proceeds the same energy barriers (Supplementary Fig. 60). Therefore, for the MFI zeolites with similar Si/Al ratios (e.g. *s*-ZSM-5 vs *n*-ZSM-5), the diffusion efficiency of olefin within one zeolite crystal is not determined by the energy barriers, but strongly related to the number of adsorption sites within the zeolite micropores (e.g. proton sites within aluminosilicate zeolites and silanol sites within siliceous zeolites). Considering the MFI zeolites with the same components (e.g. Si/Al ratios), the number of adsorption sites is linearly related to the length of the microporous channels in one zeolite crystal. According to the theoretical simulations, the efficacy of olefin diffusion in one MFI zeolite crystal is strongly related to the distance of diffusion channels, which could be adjusted by the zeolite crystal morphology (e.g. *s*-ZSM-5 vs *n*-ZSM-5).



Supplementary Fig. 61. Model showing 128 ethene molecules adsorbed in *s*-ZSM-5 structure with 4-unit layers. The number of ethene molecules at 32, 64 and 256 were used for calculating in *s*-ZSM-5 models with 1-, 2- and 8-unit layers, respectively.



Supplementary Fig. 62. (a) The initial state of 20 ethene molecules adsorbed on Na-FeC_x surface with zeolite micropores containing reserved ethene molecules and zeolite with free micropores. (b) Scheme showing ethene molecules desorbing from Na-FeC_x surface to the zeolite micropores containing reserved ethene molecules and zeolite with free micropores. Ethene molecules: Blue-white balls. Na-FeC_x: Black-purple surfaces. *s*-ZSM-5: Gray grids. (c) Percentages of ethene molecules adsorption on Na-FeC_x surface during the desorption process over zeolite micropores containing ethene molecules and zeolite with free micropores.

Note: 20 ethene molecules were localized on the Na-FeC_x surface, and their desorption processes were qualitatively predicted as a function of desorption time by the MD simulation at 260 °C. In this case, the number of ethene molecules on the Na-FeC_x surface maintained a quantitative equilibrium of about 69%, much higher than 48% in the case using zeolite with free micropores (Supplementary Fig. 62). These results would explain the requirement of zeolite with short *b*-axis thickness for the efficient LT-FTO over Na-FeC_x/zeolite catalysts. In fact, the enhancement of molecule diffusion by employing nano-sized zeolite crystals (and zeolite containing mesopores and/or macropores) in the catalyst granules has been extensively reported²⁸⁻³³ and industrially applied^{28,32-35}. These results might help to understand zeolite with short *b*-axis thickness benefiting the rapid olefin escape compared with the general zeolites.

Supplementary Tables

Supplementary Table 1. Catalytic data of various catalysts in the LT-FTO.^[a]

Catalyst	CO	CO ₂	Selectivity (%)						o/a
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₋₁₀ ⁰	C ₅₋₁₀ ⁼	C ₁₀ ⁺ & C-oxy	
Na-FeC _x	1.9	30.3	10.8	6.7	33.0	7.7	32.0	9.8	4.4
Na-FeC _x ^[b]	25.6	45.3	9.2	4.5	23.3	8.0	32.7	22.3	4.5
Na-FeC _x /n-ZSM-5	45.1	38.5	3.2	47.7	1.0	43.6	4.3	0.2	0.1
Na-FeC _x /s-ZSM-5	82.5	46.6	3.0	8.9	29.1	10.9	39.9	8.2	3.5
Na-FeC _x /SSZ-13	39.8	49.0	4.2	7.3	27.1	4.9	24.6	31.9	4.2
Na-FeC _x /Beta	32.5	49.5	3.5	7.2	27.8	4.6	21.1	35.8	4.1
Na-FeC _x /MOR	62.2	46.5	4.5	6.7	23.5	4.8	42.0	18.5	5.7
Na-FeC _x /SAPO-34	67.1	46.6	5.3	5.1	18.6	5.4	23.6	42.0	4.0
Na-FeC _x /ZSM-22	74.1	49.2	3.6	10.6	28.7	12.8	29.1	15.2	2.5

[a] Reaction conditions: weight ratio of zeolite/Na-FeC_x at 3, CO/H₂/Ar at 45/45/10, 2400 mL·h⁻¹·g⁻¹, 2.0 MPa, 260 °C, granule mixture of zeolite and Na-FeC_x.

[b] Reaction temperature at 300 °C.

Supplementary Table 2. Structural parameters of various samples.

Catalyst	S _{BET} (m ² g ⁻¹)	V (cm ³ g ⁻¹)	Si/Al ratio by ICP ^[a]
<i>n</i> -ZSM-5	388.7	0.13	30.5
<i>s</i> -ZSM-5	337.3	0.21	26.2
SSZ-13	542.9	0.27	12.6
Beta	471.1	0.18	13.1
MOR	502.7	0.21	10.4

[a] The Si/Al ratio was obtained by ICP analysis with error bounds at $\pm 3\%$, which is obtained by repeating the measurement for multiple times.

Supplementary Table 3. Catalytic data of the Na-FeC_x/s-ZSM-5 catalyst with different zeolite/Na-FeC_x weight ratios of 2, 3, and 4.^[a]

Zeolite/Na-FeC _x	CO	CO ₂	Selectivity (%)						o/a
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₋₁₀ ⁰	C ₅₋₁₀ ⁼	C ₁₀ ⁺ & C-oxy	
2	66.8	47.4	3.6	7.8	30.8	8.4	37.0	12.4	4.2
3	82.5	46.6	3.0	8.9	29.1	10.9	39.9	8.2	3.5
4	53.2	45.4	2.4	7.3	25.4	9.1	42.2	13.6	4.1

[a] Reaction conditions: CO/H₂/Ar at 45/45/10, 2400 mL·h⁻¹·g⁻¹, 2.0 MPa, 260 °C, granule mixture of zeolite and Na-FeC_x.

Supplementary Table 4. Data showing the performances of *s*-ZSM-5 zeolite for 1-butene isomerization/cracking.^[a]

Item	Tempt. (°C)	Conv. (%)	Product selectivity (%)						
			CH ₄	C ₂	C ₃ ⁼	C ₃ ⁰	i-C ₄ ⁼	n-C ₄ ⁰	i-C ₄ ⁰
1-butene	260	97.8	0	0	0.5	<0.1	99.1	0.1	0.3

[a] Reaction conditions: 0.2 g of *s*-ZSM-5, flow rate at 20 mL/min, feed gas was 1% of C₄H₈, 0.1 MPa, 260 °C.

Supplementary Table 5 Data showing the performances of *s*-ZSM-5 zeolite for 1-hexene isomerization/cracking.^[a]

Item	Temp. (°C)	Conv. (%)	Product selectivity (%)								
			CH ₄	C ₂	C ₃ ⁼	C ₃ ⁰	n-C ₄ ⁼	i-C ₄ ⁼	n-C ₅ ⁼	i-C ₅ ⁼	i-C ₆ ⁼
1-hexene	260	61.9	0	0	1.4	0	0.7	2.4	1.1	2.9	91.5

[a] Reaction conditions: 0.2 g of *s*-ZSM-5, 1-hexene bubbling with a pure N₂ flow at 20 mL/min at 120 °C, 0.1 MPa, 260 °C.

Supplementary Table 6. Data showing the performances of *s*-ZSM-5 zeolite for 1-octene isomerization/cracking.^[a]

Item	Tempt. (°C)	Conv. (%)	Product selectivity (%)												
			CH ₄	C ₂	C ₃ ⁼	C ₃ ⁰	n- C ₄ ⁼	i- C ₄ ⁼	n- C ₅ ⁼	i- C ₅ ⁼	n- C ₆ ⁼	i- C ₆ ⁼	n- C ₇ ⁼	i- C ₇ ⁼	i- C ₈ ⁼
1-octene	260	51.8	0	0	0.3	0	0.2	0.6	0.4	0.7	0.6	1.2	1.1	1.5	93.4

[a] Reaction conditions: 0.2 g of *s*-ZSM-5, 1-octene bubbling with a pure N₂ flow at 20 mL/min at 120 °C, 0.1 MPa, 260 °C.

Supplementary Table 7. Catalytic data of various catalyst in LT-FTO.^[a]

Catalyst	CO	CO ₂	Selectivity (%)						o/a
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₋₁₀ ⁰	C ₅₋₁₀ ⁼	C ₁₀ ⁺ & C-oxy	
Na-FeC _x /n-S1	42.7	46.7	5.0	3.2	13.8	8.0	45.3	24.7	5.3
Na-FeC _x /n-S1-OH	57.5	47.1	6.4	10.0	16.2	20.4	19.9	27.1	1.2
Na-FeC _x /s-S1	50.2	43.8	2.6	4.1	21.4	2.9	24.7	44.3	6.6
Na-FeC _x /s-S1-OH	66.5	47.7	3.0	7.6	32.5	8.3	36.2	12.4	4.3

[a] Reaction conditions: weight ratio of zeolite/Na-FeC_x at 3, CO/H₂/Ar at 45/45/10, 2400 mL·h⁻¹·g⁻¹, 2.0 MPa, 260 °C, and granule mixture of zeolite and Na-FeC_x.

Supplementary Table 8. Catalytic data of the Na-FeC_x/*s*-S1-OH catalyst with H₂/CO ratios of 1, 2, and 3.^[a]

H ₂ /CO	CO Conv. (%)	CO ₂ Sel. (%)	Selectivity (%)						o/a
			CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₋₁₀ ⁰	C ₅₋₁₀ ⁼	C ₁₀ ⁺ & C-oxy	
1	66.5	47.7	3.0	7.6	32.5	8.3	36.2	12.4	4.3
2	81.3	45.2	4.3	8.8	36.8	8.6	32.6	8.9	4.0
3	82.6	37.1	6.0	9.2	39.3	9.2	30.0	6.3	3.8

[a] Reaction conditions: weight ratio of *s*-S1-OH/Na-FeC_x at 3, 2400 mL·h⁻¹·g⁻¹, 2.0 MPa, 260 °C, granule mixture of *s*-S1-OH and Na-FeC_x.

Supplementary Table 9. Catalytic data of the Na-FeC_x/*s*-ZSM-5 catalyst with different mixture manners of Na-FeC_x and *s*-ZSM-5.^[a]

Mixture manner	CO	CO ₂	Selectivity (%)						o/a
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₋₁₀ ⁰	C ₅₋₁₀ ⁼	C ₁₀ ⁺ & C-oxy	
Powder-mixing	21.5	20.2	18.8	17.2	23.8	18.3	21.9	<1	1.3
Dual-bed	13.8	47.6	7.0	13.9	24.6	14.8	30.7	9.0	1.9
Granule-mixing	82.5	46.6	3.0	8.9	29.1	10.9	39.9	8.2	3.5

[a] Reaction conditions: weight ratio of *s*-ZSM-5/Na-FeC_x at 3, CO/H₂/Ar at 45/45/10, 2400 mL·h⁻¹·g⁻¹, 2.0 MPa, and 260 °C.

Supplementary Table 10. Catalytic data of the Na-FeC_x/*s*-ZSM-5 catalysts with different proximities between the Na-FeC_x and *s*-ZSM-5 zeolite in the LT-FTO.^[a]

Na-FeC _x / <i>s</i> - ZSM-5 proximity	CO Conv. (%)	CO ₂ Sel. (%)	Selectivity (%)						o/a
			CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₋₁₀ ⁰	C ₅₋₁₀ ⁼	C ₁₀ ⁺ & C-oxy	
10-20 mesh	77.1	49.4	4.4	8.6	27.8	9.2	42.6	7.4	4.0
20-40 mesh	82.5	46.6	3.0	8.9	29.1	10.9	39.9	8.2	3.5
40-60 mesh	88.3	45.8	7.1	11.4	38.0	7.3	31.4	4.8	3.7
80-100 mesh	69.8	42.8	11.0	10.0	36.7	6.4	31.0	4.9	4.1
Powder-mixing	21.5	20.2	18.8	17.2	23.8	18.3	21.9	<1	1.3

[a] Reaction conditions: weight ratio of *s*-ZSM-5/Na-FeC_x at 3, CO/H₂/Ar at 45/45/10, 2400 mL·h⁻¹·g⁻¹, 2.0 MPa, 260 °C.

Supplementary Table 11. Catalytic data of the Na-FeC_x/*s*-ZSM-5 catalyst with GHSV of 2400, 4800, and 9600 mL·h⁻¹·g⁻¹ [a]

GHSV (mL·h ⁻¹ ·g ⁻¹)	CO	CO ₂	Selectivity (%)						o/a
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₋₁₀ ⁰	C ₅₋₁₀ ⁼	C ₁₀ ⁺ & C-oxy	
2400	82.5	46.6	3.0	8.9	29.1	10.9	39.9	8.2	3.5
4800	77.2	48.2	2.8	8.7	36.7	9.9	37.1	4.8	4.0
9600	43.6	48.1	3.2	7.7	41.3	8.1	36.8	2.9	4.9

[a] Reaction conditions: weight ratio of *s*-ZSM-5/Na-FeC_x at 3, CO/H₂/Ar at 45/45/10, 2.0 MPa, 260 °C, and granule mixture of *s*-ZSM-5 and Na-FeC_x.

Supplementary Table 12. Catalytic data of the Na-FeC_x/s-S1-OH catalyst with GHSV of 2400, 3600, and 4800 mL·h⁻¹·g⁻¹.^[a]

GHSV (mL·h ⁻¹ ·g ⁻¹)	CO	CO ₂	Selectivity (%)						o/a
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₋₁₀ ⁰	C ₅₋₁₀ ⁼	C ₁₀ ⁺ & C-oxy	
2400	66.5	47.7	3.0	7.6	32.5	8.3	36.2	12.4	4.3
3600	40.7	45.6	3.4	7.2	30.4	8.6	37.0	13.4	4.2
4800	29.9	42.3	3.7	6.9	28.3	8.7	37.8	14.6	4.2

[a] Reaction conditions: weight ratio of s-S1-OH/Na-FeC_x at 3, CO/H₂/Ar at 45/45/10, 2.0 MPa, 260 °C, and granule mixture of s-S1-OH and Na-FeC_x.

Supplementary Table 13. Intensities of band in IR spectra of C₂H₄ and the residue percentage in 0-20 min over various catalysts.

Time (min)			0	4	8	12	16	20
<i>s</i> -ZSM-5	V _s =CH ₂	Strength ^[a]	0.1206	0.0226	0.0134	0.0041	0.0013	0.0000
	2986 cm ⁻¹	Percentage (%) ^[b]	100	6.8	4.0	1.2	0.4	0.0
	V _{as} =CH ₂	Strength	0.1670	0.0329	0.0118	0.0004	0.0000	0.0000
	1445 cm ⁻¹	Percentage (%)	100	9.0	3.2	0.1	0.0	0.0
Na-FeC _x	V _s =CH ₂	Strength	0.3238	0.2974	0.1988	0.0941	0.0502	0.0226
	2986 cm ⁻¹	Percentage (%)	100	91.8	61.4	29.1	15.5	7.0
	V _{as} =CH ₂	Strength	0.3856	0.3486	0.1825	0.0586	0.0333	0.0243
	1445 cm ⁻¹	Percentage (%)	100	90.4	47.3	15.2	8.6	6.3
Na-FeC _x / <i>n</i> -ZSM-5	V _s =CH ₂	Strength	0.3638	0.3174	0.0986	0.038	0.0153	0.0053
	2986 cm ⁻¹	Percentage (%)	100	89.6	61.4	30.6	15.1	8.7
	V _{as} =CH ₂	Strength	0.3767	0.3174	0.0986	0.038	0.0153	0.0053
	1445 cm ⁻¹	Percentage (%)	100	91.0	43.2	16.5	6.0	3.0
Na-FeC _x / <i>s</i> -ZSM-5	V _s =CH ₂	Strength	0.3327	0.294	0.1493	0.0547	0.0281	0.0105
	2986 cm ⁻¹	Percentage (%)	100	88.4	44.9	16.4	8.4	3.2
	V _{as} =CH ₂	Strength	0.3638	0.3174	0.0986	0.038	0.0153	0.0053
	1445 cm ⁻¹	Percentage (%)	100	87.3	27.1	10.4	4.2	1.5

[a] The value of absorbance intensity in the IR spectra.

[b] Calculated from the residue percentage of 2986 cm⁻¹ and 1445 cm⁻¹ bands in 0-20 min, respectively.

Supplementary Table 14. Intensities of band in IR spectra of C₃H₆ and the residue percentage in 0-20 min over various catalysts.

Time (min)			0	4	8	12	16	20
<i>s</i> -ZSM-5	V _{as} =CH ₃	Strength ^[a]	0.4280	0.033	0.0243	0.0192	0.0088	0.0096
	2952 cm ⁻¹	Percentage (%) ^[b]	100.0	4.1	3.0	2.4	1.1	1.2
	δ=CH ₂	Strength	0.4503	0.0237	0.0142	0.0020	0.0028	0.0026
	910 cm ⁻¹	Percentage (%)	100.0	3.0	1.8	0.3	0.4	0.3
Na-FeC _x	V _{as} =CH ₃	Strength	0.7632	0.6942	0.4001	0.1564	0.0589	0.0265
	2952 cm ⁻¹	Percentage (%)	100.0	91.0	52.4	20.5	7.7	3.5
	δ=CH ₂	Strength	1.0476	0.9437	0.5867	0.2647	0.1069	0.0520
	910 cm ⁻¹	Percentage (%)	100.0	90.1	56.0	25.3	10.2	5.0
Na-FeC _x / <i>n</i> -ZSM-5	V _{as} =CH ₃	Strength	0.8982	0.8351	0.5205	0.2532	0.1469	0.1115
	2952 cm ⁻¹	Percentage (%)	100.0	93.0	57.9	28.2	16.4	12.4
	δ=CH ₂	Strength	0.9723	0.9470	0.6060	0.2788	0.1117	0.0653
	910 cm ⁻¹	Percentage (%)	100.0	97.4	62.3	28.7	11.5	6.7
Na-FeC _x / <i>s</i> -ZSM-5	V _{as} =CH ₃	Strength	0.7971	0.7065	0.3369	0.0871	0.0299	0.0161
	2952 cm ⁻¹	Percentage (%)	100.0	88.6	42.3	10.9	3.8	2.0
	δ=CH ₂	Strength	0.7967	0.7365	0.4355	0.1443	0.0411	0.0275
	910 cm ⁻¹	Percentage (%)	100.0	92.4	54.7	18.1	5.2	3.5
Na-FeC _x / <i>s</i> -S1-OH	V _{as} =CH ₃	Strength	0.9182	0.4996	0.1742	0.0682	0.0304	0.0127
	2952 cm ⁻¹	Percentage (%)	100.0	54.4	19.0	7.4	3.3	1.4
	δ=CH ₂	Strength	1.0715	0.6487	0.2802	0.1181	0.0452	0.0257
	910 cm ⁻¹	Percentage (%)	100.0	60.5	26.2	11.0	4.2	2.4

[a] The value of absorbance intensity in the IR spectra.

[b] Calculated from the residue percentage of 2952 cm⁻¹ and 910 cm⁻¹ bands in 0-20 min, respectively.

Supplementary Table 15. Catalytic performances of Na-FeC_x with different Na loading in the LT-FTO tests.^[a]

Catalyst	Na	CO	CO ₂	Selectivity (%)						C ₁₀ ⁺ & C-oxy	o/a
	loadings (wt%)	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₋₁₀ ⁰	C ₅₋₁₀ ⁼			
5.35NaFeC _x	5.35	1.9	30.3	10.8	6.7	33.0	7.7	32.0	9.8		4.4
4.08NaFeC _x	4.08	9.1	38.3	8.7	7.7	31.4	6.5	31.9	13.8		4.4
4.08NaFeC _x /s-ZSM-5	4.08	77.7	46.4	2.9	8.5	33.5	11.3	37.8	6.0		3.6
1.30NaFeC _x	1.30	14.6	43.7	13.2	13.4	31.0	12.7	23.0	6.7		2.0
Na-free FeC _x	<0.04	17.3	38.7	18.6	12.5	28.9	12.2	20.8	7.0		1.8
Used-5.35NaFeC _x	5.19	6.0	34.2	9.4	6.8	30.0	7.3	33.5	13.0		4.5

[a] Reaction conditions: weight ratio of zeolite to Na-FeC_x at 3, H₂/CO at 1, 2400 mL·h⁻¹·g⁻¹, 2.0 MPa, 260 °C, granule-mixing of the zeolite and Na-FeC_x individuals.

Note: Discussion to exclude the Na amount change on influencing catalytic activity. The FeC_x catalysts have been extensively used for the classical Fischer-Tropsch synthesis reactions at low temperatures (e.g. 240-280 °C), giving high CO conversion but poor selectivity to olefins. In these cases, the reactions followed a classical Fischer-Tropsch route that converts syngas to alkanes³⁶⁻³⁹. With regard to the Fischer-Tropsch synthesis to olefins (FTO), more alkaline species were necessary to reduce the catalyst hydrogenation activity and weaken the olefin adsorption on Na-FeC_x surface, thus to hinder the deep hydrogenation and selectively obtain olefins^{1,40-44}. These Na-FeC_x catalysts for FTO processes have to work at higher reaction temperatures to obtain desired CO conversions (e.g. higher than 320 °C) because of their reduced activity compared with the FeC_x catalysts for classical Fischer-Tropsch synthesis. In the previously tested catalysts for FTO, the alkaline modified FeC_x exhibited extremely low CO conversions at temperatures lower than 300 °C^{41,43,45}, in good agreement with the performance of Na-FeC_x catalyst in this work. This Na-FeC_x catalyst is synthesized following the classical method described in the previous reference^{1,46,47}, and showed similar performances to that in literatures under their described reaction conditions (e.g. at 340 °C)¹.

We rationally synthesized the Na-FeC_x catalyst with Na loading at 4.08%, denoted as 4.08Na-FeC_x, which exhibited a higher CO conversion (9.1%, Supplementary Table 15) than the Na-FeC_x catalyst (CO conversion of 1.9%), but still far to meet the level using zeolite promoter. After mixing with the s-ZSM-5 promoter (4.08Na-FeC_x/s-ZSM-5), the CO conversion could

reach as high as 77.7% with olefins as a dominant product. The Na-FeC_x catalyst with Na loading at 1.3% (1.30Na-FeC_x) showed CO conversion at 14.6%. Even for the Na-free FeC_x catalyst (Na loading <0.04%), the CO conversion was 17.3% with abundant alkane products, in good agreement with the phenomenon in classical Fischer-Tropsch synthesis. These results confirm that the sodium loading to influence the catalytic activity, but still much lower than that using zeolite promoters.

We also evaluated the used Na-FeC_x catalyst (after reaction, the zeolite individual of used Na-FeC_x/S-ZSM-5 catalyst was removed to obtain the used Na-FeC_x, and reused in the next run) in the LT-FTO reaction process, showing the CO conversion at 6.0%.

Supplementary Table 16. Parameter of the force fields for zeolites and ethene in theoretical simulation.

Atom type	ϵ (kcal/mol)	σ (Å)	q (e)
Oa [Si-Oa-Al] ¹¹	0.115	3.600	-0.840
Oh [Si-Oh-H] ¹²	0.122	3.090	-0.675
Os [Si-Os-Si] ¹⁰	0.105	3.300	-0.750
Ha [Al-O-Ha] ¹¹	0.000	0.000	0.500
Hs [Si-O-Hs] ¹²	0.015	0.970	0.400
Sh [Si-O-H] ¹²	0.093	3.700	1.400 ^[a]
Si [O-Si-O] ¹⁰	0.044	2.300	1.500
Al [O-Al-O] ¹¹	0.044	2.300	1.360
CH ₂ [CH ₂ -CH ₂] ⁹	0.169	3.675	0.000

[a] The charge has been adjusted to keep the total charge equal 0.

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